

A Promised Grant and Stated Attitudes toward Early Marriage: Experimental Evidence from Burkina Faso*

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Abstract

We evaluate a women empowerment program in rural Burkina Faso. The program was randomized at the village level and included three experimental arms and a control group. Two of the experimental arms delivered the same girl-targeted content to out-of-school girls and young wives, and one also carried an announced \$400 grant, undisbursed when outcomes were measured. Stated attitudes toward early marriage moved sharply only where the grant was announced: the index-level announcement margin is 0.26 standard deviations. The response extends beyond participants: girls who never attended a session moved about 70 percent as much as attendees, and girls too young for grant eligibility moved as much as eligible girls. Household heads moved in parallel. Because no cash had been paid, the reach beyond participants cannot be an income spillover. Incremental effects on marriage, fertility, and income above universal community sensitization were mostly null, estimated under acute insecurity. Whether the stated-attitude response reflects normative change or affirmation of an announced program is not separable in these data.

Keywords: Safe spaces; Adolescent girls; Income-generating activities; Social Norms; Local Economies
JEL codes: J12, J13, J16

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1 Introduction

Early marriage remains common in many low-income settings, with significant implications for human capital accumulation, intergenerational welfare, and economic development in general (Field and Ambrus, 2008; Roychowdhury and Dhamija, 2021; Chari et al., 2017; Sunder, 2019; Garcia-Hombrados, 2018; Wilson, 2022). Governments and international organizations increasingly target adolescent girls with empowerment programs that combine life skills and business training. Yet the impacts of such interventions on marriage and fertility remain, by and large, ambiguous.

In this paper, we exploit a design feature of a large-scale randomized trial—the Sahel Women’s Empowerment and Demographic Dividend (SWEDD) program in Burkina Faso, targeting out-of-school adolescent girls and young wives—in which the same girl-targeted curriculum was delivered with and without an announced \$400 grant, undisbursed at the time outcomes were measured. The announcement, not the curriculum, is what moved outcomes: normative attitudes toward early marriage shifted by roughly 20 percentage points only in the announced-grant arm, and the shift ran through girls who never attended a program session as much as through participants—the pattern which a village-level response to an announced resource inflow predicts, and one that curriculum internalization alone cannot produce. Incremental effects on marriage, fertility, and income generation above the community sensitization that reached every study village were, by contrast, mostly null at follow-up.

We organize the interpretation with a simple Becker-type bride-price marriage-market model (Becker, 1973, 1974). In our model, supply-side reductions in the willingness to marry young and demand-side increases in the value of trained brides can offset. We also characterize—as exploratory evidence—how treatment responses vary with local norms, economic infrastructure, and security, a pattern that underscores the importance of characterizing local contexts in program evaluations.

A growing body of research examines how interventions that expand women’s human capital and economic opportunities affect their decisions on marriage and fertility. Specifically, programs offering schooling, vocational training, or cash transfers have been shown to delay marriage and childbearing in some contexts (Baird et al., 2011; Jensen, 2012; Bandiera et al., 2020; Ingwersen et al., 2019; Amin et al., 2018), while failing to do so in others (Buchmann et al., 2023; Edmonds et al., 2023; Austrian et al., 2020; Chow and Vivalt, 2022). One possible explanation is that these interventions alter both sides of the marriage market. In settings where bride price transfers flow from the groom’s family to the bride’s, the skills and economic potential that empowerment programs confer can make young women more attractive spouses, offsetting any supply-side reduction in early marriage; Ashraf et al. (2020) document precisely this capitalization of female human capital into bride-price values, showing that education investments respond to the custom. We formalize this logic in a simple Becker-style model, where training shifts both the supply and demand for young brides, resulting in ambiguous effects on marriage age in equilibrium. The model helps rationalize the mixed empirical findings that characterize the existing literature.

We test these predictions using a three-arm randomized controlled trial of the SWEDD program in 162 villages in two regions of Burkina Faso, targeting out-of-school girls and young wives. The

arms layered community safe-space training, business and vocational skills training, and a \$400 cash grant. The grant was not disbursed at the time of the follow-up survey, so results for the full-package arm capture the impact of expecting a transfer, not receiving one. Details are provided in Section 2. We estimate ITT effects via ANCOVA with PDS LASSO-selected controls, then apply the generic machine learning framework of Chernozhukov et al. (2025) to characterize how local social norms, economic infrastructure, and access to reproductive health services shape heterogeneous treatment responses.

This paper makes three contributions to the literature on gender, human capital, and marriage markets in developing countries. First, it estimates the effect of assignment to an arm carrying an announced future transfer, relative to the same content offered without it, inside a government-implemented program at scale. The estimand is an arm contrast: the two arms differ by the promise and by the higher take-up the promise induced (33 against 24 percent), and we show in Section 6.1.2 that the take-up difference can account for at most a tenth of the contrast. Because the grant was disbursed only after outcomes were measured, the contrast is a response to a credible announcement rather than to cash. The announcement roughly doubled the curriculum-only point effect on the advocacy outcomes: a pre-oriented index of the four normative outcomes gives an announcement margin of +0.26 SD (randomization-inference $p = 0.03$). Across the four components the smallest familywise-adjusted p -value is 0.08. We are not aware of close experimental antecedents on announced-but-not-yet-received transfers moving stated normative attitudes.

The response extends beyond participants. Two-thirds of the arm never attended a program session, yet these girls' attitudes moved about 70 percent as much as attendees', and girls too young for grant eligibility moved as much as eligible girls. Household heads—reached for 86 percent of the baseline roster, including for girls who had left the household—moved by 17 to 24 percentage points with the same announcement-specific pattern, formally tested and surviving familywise correction at the head level. The heads' omnibus gender-norms index, by contrast, stays flat. Because no cash had been paid at measurement, the reach beyond participants cannot be an income spillover—the channel the general-equilibrium cash-transfer literature identifies. These contrasts establish community-wide exposure to the arm; they do not identify transmission from participants to others. This anchors a direct audit of survey response bias.

Second, the paper provides a transparent estimand under adverse conditions: community sensitization reached every study village and follow-up coincided with an acute deterioration of security, so the experiment identifies the incremental effects of girl-targeted offers above universal outreach in a conflict-affected setting. Under that estimand, marriage and fertility effects are mostly null, with ex-post minimum detectable effects reported so the nulls can be read quantitatively. Mental health, measured by the GHQ-8 index (Goldberg et al., 1988) specified in the pre-analysis plan, shows no broad average deterioration. What we find is a marginal self-worth decline concentrated in the most conservative villages and affecting non-participants as much as participants; we probe its mechanism but cannot settle it.

Third, we document—as exploratory, design-based evidence—a channel-specific heterogeneity

pattern. In assignment-by-village-structure interactions that use no machine-learning first stage, assignment raises take-up most in villages with permissive norms and stronger economic infrastructure, while reduced-form gains on pregnancy prevention and income generation are largest where the take-up response is weakest. Decomposing the village-level take-up and outcome CATEs onto the same fixed set of baseline structural predictors corroborates the pattern. This opposite-signed pattern is consistent with participation barriers—social norms on the pregnancy channel, economic infrastructure on the income-generation channel. The marriage-market delay evidence suggests a different relationship: within villages in the announced-grant arm, girls with larger predicted increases in age at marriage also have higher predicted take-up, a pattern consistent with positive selection into participation. Because the moderators are not randomized and observed non-participation mixes individual choice with implementation disruption, the design cannot identify returns to participation or the effect of externally raising participation; we therefore present the pattern as hypothesis-generating.

On the income channel the barrier, reading is moreover observationally equivalent to substitution—the program’s content standing in for what strong infrastructure would otherwise provide—with opposite policy implications. The pattern motivates cross-randomizing candidate complements (norm-change interventions; infrastructure) with girl-targeted programs, which is also the only way to separate the two readings. It also offers one explanation for why null average effects can coexist with large predicted gains for the most constrained girls.

A substantial body of literature examines the determinants of early marriage and assesses the effectiveness of interventions to reduce it. Poverty, limited education and economic prospects, and stringent gender norms have been identified as factors perpetuating early marriage in developing areas (Wahhaj, 2022). Natural experiments using exogenous income or demographic variations show that not only does the supply of young brides available in the marriage market respond to economic circumstances (Khan, 2024), but the direction of the response depends on local marriage norms (Corno and Voena, 2023; Corno et al., 2020; Trinh and Zhang, 2021). When marriage requires a dowry, adverse income shocks contract the supply of brides, as liquidity-constrained families postpone their daughters’ marriages. Conversely, where a bride price is the norm, these shocks expand the supply of young brides, as families turn to early marriage as a coping mechanism. These opposing responses to similar economic shocks—contraction of bride supply in dowry systems versus expansion in bride price systems—illustrate the shifts in the supply of young brides in the marriage market, thus motivating our conceptual framework.

A series of recent studies explore the effectiveness of interventions designed to delay early marriage. Overall, these interventions have produced different results, depending on whether they targeted in-school or out-of-school populations. Programs that sought to keep girls enrolled in formal education—through fee elimination, conditional cash transfers, transportation subsidies, merit scholarships, or school expansion—have consistently demonstrated success in delaying marriage (Baird et al., 2011; Duflo et al., 2015; Giacobino et al., 2024; Ingwersen et al., 2019; Garcia-Hombrados, 2018). Arguably, these education-related interventions operate through clear mechanisms: they

raise the opportunity cost of early marriage by keeping girls enrolled in school, provide a visible marker of continued investment in human capital, and often carry explicit conditionality that directly penalizes early marriage or dropout.

In contrast, interventions focusing on out-of-school girls—presumably the population most vulnerable to early marriage—have yielded somewhat mixed results. Specifically, safe spaces alone, which provide empowerment training and information on reproductive health, consistently fail to delay marriage (Buchmann et al., 2023; Boulhane et al., 2024). Programs that combine safe spaces with vocational training and income-generating activities show context-dependent effects: positive impacts in Uganda (Bandiera et al., 2020), null effects in Tanzania (Buehren et al., 2024, 2017), and heterogeneous results across treatment combinations in Côte d’Ivoire (Boulhane et al., 2024). This pattern suggests that once girls have left the formal education system, interventions face greater challenges. They must compete with existing marriage market pressures, operate within communities where early marriage may be the norm, and rely on voluntary participation rather than institutionalized engagement. Understanding when out-of-school interventions produce heterogeneous effects is, therefore, critical for designing policies that reach the most vulnerable populations.

The closest antecedent to our identifying contrast is Buchmann et al. (2023): in Bangladesh, a safe-spaces empowerment curriculum alone did not delay marriage while a conditional incentive did. Our design contains the same comparison inside one trial (the same content offered with and without a \$400 grant), but at the *announcement* margin: the grant had not been paid when outcomes were measured. Therefore the contrast excludes the payment; leaving the assignment to the promise, together with the higher take-up the promise induced. The nearest promise-based antecedent reports the mirror image of our pattern: evaluating Haryana’s Apni Beti Apna Dhan bonds (payable at 18 if the girl remained unmarried), Biswas and Das (2026) find that the promise reduced child marriage and raised education in the long run but did not alter gender norms, whereas our announcement moves stated attitudes with (as yet) no marriage behavior. However, the two designs differ. Biswas and Das (2026) condition the payment on marriage postponement over a decade, while we announce an unconditional windfall measured within eighteen months. Thus, the pair brackets which margin a promise moves at which horizon. Online Appendix Table S21 places the three designs side by side; we do not attempt a unified model of conditional and unconditional promises, whose comparative statics would rest on the unobserved bride-price primitives of Appendix A.

The mechanism we uncover is consistent with two strands of literature. Announced future transfers changing current behavior connects to work on anticipation and expectations (Bianchi and Bobba, 2013; Attanasio et al., 2012; Macours and Vakis, 2014); and effects reaching girls and households that never engaged with the program connect to the general-equilibrium and spillover literature on cash transfers (Egger et al., 2022; Angelucci and De Giorgi, 2009; Cunha et al., 2019). We lack the designs that identify spillover channels directly (Baird et al., 2018)—the announcement varied at the village level, so we observe reach beyond participants but cannot identify how it traveled.

On the norms side, three strands anchor our reading. Ashraf et al. (2020) provide the bride-price economics on which our framework builds. Bau (2021) shows that policy can move marriage-linked cultural practices, and the media literature (Jensen and Oster, 2009; La Ferrara et al., 2012) supplies the magnitude benchmarks for normative change against which we compare our estimates.

Bandiera et al. (2020) evaluate NGO-run adolescent clubs against a no-program control and find delays in marriage and childbearing. Their estimand differs from ours on both margins. Our comparison villages received community sensitization, so our nulls are incremental to universal outreach rather than to no program, and our implementer is a government rather than an NGO.

The study most directly related to ours is the multi-country evaluation of the same generation of SWEDD adolescent-girls clubs (Bertelli et al., 2026), which establishes the cross-country pattern of average impacts. That evaluation pools the safe-spaces component (T1) across four countries, including Burkina Faso. The two arms that carry this paper’s contributions, T2 (training) and T3 (training plus announced grant), were implemented only in Burkina Faso and are analyzed only here, as are the baseline village questionnaire and the village-level characteristics.

Ours differ from Bertelli et al. (2026) in two significant ways. In Bertelli et al. (2026), the strongest predictors of treatment-effect variation, conditional on individual and household characteristics, are the country indicators themselves. This is a residual attributed to implementation quality and community receptivity but not measured. Here, a baseline village questionnaire and a fixed set of village-level moderators collected before rollout locate that heterogeneity across villages within one institutional and marriage-market context, and the evidence comes from government implementation at scale under acute insecurity.

2 Experimental Design and Setting

2.1 Program description

The SWEDD project is a regional initiative implemented in seven countries¹. The program received financial support from the World Bank and technical assistance from the United Nations Population Fund (UNFPA) Regional Office for West and Central Africa. It aimed at accelerating the demographic transition by addressing both supply- and demand-side barriers to family planning and sexual and reproductive health. To achieve this goal, the project focused on adolescent girls and young women, primarily between the ages of 10 and 24, who are particularly vulnerable to early marriage, teenage pregnancy, and school dropout. Its key objectives include strengthening human capital and learning outcomes, reducing child marriage and early pregnancy, enhancing women’s financial autonomy, and improving access to quality reproductive, maternal, newborn, child, and nutritional health (RMNCNH) services. By advancing these goals, the program seeks to reduce dependency ratios and increase young women’s participation and productivity in the labor market, enabling countries to harness the full potential of the demographic dividend.

¹These countries are Benin, Burkina Faso, Chad, Côte d’Ivoire, Mali, Mauritania, and Niger.

In Burkina Faso, the program included two sub-projects: *Sukaabé Rewlé* (literally, “Protecting the Girl Child”), and *Entreprendre au Féminin* (Women in Business). Hereon, we refer to *Sukaabé Rewlé* as “Safe Spaces” and *Entreprendre au Féminin* as “Female Entrepreneurship”. The Safe Spaces Intervention aimed to reduce child marriage by: (i) strengthening adolescent girls’ knowledge and skills in reproductive health, life skills, rights, and harmful traditional practices; (ii) increasing parents’ and communities’ awareness of reproductive health and harmful social norms; and (iii) supporting families with girls at risk of or affected by child marriage. The Female Entrepreneurship intervention sought to enhance the economic autonomy of out-of-school and unschooled adolescent girls and young women by: (i) developing their entrepreneurial and leadership capacities; (ii) expanding employment opportunities; and (iii) improving access to financing for income-generating activities. The rationale was to address both social norms and economic constraints that sustain early marriage.

The Safe Space intervention focused on creating safe spaces for adolescent girls and young women, where participants received training on life skills, sexual and reproductive health, gender norms and stereotypes, nutrition, gender-based violence, and women’s and children’s rights.

Specifically, out-of-school girls aged 10–19 and married girls and young women under 24 met twice a week—about eight sessions per month over roughly five months beginning in mid-June 2019—in community venues, guided by locally respected female mentors.² The curriculum focused on two main areas: (i) life skills and health, including self-esteem, leadership, contraceptive use, and menstrual hygiene; and (ii) broader well-being and rights, covering issues like gender norms, nutrition, and gender-based violence. Materials were adapted for two age groups (9–12, and 13 and older), and girls participated in selecting meeting sites to ensure safety and privacy.

Mentors were women from the same communities, selected by the implementing partner, a local Non-Governmental Organization known as the *Association Songui Manegre Aide Au Developpement Endogene* (ASMADE). Across all sites, mentors delivered more than 7,000 sessions and covered 26 of the curriculum’s 32 modules; the six financial-capability modules were not included in the initial mentor training. Participants and their families gave final approval to ensure that mentors were trusted role models. While the preferred profile included women aged 25–45 with at least a ninth-grade education, some mentors lacked formal schooling due to limited availability, which may have affected the quality of delivery in a few locations.

The Female Entrepreneurship intervention targeted beneficiaries of safe spaces who were at least 14 years old at the start of the program. The entrepreneurship training curriculum included two components: (i) financial and business skills—such as entry costs, business planning, management, and cost tracking—and (ii) basic vocational training tailored to local markets. Participants were trained in various skills, including soap making, hairdressing, and drying local fruits and vegetables. A subset of participants in this intervention was scheduled to receive a lump sum grant of \$400 per beneficiary, intended to support income-generating activities. However, the grant was not yet

²The lower age limit was 10 for unmarried out-of-school girls but 8 for girls already married at baseline; 13 girls aged 9 or younger who were married at baseline appear in the sample. The younger curriculum age band (9–12) accommodates these participants.

disbursed at the time of the follow-up survey; it was disbursed after the survey was completed. The announced transfer was therefore a genuine commitment that was eventually honored, but every outcome in this paper is measured before any girl had received it, so the full-package estimates capture responses to the announcement alone. No survey round after the disbursement exists, so post-disbursement effects cannot be measured in these data.

Across all SWEDD villages, including both treated and control ones, the program also engaged influential community figures—including religious leaders, village chiefs, and women leaders—to promote messages around gender equality, family planning, and girls’ empowerment. These leaders received training on topics such as girls’ education, gender-based violence, birth spacing, and the risks associated with early pregnancy and marriage. They then disseminated key messages through multiple channels: sermons, community meetings, and educational talks with parents. The intervention also reached a broader group of indirect beneficiaries, including girls’ families, local authorities (e.g., councilors, prefects, governors), and professionals in health, education, and social services. Leaders from civil society, community-based organizations, and NGOs, along with the wider community, were also included in outreach efforts.

Estimand and the lack of pure controls. Because the community-sensitization arm reached every SWEDD village, including those randomized to control, the experimental contrast does not identify the gross effect of the program relative to zero exposure. Instead, our intent-to-treat estimates identify the *incremental* effect of the randomized girl-targeted components (Safe Spaces in T1, Safe Spaces + vocational training in T2, the full package + announced grant in T3) relative to a common outreach environment.

The implications for interpretation are twofold. First, if outreach alone shifted norms or behavior in control villages, our ITT understates the gross effect of the SWEDD program; the null incremental effects on marriage and fertility documented below are therefore consistent with a substantively meaningful sensitization channel that we cannot identify experimentally. Second, we cannot rule out by design an alternative reading: that outreach in treatment villages crowded out take-up of the girl-targeted components, generating attenuated ITTs even where the components would otherwise have mattered. The participation-barrier evidence in Section 6.3 speaks directly to take-up dilution and shows that low participation is most pronounced in conservative villages where the predicted gains are largest. A within-village competition between the outreach and girl-targeted channels would generate this same pattern only under a specific condition: conservative gatekeepers must treat community sensitization as sufficient engagement and restrict further girl-targeted participation. The village-level evidence is consistent with that reading but does not by itself adjudicate between the two mechanisms.

Two pieces of evidence bear on the related distinction between offsetting channels and diminishing returns above a common outreach base. First, the pre-post comparison in control villages is only partially informative, and we state its limits. The share of girls who actively advocate early marriage is essentially unchanged between baseline and follow-up (0.446 versus 0.439), while the

share who disagree that early marriage is appropriate rises by about 15 percentage points (0.369 to 0.521). The count measures cannot be compared across waves because the baseline instrument recorded only whether any disadvantage (advantage) was mentioned—the baseline variables are binary—whereas the follow-up variables are counts. The comparable binary margins are therefore mixed (advocacy flat, disagreement higher), and the comparison cannot separate outreach from aging, instrument, and secular trends, so it neither establishes nor rules out an outreach effect on norms in control villages.

Second, if control girls had already harvested the outreach gains, incremental effects on normative attitudes should be attenuated. Instead, the full package produces large normative shifts relative to the outreach-only control (Section 6.1.2). That rules out full saturation of this margin by follow-up, though it does not bound how far outreach had already moved the control group.

2.2 Sampling

The study covers 11 provinces in two regions. Initially, we randomly selected 200 villages from these 11 provinces (Kazianga and Combarry, 2022). However, the security situation at the time of the baseline survey led to the exclusion of 25 villages, which the data collection teams could not reach. Moreover, during random allocation, two other villages with fewer than 10 adolescent girls were excluded, leaving us with 173 villages at the start of program implementation. Additional villages were dropped during the follow-up survey, again for security reasons, leaving 162 villages observed at both survey rounds. The analysis focuses on these 162 villages after establishing that the villages we dropped due to safety concerns did not differ significantly from the remaining ones in terms of their baseline characteristics. Online Appendix Figure S1 summarizes the participant flow—village selection, the baseline roster of 7,719 girls, assignment counts by arm, and follow-up counts—under a single denominator convention: all attrition percentages in the paper are relative to the baseline analysis roster.

In each sampled village, a household-level census—administered by INSD, the survey firm, immediately before the baseline survey—first identified all adolescent girls and young women (AGYW), recording age, marital status, school enrollment, and future schooling plans. Eligible individuals were out-of-school girls aged 10 to 19 and married girls and young women younger than 24—with a lower age limit of 8 for those already married—with no plans to return to school at the time of the census. Consent to participate in the program was sought at this stage from all eligible AGYW and their parents,³ and all households with at least one eligible person constituted the universe from which the survey sample was drawn.

A total of 8,327 adolescent girls and young women and their 7,241 households were surveyed. Two populations must be distinguished. Program eligibility was defined by the implementer: out-of-school girls aged 10 to 19 and married girls and young women under 24. The analysis roster is broader: it retains married girls and young wives aged 24 or younger and unmarried girls aged 12 to 24, without conditioning on schooling status, because the field survey cast a wider net than

³Consent was sought from parents for individuals younger than 18.

the program rules and because trimming the roster on characteristics that the program could itself affect would compromise the intention-to-treat comparison. It excludes the 608 baseline interviews (7.3 percent) that fall outside these bounds: unmarried girls younger than 12 or older than 24, and married respondents older than 24. The resulting baseline analysis roster of 7,719 girls in 173 villages is the denominator for the arm counts, the balance and attrition analyses, and the worst-case bounds throughout the paper.

Appendix Figure D.1 summarizes the study timeline. Baseline data collection took place between October and November 2018. In December 2018, a public lottery was held in each region to randomly assign villages to experimental arms. The interventions were conducted from mid-June to December 2019 for safe space training and from October to December 2019 for entrepreneurship training. The follow-up survey that concluded this process took place between May and July 2021, a year and a half after the end of the training activities.

The baseline survey included three modules: (i) households, (ii) adolescent girls, and (iii) villages. Information collected in the household module covered household structures and household heads’ perceptions of early marriage and childbirth. The adolescent girls’ module, administered separately to each girl, collected information on marriages, life skills, and educational and employment aspirations, among others. The village module, on the other hand, collected information on social norms related to marriage in villages, socioeconomic infrastructure, and access to basic public goods and services. The follow-up surveys covered all the baseline modules. Furthermore, in the household module, household heads provided information on adolescent girls’ marriages and childbirth even when they had left the household at the time of the follow-up survey. The adolescent module was also modified to include questions about participation in program activities and questions about psychosocial well-being and mental health.

2.3 Experimental design and data collection

We developed the research design jointly with Burkina Faso’s SWEDD program and the Ministry of Women, National Solidarity, Family, and Humanitarian Action. The experimental design exploited the rollout of the two program interventions described in subsection 2.1. We randomly selected villages to receive either Safe Spaces (SS) only, Safe Spaces and Female Entrepreneurship without the grant, or Safe Spaces and Female Entrepreneurship with the grant. Appendix Figure D.2 summarizes the design, and Appendix Figure D.3 maps the study villages by experimental arm.

Villages were randomly assigned within each province to one of four groups. These included (i) safe spaces only, (ii) safe spaces and female entrepreneurship training, (iii) the nominal “full package” consisting of safe spaces, entrepreneurship training, and an announced financial subsidy, and (iv) a pure control group. As we discussed earlier, the grant was not disbursed at the time of the follow-up survey, so the second and third groups differed only in the promise of a future transfer. We nonetheless analyze the two arms separately throughout, because the announced grant may itself have affected participation incentives and behavior; the evidence below bears this out (take-up and normative attitudes differ significantly between the two arms; see Table S14). Of the three

arms, only safe spaces (T1) was offered beyond Burkina Faso and enters the pooled four-country evaluation (Bertelli et al., 2026); the T2 and T3 arms and the baseline village questionnaire are specific to Burkina Faso and to this paper.

Randomization was conducted publicly after baseline data collection, with village representatives invited to the regional capitals of Dédougou and Fada N’Gourma to witness the lottery (Appendix Figure D.1). Both the baseline and follow-up surveys were fielded by the Institut National de la Statistique et de la Démographie (INSD), the national statistical office, independently of program implementation. The questionnaire did not mention the grant in either round; Section 6.1.2 discusses the implications for survey response bias.

The evaluation was registered with the Registry for International Development Impact Evaluations after the interventions were delivered but before any follow-up data were collected; Online Appendix S2 details the registration—including the pre-specified outcome domains and village-level heterogeneity dimensions—and maps every analysis reported in the paper to its registration status. Two registration facts bear directly on how the results should be read: the early-marriage attitude domain is registered, but the T2-versus-T3 announcement contrast that this paper centers is not among the registered hypotheses, so the announcement margin is a post-hoc contrast within a registered outcome domain, and the index that summarizes it was constructed after the component results were known. Throughout the paper, *announcement margin* denotes the difference between the announced-grant arm and the training-only arm. It is the effect of assignment to the promise together with whatever differences in delivery, take-up, and salience accompanied the promise; Section 6.1.2 quantifies the take-up component.

Implementation fidelity. Adherence to the implementation plan was imperfect, and the implementing NGO’s final activity report documents the disruptions.⁴ Insecurity forced the relocation of 22 planned host villages to other zones and the within-zone displacement of 13 further sites. Within the evaluation regions, nine spaces in Sourou province were relocated, and the Est region operated 59 spaces against 72 initially planned. From mid-October 2019, four spaces—all in the Est evaluation region (Gnagna and Tapoa provinces)—suspended operations because of insecurity, one after being vandalized.

Delivery was also incomplete along the curriculum margin. Mentors covered 26 of the 32 modules, the six financial-capability modules having been left out of the initial mentor training. That gap is relevant to the null finance effects reported in Section 6.1.2. Finally, while every space had two assigned mentors, mentors’ limited formal schooling was cited as an obstacle to delivery quality in some locations.

These disruptions matter for interpretation. Measured take-up conflates girls who chose not to participate with girls whose local safe space was relocated, displaced, or suspended. To the extent that disruptions correlate with village-level security and infrastructure, part of the cross-village

⁴ASMADE, *Rapport définitif du sous-projet Sukaabé Rewlé*. Nationally, the sub-project operated 193 safe spaces across 21 provinces in five regions, staffed by 386 mentors (two per space); the two impact-evaluation regions, Boucle du Mouhoun and Est, hosted 137 of these spaces across the 11 study provinces.

variation in participation documented in Section 6.3 may reflect variation in program availability rather than in demand for participation alone.

3 Conceptual Framework

We organize interpretation with a simple model of adolescent empowerment in a bride-price marriage market, developed formally in Appendix A. The model has two parts. A marriage-market module notes that girl-targeted programs move both sides of the market: training raises the value of remaining unmarried, but in bride-price settings it can also raise the perceived value of a trained bride (Becker, 1973, 1974; Ashraf et al., 2020)—two margins that can offset, so null average effects on marriage timing are consistent with successful human-capital transmission rather than only with program failure (Corno et al., 2020; Corno and Voena, 2023). A participation module notes that program gains arrive only through take-up, and the villages where predicted gains are largest may be those where participation is hardest—the pattern the exploratory heterogeneity analysis of Section 6.3 examines. The framework yields one signed moderator prediction (SRH access), one joint sign-reversal prediction across outcome families (exogamy, in which the marriage-delay leg is tied to male exogamy and the IGA leg to female exogamy), and sign-ambiguous comparative statics for two other moderators (Appendix A.6). Its empirical status, reported below, is mixed: the signed SRH-access prediction fails in the data, and the exogamy sign-reversal is supported as a joint prediction only in the training-only arm, so the framework functions in this paper as an organizing device rather than as a tested structure.

4 Empirical Strategy

4.1 Econometric Specifications

We start with the following standard ANCOVA specification.

$$y_{iv1} = \alpha_0 + \sum_{d=1}^3 \beta_d T_{dv} + \alpha_1 y_{iv0} + X'_{iv0} \alpha_2 + \varepsilon_{vi}, \quad (1)$$

where y_{iv1} is the outcome at follow-up for individual i living in village v , T_d for $d = 1, 2, 3$ are treatment indicators, y_{iv0} is outcome y measured at baseline, X_{iv0} is a set of baseline controls that include strata fixed effects, age dummies, and household and individual characteristics, and we cluster standard errors at the village level (unit of randomization). The omitted category should be interpreted as “outreach only” because village-wide sensitization occurred in all study villages during the program period. Coefficients on treatment assignment therefore identify ITT effects of the treatment arms relative to outreach-only exposure. Accordingly, we refer to outreach-only villages as the control arm.⁵

⁵The experimental design does not separately identify the standalone effect of outreach.

The controls in X_{iv0} are selected from 65 candidate baseline household and individual characteristics by the post-double-selection (PDS) LASSO (Belloni et al., 2012, 2014), which retains the union of covariates predicting the outcome and covariates predicting assignment.⁶ The results do not hinge on this choice. Online Appendix Table S5 re-estimates the main ITTs with no covariates beyond strata fixed effects and with a simple prespecified set (strata, age dummies, baseline outcome). All three specifications produce closely similar coefficients.

The decision to focus the heterogeneity analysis on village-level characteristics is rooted in the central hypothesis that local conditions are determinants of how the intervention shifts outside-option and marriage-market returns. Household- and individual-level characteristics may also shape heterogeneity, and we examine them in 6.3. They are not, however, the main objective of the theory-guided heterogeneity analysis.

4.2 Heterogeneity Analysis

We now turn to an analysis of heterogeneity across village-level characteristics. The theory sketched in Appendix A suggests that treatment effects could vary from negative to positive across village-level characteristics while producing a null average treatment effect. We use the following specification to investigate village-level treatment heterogeneity, using PDS LASSO to select the covariates:

$$y_{iv1} = \alpha_0 + \sum_{d=1}^3 \beta_d T_{dv} + \sum_{d=1}^3 \sum_{k=2}^3 \phi_{dk} T_{dv} \mathbb{1}(Z_{i,0}^v \in \tau_k) + \alpha_1 y_{iv0} + X'_{iv0} \alpha_2 + \varepsilon_{vi}, \quad (2)$$

which extends equation (1) to allow for heterogeneous treatment effects, where $Z_{i,0}^v$ is a village-level heterogeneity dimension. We interact indicators for the second and third terciles of $Z_{i,0}^v$ with all treatment arms, taking the first tercile as the reference category. The tercile indicators $\mathbb{1}(Z_{i,0}^v \in \tau_k)$ enter as levels as well: all variables are residualized on strata effects and on these tercile dummies before estimation, so that β_d is the effect of arm d in the first tercile and ϕ_{dk} is the differential effect of arm d in tercile k .⁷ We consider four dimensions of heterogeneity: social norms, access to public transportation, access to reproductive health services, and economic infrastructure. The interactions $T_d \mathbb{1}(Z_{i,0}^v \in \tau_k)$ can now be used as LHS variables for Step 2 to select additional covariates. Cilliers et al. (2024b) note that terciles $Z_{i,0}^v$ can predict outcomes and treatments in Step 2. However, these level variables should be partialled out to prevent spurious correlations with other covariates, which could lead to excessive control selection.

⁶The method was developed for high-dimensional settings (Belloni et al., 2016); our application is low-dimensional (65 candidates against several thousand observations). We use it as a principled, data-driven selection rule that minimizes researcher discretion and keeps the number of selected covariates well below the number of clusters (Cameron and Miller, 2015), treating approximate sparsity as a maintained assumption. Implementation follows the practical guidance of Cilliers et al. (2024a): strata fixed effects are partialled out, and the baseline outcome is forced into the regression rather than subjected to selection.

⁷The choice of terciles is somewhat arbitrary, but we use this specification for two reasons. First, terciles allow treatment effects to vary nonlinearly across low-, medium-, and high-constraint environments without imposing a strict linear interaction with the underlying index. Second, terciles preserve statistical power relative to finer choices and keep the heterogeneity results comparable across the outcomes and heterogeneity dimensions.

The set of selected controls ($\bar{\mathbf{I}}$) is from Step 1 and repeating Step 2 for each treatment arm and its interaction with the second and third terciles of $Z_{i,0}^v$. Strata effects and levels of $1(Z_{i,0}^v \in \tau_k)$ are partialled out, and we again pre-standardize to unit variance. To aid in the interpretation of (2), we note that each of our heterogeneity dimensions has been structured so that a higher value corresponds to more favorable conditions. That is, the first tercile represents the most restrictive conditions and the third tercile represents the least restrictive conditions. Thus, within each arm d , the implied effect in tercile k is $\beta_d + \phi_{dk}$ (with $\phi_{d1} \equiv 0$ for the reference tercile). The pattern $\hat{\beta}_d > \hat{\beta}_d + \hat{\phi}_{d2} > \hat{\beta}_d + \hat{\phi}_{d3}$ —equivalently, $0 > \hat{\phi}_{d2} > \hat{\phi}_{d3}$ —is monotone heterogeneity in which the arm effect is largest where conditions are most restrictive; the reverse ordering, $0 < \hat{\phi}_{d2} < \hat{\phi}_{d3}$, is monotone heterogeneity in which the arm effect grows with a more supportive environment. Either ordering is a reduced-form pattern in assignment effects. It does not by itself reveal whether the gradient arises from differences in exposure, in conditional returns, or both, and it says nothing about the sign of the effect in any tercile. A reading in terms of binding constraints being removed (the first ordering) or of complementarity between the program and its environment (the second) therefore requires separate evidence on take-up across terciles and on the sign of the tercile-specific effects, which is the purpose of the participation-barrier decomposition in Section 6.3.

4.3 Data driven heterogeneity analysis

In addition to the tertile interactions, we apply the generic machine learning (ML) framework of Chernozhukov et al. (2025), which adds a formal test for the *presence* of heterogeneity and flexibility about its form. Proxy CATE scores are fitted on repeated sample splits (200 splits, four learners, median aggregation) and three targets are estimated on held-out data: the best linear predictor of heterogeneity (BLP), sorted group average treatment effects (GATES), and classification analysis (CLAN) on indexes of local conditions. Online Appendix S8 states the estimating equations, weights, aggregation, and goodness-of-fit criteria, and Appendix C reports the estimates; Section 6.3 summarizes what they contribute.

5 Data, Balance and Outcomes

5.1 Variable construction

5.1.1 Individual and household level variables

Based on the program objectives, we construct main and intermediate outcomes. Main outcomes of interest are those reflecting the core objectives of the SWEDD program: delaying marriage, improving well-being, and improving economic autonomy of adolescent girls and young wives in Burkina Faso. Specifically, we focus on (i) marital status, both conditional and unconditional on baseline status; (ii) childbearing status, including whether she has ever been or is currently pregnant; (iii) emotional violence experienced; (iv) participation in income-generating activities; (v) mental health; and (vi) attitudes toward early marriage, measured by the number of perceived

advantages and disadvantages. This set of variables captures the proximate goal of reducing early marriage and childbearing, improving broader agency and well-being, increasing economic participation, and altering perceptions toward early marriage. These outcomes also naturally connect to the conceptual framework of Appendix A: marriage timing and fertility are equilibrium outcomes that respond to the net margin $\Pi_v - \Omega_v$, mixing the marriage-market and outside-option channels, while IGA is the more direct outcome on the outside-option side.

Marriage, pregnancy, and emotional violence are recorded as dummy variables; the attitude outcomes toward early marriage appear in two forms—counts of perceived advantages and disadvantages, and indicator versions derived from those counts—and we state which form each table uses.⁸ For outcomes like IGA, we construct indices following the approach of Anderson (2008), yielding standardized measures that aggregate multiple survey items into a single composite. The set of main outcomes is largely consistent with other analyses of the SWEDD program (Bertelli et al., 2026; Bandiera et al., 2020). As such, we also evaluate the program’s effect on intermediate outcomes—potential mechanisms through which the program might affect marriage, pregnancy, and IGA. These include finance, gender attitudes, aspirations, sexual and reproductive health (SRH) knowledge, SRH behavior, menstrual hygiene management (MHM), attitudes toward decision making power, and socio-emotional skills. All intermediate outcomes are Anderson indexes; more information about the construction of these indexes can be found in Table S1.

Sign convention for marriage and pregnancy outcomes. The marriage, pregnancy, and emotional-violence indicators appear in two algebraically equivalent codings. The ITT and 2SLS specifications in Section 6.1 use the original “ever-X” indicators for comparability with the existing literature (a negative coefficient is the desired direction); the GATES, CLAN, and participation-barrier displays in the online appendix (summarized in Section 6.3) use the complement ($1 - \text{ever-X}$), so that quintile orderings and CLAN δ estimates are monotonic in the desired direction (a positive value is desired). Row labels make the active coding explicit (“Ever Married” vs. “Never Married,” and analogously for the other outcomes); the single exception is the early-pregnancy indicator (“Preg. bf18”), which is never inverted because its low base rate makes the un-inverted indicator more informative. With these conventions, all outcomes other than *perceived early-marriage advantages* and *Preg. bf18* are oriented so that higher values correspond to the more-preferable direction.

Analysis populations and conditional outcomes. Most outcomes are defined for every girl observed at follow-up, but two are defined only within a subpopulation whose membership can itself respond to treatment: *age at first marriage* is observed only for girls married by follow-up, and the *couple decision-making* index is defined only for girls currently in a union. (*Pregnancy before age*

⁸For attitudes toward early marriage, we abbreviate advantages and disadvantages EMA and EMD, respectively. The variable “EMD > 1” takes a value of one if the individual perceives more than one disadvantage of a girl marrying before the age of 18. Emotional violence is a dummy equal to one if any emotional violence was endured in the 12 months preceding the follow-up survey, so a positive coefficient indicates an adverse effect; the GATES, CLAN, and participation-barrier displays use the complement (“No Emo. Viol.”), per the sign conventions stated below.

18 is unconditional: never-pregnant girls past age 18 are coded zero, and never-pregnant girls still under 18 are treated as right-censored and left missing, so the row is defined for all non-attrited girls with an observable outcome.) Because treatment could shift entry into these subpopulations, effects on the conditional outcomes are post-treatment-selected: they compare treated and control girls within groups whose composition may differ by arm, and we read them as descriptive complements rather than experimentally identified effects. The causal weight in each domain rests on the unconditional counterparts—ever married, never pregnant, and the unconditional agency indices—which are defined for every girl observed at follow-up. Pregnancy before age 18 is unconditional in the marital-status sense but, because of the right-censoring just described, is defined only for girls whose outcome is observable by follow-up rather than for the full follow-up sample.

5.1.2 Village level variables

To characterize local marriage markets, we administered a baseline village questionnaire to knowledgeable local informants covering local marriage norms, economic opportunities and infrastructure, and access to markets, transportation, schools, health services, and contraceptives.⁹ These village-level characteristics map naturally to the framework in Appendix A. Local conditions shape both girls’ outside options and the constraints governing program participation and marriage-market behavior. They impose spatial and access-related constraints on program effectiveness as well as social-normative ones.

To improve interpretability, we summarize these variables into four indices, following Anderson (2008). These indices include social norms, access to transportation, access to SRH (birth control methods), and economic infrastructure. We orient each index such that higher values are more desirable¹⁰. Table S2 contains the full list of underlying components.

Social-norms index. Because the social-norms index is the linchpin moderator throughout the heterogeneity analysis (Comparative Static 2, Section 6.2), we describe its construction in the main text. The index aggregates ten village- and household-level items capturing the local marriage regime: four indicators of permitted cross-group unions; an indicator that there is no social cost to renegeing on a marriage promise; the village-level and ethnic-group-recalled ideal ages at marriage; an indicator that the adolescent girl has the final say in her own marriage decision; the household head’s score on women’s autonomy; and the household-head-reported minimum age at which a girl should marry (Table S2). Each item is oriented so that higher values reflect more permissive norms and aggregated with the inverse-covariance-weighted procedure of Anderson (2008), which downweights components loading on a common factor and upweights components contributing distinct information about the same latent norm.¹¹ Tertiles of the index used in Table S36 are

⁹The village questionnaire recorded the respondents’ positions in one of the following categories: village chiefs, delegates, religious leaders, government authorities, and other community respondents.

¹⁰For instance, more progressive social norms will have higher values.

¹¹An exploratory factor analysis on the ten standardized items returns one dominant factor accounting for the bulk of the cross-village variance, consistent with treating the index as a single-norm construct; the inverse-covariance index is not equivalent to the first factor, however, because Anderson weighting uses all components rather than

computed from the village-level mean.

A baseline imbalance in the full sample ($p = 0.034$; $p = 0.289$ in the non-attrited analysis sample) on one of the ten components—the ideal age at marriage as recalled by ethnic-group informants (`ideal_age_mar_eth_w0`)—warrants attention because the imbalanced variable enters the index used as a heterogeneity moderator. The imbalance is on a continuous variable that ranges from 16.7 to 17.6 years across treatment arms and is confined to the full baseline sample. Two checks address it. First, adding the recalled ideal age directly to the design-based interaction specifications of Section 6.2—as a baseline control and, more stringently, interacted with assignment as well—leaves the social-norms interactions essentially unchanged.¹² Second, the machine-learning analysis of Section 6.3 includes the recalled ideal age as a separate baseline covariate alongside the norms index, so the CATE estimates condition on the imbalanced component by construction.

5.2 Sample characteristics and balance

Table B.1 and Table B.2 in Appendix B display balance statistics for the non-attrited and full baseline samples, respectively. Each table reports means and standard deviations for the full sample and by treatment arm, along with p -values from F -tests of joint equality across arms. All specifications control for strata fixed effects and cluster standard errors at the village level. The set of variables tested is fixed by the study’s registration: every outcome collected at baseline and every variable named in the registration’s heterogeneity dimensions enters the balance and attrition battery, while variables not collected at baseline—program take-up, vocational and professional training, socio-emotional skills, and the mental-health module, which was introduced at follow-up—are necessarily excluded. The candidate controls for PDS-LASSO selection (Section 4.1) are not restricted to this registered set, since selection serves precision rather than pre-specified hypothesis testing.

The full baseline sample comprises 7,719 adolescent girls and young women across 173 village clusters (Table B.2). The non-attrited analysis sample includes 4,936 individuals across 162 clusters (Table B.1). The distribution across treatment arms in the analysis sample is as follows: Control (1,077), T1: Safe Spaces (1,108), T2: Safe Spaces + Training (1,232), and T3: SS + Training + Announced Grant (1,519).

Panel A1 presents the main outcomes measured at baseline. In the non-attrited sample, 66 percent of participants had ever been married and 59 percent had ever been pregnant. The marriage margin is where the arms differ appreciably: T2 villages exhibit lower marriage rates (56 percent) relative to the control (68 percent), though the joint test does not reject in the analysis sample ($p = 0.105$). In the full baseline sample the same contrast is sharper ($p = 0.055$) and extends to the marriage-timing baselines (age at marriage $p = 0.037$, married before 18 $p = 0.061$). The pattern is confined to the full sample: none of the timing variables rejects in the non-attrited analysis

projecting onto the leading eigenvector.

¹²For the full-package arm, the take-up interaction moves from +0.069 to +0.068 with the baseline control and +0.068 when its assignment interaction is added; the never-pregnant interaction moves from −0.081 to −0.081 and −0.075. Specifications and inference follow Table C.1 (strata fixed effects, village-clustered standard errors).

sample. The T2 baseline-marriage imbalance and its implications are examined directly in Online Appendix Table S22. Panel A2 reports intermediate outcomes, which are generally well-balanced. SRH behavior shows some imbalance ($p = 0.077$ in the non-attrited sample; $p = 0.010$ in the full baseline), as does the count of perceived advantages of early marriage ($p = 0.062$).

Panels B through D present balance on covariates at the adolescent, household, and village levels. Adolescent characteristics (Panel B) are largely balanced: participants average 19 years of age, and 16 percent have ever attended school. One exception is the savings indicator, which differs significantly across arms ($p = 0.006$), with control villages exhibiting higher rates (25 percent) than T1 villages (12 percent). Household characteristics (Panel C) show no imbalance significant at the 10 percent level; average household size is 7.3 members, and 30 percent of household heads practice polygamy. Among village-level characteristics (Panel D), the ideal marriage age according to ethnic norms is balanced in the non-attrited sample ($p = 0.289$), ranging from 16.8 years in T1 villages to 17.4 years in T3 villages; in the full baseline sample this variable is imbalanced ($p = 0.034$), a point taken up in Section 4.3.

Overall, across the 53 variables tested in the non-attrited sample, four reject the null of balance at the 10 percent level (7.5 percent), and two reject at the 5 percent level (3.8 percent), consistent with what one would expect from random variation alone. In the full baseline sample the rejection rate is modestly higher: seven at the 10 percent level (13.2 percent) and four at the 5 percent level (7.5 percent), concentrated on the marriage margin discussed above. It nonetheless remains within reasonable bounds.

Table B.3 reports omnibus orthogonality tests, which regress each treatment indicator on all baseline covariates and test their joint significance. Following Kerwin et al. (2024), we compute p-values using randomization inference rather than sampling-based inference, permuting treatment assignments at the cluster level within strata. This approach has the correct size even with many covariates relative to observations, whereas standard F-tests substantially over-reject the null¹³. In the full baseline sample, no comparison rejects the null of joint orthogonality (all randomization-inference $p \geq 0.12$): the covariate vector shows no system-wide imbalance. Nor does any comparison reject in the non-attrited sample (all randomization-inference $p \geq 0.16$). An omnibus test can fail to reject while particular variables differ across arms, and the variable-level marriage imbalances noted above remain relevant; they are handled in the outcome analysis through baseline conditioning. We likewise do not treat the follow-up sample as randomly selected: the attrition analysis in the next subsection rejects equality of attrition rates across arms, and every main estimate is accompanied by differential-attrition bounds.

5.3 Attrition

Appendix Table B.4 reports tests for differential attrition by treatment arm at the adolescent, household, and village levels. Overall attrition was substantial. Approximately 36 percent of the baseline sample did not complete the follow-up survey, primarily due to security-related displace-

¹³We also report the sample-based F-Statistic for comparison.

ment during the 2019–2021 period. Attrition is not uniform across arms at the adolescent level: the F -statistic for the joint test $T1 = T2 = T3$ is 3.38 ($p = 0.036$), driven by modestly higher attrition in T1 (+4.0 percentage points) and lower attrition in T3 (−5.5 percentage points) relative to control. No individual coefficient attains statistical significance, and the spread is between treated arms rather than between treatment and control, but the joint rejection is why we do not rest the main results on the assumption of equal attrition and report differential-attrition bounds for every main outcome (Section 6.1). The direction matters for those bounds, and it is comparison-specific. T3—the arm carrying the paper’s positive findings—retains *more* girls than control, so the Lee trimming is applied to the T3 sample, and the monotonicity assumption runs in the direction the observed attrition gap implies. T1 retains *fewer* girls than control, so for that comparison the trimming is applied to the control sample and the monotonicity assumption is oriented the other way; T2 sits between the two with an attrition gap close to zero. At the household and village levels, differential attrition is negligible (F -statistics of 0.998 and 0.034, respectively; Table B.4).

The observed imbalances, while limited in number and magnitude, are absorbed by the ANCOVA specification, which conditions on baseline outcomes and LASSO-selected covariates; this improves precision among observed cases but does not recover missing outcomes or correct selective follow-up. That work is done by the bounds and the attrition audit. Because attrition is both substantial and not equal across arms, we do not treat the follow-up sample as if it were randomly selected: every main outcome is accompanied by marginal Lee bounds and covariate-conditional sharp bounds, and the heterogeneity results are subjected to a dedicated attrition audit (Online Appendix S11.10).

6 Results

6.1 Intent to treat

Tables 1 and 2 report ITT estimates from equation (1), with PDS LASSO-selected controls and village-clustered standard errors. Outcomes measured at baseline are estimated with the baseline value y_{iv0} included; outcomes observed only at follow-up (take-up, vocational and professional training, socio-emotional skills, and mental health) use the same specification without that term.

The results carry different evidential weight across domains, and we state the ordering before the estimates. The paper is on firmest ground for stated attitudes toward early marriage, where effects are large, replicated in the household-head module, and one-sided under differential-attrition bounds. It is on weaker ground for marriage, fertility, and income: the follow-up loss of 36 percent is common to all arms, loads on the marriage domain itself (every exited girl with a head-reported marital status is recorded as married), and is bounded only by worst-case intervals that span zero, so the behavioral nulls are weakly informative about behavior.

6.1.1 Main outcomes

The ITT estimates on marriage and fertility outcomes are broadly null across all three treatment arms. Across the nine main outcomes (variable definitions in Online Appendix S5), no average effect survives the Holm family-wise correction. Ever-married rates are essentially unchanged relative to the control mean of 80 percent, with RI p -values of 0.58, 0.83, and 0.95 for T1, T2, and T3, respectively. Marriage before age 18, ever-pregnant, current pregnancy, and pregnancy before age 18 are similarly unaffected across all arms, with no RI p -value below 0.24 for any of these outcomes.¹⁴

One margin moves at borderline significance but is read descriptively. Among girls married by follow-up—a post-treatment-selected subpopulation, as the analysis-populations discussion above records—age at first marriage is 0.22 years higher under the announced-grant arm (RI $p = 0.058$, Holm $p = 0.520$). Because marriage itself could respond to treatment, we do not treat this as an experimentally identified effect, and the row carries no statistical significance in Table 1.

The mental-health index is the GHQ-8 (Goldberg et al., 1988) specified in the pre-analysis plan. It shows no robust average effect in any arm. The T3 estimate is -0.090 (RI $p = 0.560$, Holm $p = 1.000$)¹⁵.

Because the \$400 grant was not disbursed at the time of follow-up, T3 estimates throughout are effects of assignment to the announced-grant offer. No girl had received cash, so the estimates cannot reflect cash receipt. Whether they operate through anticipation of the transfer or through other features bundled with the announcement is an interpretive question taken up in Section 6.1.2.

Several accounts of these nulls are observationally equivalent in the data. One is the framework’s offsetting of the marriage-market and outside-option channels (Appendix A), under which the average effect is near zero even when treatment raises human capital. Others require no offsetting: effective dosage was low (take-up of 20 to 33 percent, with many knowledge and skills outcomes null after multiplicity adjustment); the common outreach baseline may have compressed the incremental margin; and true effects may simply be smaller than the design can detect, as the minimum detectable effects below quantify. We do not prefer the offsetting account over these alternatives. The heterogeneity analysis in Section 6.2 asks whether treatment effects vary with the village-level moderators of the comparative statics in Appendix A and Appendix A.6, which is where the framework has testable content.

A further caveat is that marriage markets straddle the randomization unit. A substantial share of grooms originate outside the study villages (the male-exogamy margin of Appendix A), so part of any supply-side response may diffuse across village boundaries and attenuate village-level contrasts. The exogamy heterogeneity analysis in Section 6.3 exploits precisely this variation, but the average

¹⁴Randomization-inference p -values for the main and intermediate ITT tables and the focal interaction family use 10,000 within-strata permutations of the four-category village assignment with the selected controls held fixed; auxiliary and mechanism tables use 1,000 permutations, and every table note states its own draw count.

¹⁵Alternative scorings (the GHQ-12 Likert index, a caseness count, and a two-item self-worth subindex) are reported in Online Appendix S5 (see Section 6.1.2). The only movement among them is a marginal two-item self-worth decline (-0.24 SD, $p = 0.066$, RI $p = 0.054$) that does not survive multiplicity correction.

ITTs should be read as net of any such cross-village equilibrium adjustment.

Minimum detectable effects. The informativeness of these nulls depends on what the design could have detected. Online Appendix Table S7 reports ex-post minimum detectable effects (80 percent power, 5 percent two-sided tests) computed from the cluster-robust standard errors in Table 1. For ever-married, the MDEs range from 4.8 to 7.0 percentage points against a control mean of 80 percent, so proportional effects smaller than about a tenth of the base rate would escape detection. For age at marriage, the T3 MDE is 0.31 years; the observed ITT of 0.22 years (Table 1) sits below that 80-percent-power threshold but close to the five-percent significance boundary ($1.96 \times \text{SE} = 0.22$), consistent with its borderline RI p -value of 0.058. The Wald-scaled magnitude of 0.88 years (Online Appendix Table S4) rescales this same ITT by the first stage rather than adding independent evidence. The average nulls therefore rule out large incremental effects but cannot distinguish a true zero from modest effects diluted by incomplete take-up. The heterogeneity analyses of Sections 6.2 and 6.3 provide suggestive—though not conclusive—evidence on this distinction. A complementary post-mortem power analysis (Appendix E) shows that these precision limits are a feature of the design rather than of attrition: retaining the full baseline sample would have tightened the standard errors by only 4–11 percent.

Attrition robustness. Approximately 36 percent of the baseline sample is missing at follow-up due to security-related displacement. Two scope restrictions govern every bound reported in this paper, and we state them before the results. First, the Lee-type bounds address only *differential* selection between a treated arm and control: the trimmed share equals the arm–control response differential as a share of the less-attrited group’s respondents—6.8 percent for T1 (control trimmed), 0.2 percent for T2, and 9.0 percent for T3 (treated trimmed)—so the roughly 36 percent follow-up loss common to all arms lies outside them, and their estimand is the always-observed subpopulation, not the full baseline cohort. Second, for the common loss no assumption-free repair exists: Horowitz–Manski worst-case bounds that impute every missing girl at the logical extremes bracket zero for every binary outcome and arm (Online Appendix Table S11), so any statement that survives bounding is informative only under the monotone-selection assumption examined below.

With that scope fixed, we bound the identified set using the marginal Lee (2009) bounds (Online Appendix Table S9) and the covariate-conditional sharp bounds of Semenova (2025) (Online Appendix Table S10). Two conclusions follow. First, the null average effects on marriage largely persist under worst-case differential selection: covariate-conditional bounds on T3 age at marriage tightly span zero ($[-0.17, 0.74]$ years), and Curr. Pregnant bounds contain zero in every arm under Semenova. Second, the T3 mental-health bounds span zero under both variants (marginal Lee $[-0.28, 0.08]$ SD; covariate-conditional Semenova $[-0.42, 0.08]$ SD); under T2, by contrast, the marginal Lee interval lies weakly below zero ($[-0.22, -0.14]$ SD) while the Semenova interval ($[-0.28, -0.04]$) does as well, so the span-zero conclusion is specific to the announced-grant arm.

The identified sets exclude zero in a minority of further cells: under Semenova for T3 Ever Pregnant ($[+0.03, +0.28]$), Preg. bf18 ($[-0.13, -0.02]$), and T1 IGA ($[+0.16, +0.32]$), plus two near-

point-identified T2 cells (Ever Married and Preg. bf18, whose bounds nearly collapse to the point estimate). The marginal Lee bounds add a handful of arm-outcome pairs, among them the T2 marriage and pregnancy margins, T2 Mental Health as noted, the T3 under-18 margins, emotional violence under T1 and T2, and IGA under T1. Two features keep us from reading these one-sided cells as sign evidence. First, the bounded object is not the respondent-sample estimand of Table 1: each interval brackets the effect for the population that would be observed under both assignments, under the maintained selection assumption of that bounding approach. Second, sign identification is not stable across approaches: cells that exclude zero under Semenova rarely do so under Lee, and conversely, so which side of zero is excluded depends on the trimming and conditioning choice rather than on a feature the data pin down. We therefore read the one-sided cells as documenting the sensitivity of worst-case attribution to those choices.

Which girls are missing at follow-up? The household module decomposes the 36 percent girl-level loss. Household-level nonresponse accounts for 14.0 percent of the baseline roster; the remaining 22.4 percent are girls whose households were reached but who were not themselves re-interviewed. This girl-specific exit is marriage-related: among exited girls whose head reports a marital status (758 of the 1,727), every one is recorded as married—consistent with marriage-driven moves out of the natal household—so girl-level attrition is endogenous to the marriage domain, a caveat attached to every girl-reported outcome. Two facts bound the concern. Girl exit is balanced across arms (joint $F = 0.92$, $p = 0.43$), so it does not by itself generate differential selection between arms; and the household module continues to observe the exited girls’ households, which is why the head-reported outcomes (Table 6) cover 86 percent of the roster and are the study’s least attrition-exposed measurements. The common-loss caveat for girl-reported outcomes stands: the worst-case bounds on the full loss bracket zero (Online Appendix Table S11).

Sensitivity to the monotonicity assumption. Both bound families maintain the Lee (2009) monotonicity assumption that treatment assignment never reduces an individual’s selection probability ($S(1) \geq S(0)$). The principal threat is conflict-driven displacement: if girls in treated villages were differentially displaced—because the program drew unwanted attention from extremist groups or because participating households were targeted in raids—displaced girls might be both more attrition-prone and disproportionately from high-marriage-risk types, violating the assumption for the relevant subpopulation.

We view this channel as unlikely for the T3 results that drive our main bound conclusions. Community sensitization preceded girl-targeted activities in all study villages (Section 2.3), so program presence was not a salient differentiator across arms, and security-driven attrition predates and is largely orthogonal to randomization. The arm-specific attrition rates are themselves consistent with monotonicity for T3: T3 attrition is roughly 5.5 percentage points below control (Appendix Table B.4). The retention gain holds within precisely the subpopulations where displacement-driven violations would concentrate: -12.8 percentage points (SE 6.3) in the least-secure village tercile, -5.9 (SE 8.3) in villages with a terror attack within 5 km, and -8.2 (SE 6.6) in the top quintile

of displacement exposure (untabulated arm-specific retention regressions; replication package). For monotonicity to fail for a non-negligible subgroup of girls, treatment-induced exclusion would have to be masked by even larger retention gains within these same cells, which we regard as implausible.

T1 is the one arm with attrition modestly above control (+4.0 pp), and we make no T1-specific claims that hinge on monotonicity. On the mental-health index the marginal Lee bounds for T3 span zero ($[-0.28, 0.08]$; Table S9), so no sign is identified to defend for that outcome. The discussion therefore matters mainly for the outcomes whose bounds are informative, while acknowledging that no attrition-bound finding can survive arbitrary departures from monotone selection.

Attrition and the heterogeneity analysis. The bounds above apply to the average ITT; the heterogeneity analyses of Sections 6.2 and 6.3 operate on the non-attrited sample. Because displacement plausibly tracks village conditions, we check whether attrition correlates with the village-level heterogeneity dimensions. Attrition is essentially uncorrelated with the social-norms and SRH-access indices—the dimensions behind the pregnancy-channel results—but declines with the economic-infrastructure index, and the village dimensions are jointly predictive of response. Section 6.3 therefore subjects the headline interactions to a dedicated attrition and implementation audit (Online Appendix S11.10): assignment-by-moderator effects on response, inverse-probability-weighted re-estimates, high-retention subsamples, adversarial imputation bounds, and implementation-fidelity controls built from program records. The short summary is that the pregnancy-channel outcome gradient is robust to these checks—including a tercile-parameterized test showing no differential T3 response across norms terciles—while the IGA gradient reverses in high-retention villages, carries a marginally significant T3 response gap in its identifying low-infrastructure tercile, and the norms gradient in take-up is not separable from documented delivery disruption.

Compliance-adjusted magnitudes. Because take-up is incomplete (Section 6.1.2), the ITT estimates above understate the program’s effect on participants. Online Appendix Table S4 reports 2SLS coefficients that instrument individual take-up with arm-specific treatment assignment; numerically, each equals the ITT scaled by the arm-specific first stage. We read these throughout as *compliance-adjusted magnitudes* (Wald-scaled ITTs) rather than as causal complier effects. A complier (LATE) interpretation would require assignment to affect outcomes only through the measured take-up indicator. In this setting, assignment changes a bundle—safe-space content, training offers, peer and community exposure, and, under T3, an announced grant—while the take-up indicator records safe-space participation alone, so the exclusion restriction is difficult to defend for any outcome, and we do not lean on it.

On this scaling, the descriptive age-at-marriage margin corresponds to 0.88 years under T3 ($p = 0.028$; post-treatment-selected, as above); the emotional-violence ITT of 0.051—which is itself not distinguishable from zero under randomization inference (RI $p = 0.104$)—would scale to roughly 0.22 per participant if taken at face value. Mental health is omitted from this exercise: there is no average effect to scale (Section 6.1.2). For emotional violence, where non-participants

in treated villages are themselves partially exposed, the rescaled figure is best read as an upper bound on the per-participant effect. The heterogeneous Wald-scaled estimates by social-norms and SRH-access tertile are reported in Tables S34 and S35.

T2 baseline marriage imbalance. The T2 sample has a slightly lower baseline ever-married rate than the control, which the ANCOVA controls for additively. Table S22 reports two robustness checks. Adding the village-level baseline marriage rate as an additional control leaves the T2 marriage and fertility coefficients close to zero (e.g., the T2 ever-married coefficient moves from 0.019 to -0.012). Restricting the sample—both T2 and control—to women never married at baseline, so that the comparison is made on common support, likewise leaves the T2 coefficients statistically indistinguishable from zero (-0.004 , SE 0.067 for ever-pregnant; -0.026 , SE 0.019 for currently-pregnant). Both checks indicate that the T2 marriage and fertility nulls are not artifacts of the baseline marriage imbalance.

6.1.2 Intermediate outcomes

We start by documenting self-reported take-up in the first row of Table 2. Assignment substantially increased participation relative to control in all three arms: estimated take-up is 25.3 percentage points higher under T1, 19.8 percentage points higher under T2, and 29.3 percentage points higher under T3, relative to a control mean of 3.9 percent¹⁶. Nevertheless, participation remained limited in absolute terms: even under T3, roughly two-thirds of girls in the estimation sample did not participate.

T1’s higher overall take-up than T2, despite less content, reflects the composition of this indicator: it captures participation in the safe-spaces component common to all arms, and adding business training in T2—without the grant promise—may have introduced scheduling or time-commitment barriers. T3’s highest take-up is consistent with the announced cash transfer creating stronger incentives to attend. Take-up is measured on the extensive margin only: the survey records whether a girl ever participated, not the number of sessions attended, so dosage cannot be characterized, and program delivery records do not fill the gap (Section 6.3).

The training-specific uptake outcomes establish a clear two-tier first stage. Vocational training participation is flat under T1 (+2.0 percentage points, RI $p = 0.375$) but rises by 6.7 percentage points under T2 (RI $p = 0.013$) and 11.4 percentage points under T3 (RI $p < 0.001$, Holm $p = 0.006$); professional training follows the same pattern (no T1 effect; +6.3 pp under T2, RI $p = 0.005$; +7.5 pp under T3, RI $p = 0.001$, Holm $p = 0.015$). The T2 training effects do not survive the Holm correction across the 17-outcome intermediate family at the five-percent level. All arms therefore enrolled girls in safe spaces, but only T2 and T3 delivered measurable increases in formal training content: impacts under T1 are attributable solely to the safe-spaces component, while T2 and T3 effects reflect the combined influence of safe spaces and training.

¹⁶All these estimates are statistically significant at any conventional level, including with RI and multiple hypothesis correction.

Two implications follow for interpreting the outcome estimates. First, the ITT estimates capture the effects of assignment, not participation. Under the participation-only channel described in Definition 1, incomplete take-up can yield small ITTs even when effects conditional on participation are large. The heterogeneity analysis returns to this distinction by examining how predicted participation and outcome gains vary together. This is directly relevant to Definition 1, which records that ITT effects are mechanically attenuated by low take-up and that small village-level ITTs are consistent with large conditional effects when participation is selective in the marginal return g_i . Second, the negligible finance and IGA effects across all arms at follow-up indicate that measurable income gains had not appeared by follow-up despite increased program exposure; whether the training itself raised skills is not established by the knowledge and skills outcomes, which are mostly null under randomization inference.

Gender attitude scores improve significantly under both T2 (+0.323 SD, RI $p = 0.015$, Holm $p = 0.207$) and T3 (+0.428 SD, RI $p = 0.001$, Holm $p = 0.021$), with no detectable effect under T1, mirroring the training-uptake gradient. Safe spaces alone are insufficient. The T2–T3 difference on this index is not statistically distinguishable (Table S14), so the gender-attitude gains are attributable to the training tier rather than to the announcement.

The normative attitude outcomes deliver the study’s most compelling evidence of program impact. (The attitude domain is registered; the T2–T3 announcement contrast examined below is a post-hoc comparison within it; Section 2.3.) Under T3, the share of girls who disagree that early marriage is appropriate increases by 20.4 percentage points (RI $p = 0.010$, Holm $p = 0.111$) and the count of perceived disadvantages rises by 0.436 (RI $p = 0.032$, Holm $p = 0.325$). The share who actively advocate for early marriage falls by 20.6 percentage points (RI $p = 0.002$, Holm $p = 0.025$) and the count by 0.414 (RI $p = 0.002$, Holm $p = 0.029$). The two “Adv. M. Early” outcomes are the strongest normative findings in the paper: large, precisely estimated, and surviving the Holm correction across the 17-outcome intermediate family at the five-percent level (alongside gender attitudes and both training-uptake outcomes under T3).

The magnitude warrants an explicit benchmark. A 20.6-percentage-point decline on a control base of 43.9 percent is a 47 percent proportional fall in the share of girls endorsing early marriage, at a single survey round and from an announcement. For comparison, multi-year school-based curricula move gender-attitude indices by roughly a quarter of a standard deviation (Dhar et al., 2022) and sustained media exposure shifts fertility- and autonomy-linked outcomes over years (Jensen and Oster, 2009; La Ferrara et al., 2012).

A shift this large in stated attitudes is not, on the scale of these data, a large shift in behavior. We calibrate the attitude effects against the cross-sectional association between baseline attitudes and later marriage and pregnancy among girls unmarried, or never pregnant, at baseline (Online Appendix Table eftab:attitude:calibration; a descriptive benchmark, not a causal chain). The girls’ own baseline attitude items barely predict later marriage: in control villages the two binary items carry slopes of 0.11 and -0.06 , and pooled across arms 0.04 and 0.00. The household head’s minimum acceptable marriage age is the informative predictor: one additional year lowers the

probability of marriage by follow-up by 1.5 percentage points in control villages ($p < 0.01$) and by 0.4 to 0.7 points on the pooled and pregnancy margins. Pushing the T3 shifts through these slopes implies a change in the marriage margin of under one percentage point for the head item and of 0.8 to 3.4 points for the girls' items, against a T3 ITT on the same sample of 0.001 with a standard error of 0.042. Even if the observed attitude movement translated into behavior at these rates, the implied behavioral change is well inside the design's sampling error, so the null marriage and fertility estimates are what an attitude shift of this size predicts, not evidence against it.

Three considerations bear on reading a shift this large. The outcome is a stated attitude at one point in time, not a durable behavior. The announcement was a concrete, named resource for the respondent's own household rather than diffuse messaging, so a larger elasticity is not implausible. And the survey-response audit below, together with the household-head replication, is what disciplines the reading rather than the magnitude alone. That the normative shift is robust under T3 but absent under T1 and T2—despite T2 already delivering measurable training uptake—is consistent with the promise of a transfer making the economic outside option concrete even before disbursement. Appendix A.3 gives this reading a formal special case: a promised grant raises the bride side's stated reservation for an early marriage while moving realized timing only through the age-dependence of control over the grant. We treat this anticipation reading as an interpretation rather than an identified mechanism: the T2–T3 contrast bundles the grant announcement with any differences in delivery intensity, salience, and engagement that accompanied it.

A participation split sharpens the interpretation of the normative shift (Table 3), mirroring the exposure split used for the mental-health results below. Within T3, all four normative outcomes move for participants and non-participants alike: non-participating girls—two-thirds of the arm's estimation sample (66.7 percent)—are 17.2 percentage points less likely to advocate early marriage ($p = 0.002$) and report 0.36 fewer perceived advantages ($p = 0.002$), against 25.1 points and 0.51 for participants. Because non-participants dominate the arm numerically, they account for the majority—roughly 58 percent—of the aggregate effect ($0.667 \times -0.172 + 0.333 \times -0.251 \approx -0.198$, close to the -0.206 ITT); equivalently, non-participants are two-thirds of the arm carrying 58 percent of the effect, so the per-capita shift is somewhat larger among participants.

Participants shift significantly more on the two binary outcomes ($p \approx 0.03$ for the equality tests), a gradient consistent with additional direct exposure, but the aggregate shift cannot be read as internalization of curriculum content alone. The same curriculum without the announced grant (T2, Panel B) produces smaller shifts—between one-quarter and two-thirds of the T3 point magnitudes on the counts, individually insignificant among non-participants (the advocacy count reaches $p = 0.089$) and significant among participants only on the advocacy outcomes ($p = 0.060$ and 0.038)—with no detectable participant/non-participant gap. And T3 non-participants move, in point terms, more than T2 participants who actually attended the training (index contrast $+0.18$ SD, $p = 0.15$). The pattern points to a village-level response operating throughout T3 communities rather than a classroom effect confined to attendees.

Two caveats bound this reading. Participation is chosen, so the split is descriptive. And be-

cause non-participants also live in villages where the grant was announced, a village-wide reporting response to the announcement is not excluded by the split itself.

The T2–T3 comparison estimates the announcement margin as defined in Section 2.3: the announced-grant arm relative to the training-only arm (Figure 1). We summarize it with a single post-hoc index test: on a pre-oriented index of the four normative outcomes, the announcement margin is +0.265 SD (SE 0.109; randomization-inference $p = 0.029$ from 10,000 within-strata permutations; familywise $p = 0.083$ across its four components—the index was specified after the component results were known, so we report both numbers). The T3–T2 Lee trimming sets exclude zero on the index and on all four outcomes; accounting for sampling uncertainty in the bound endpoints (Imbens–Manski intervals, Table 5), the disagreement indicator remains strictly positive ($[0.02, 0.38]$), the advocacy indicator is borderline ($[-0.30, 0.00]$), and the index interval narrowly includes zero ($[-0.02, 0.74]$).

How much of this margin could the arms’ difference in realized dosage carry? Take-up was 33.3 percent in T3 against 24.3 percent in T2, a gap of 9 percentage points. If the announcement had worked only by drawing more girls into the same training, the extra effect would be that gap times the participant-level effect of the training. Using the T2 participant coefficient on the advocacy indicator (-0.134 , Table 3), the implied contribution is $0.09 \times 0.134 \approx 0.012$ on a control base of 0.44, or about 0.02 SD; using the larger T3 participant coefficient (-0.251) instead gives 0.023, about 0.05 SD. Either figure is under a tenth of the 0.26 SD index margin. Differences in salience and engagement that the announcement may have produced within the sessions are not captured by this arithmetic, which is why we describe the margin as an arm contrast rather than as the effect of the promise alone.

Within participation groups (Table 4), the gap runs in the beneficial direction for both: among participants, disagreement with early marriage is 18.1 percentage points higher under the announcement ($p = 0.004$) and the disadvantage count 0.43 higher ($p = 0.011$). Among non-participants the gaps are 13.1 points ($p = 0.069$) and 0.28, with a marginal decline on the advocacy indicator ($p = 0.076$). Because take-up itself responds to the announcement (participation is 33.3 percent of the T3 estimation sample against 24.3 percent in T2), the participant and non-participant pools are not the same strata across arms, and part of any within-status comparison reflects composition.

Three checks bound this. First, adding the baseline attitude and its missingness indicator leaves the margins essentially unchanged (Online Appendix Table S16), so the arms’ baseline attitude differences do not account for them. Second, Lee-type trimming for the announcement-induced composition—removing the 27 percent of T3 participants (or, symmetrically, the 12 percent of T2 non-participants) whom the announcement plausibly shifted across the participation margin—leaves the non-participant announcement margin one-sided on the index and on three of the four outcomes (Online Appendix Table S17). These are identified sets, and the Imbens–Manski interval for the index includes zero, so the trimming exercise bounds composition, not sampling uncertainty. Third, we do not claim the within-status margins are equal: the 95 percent confidence interval on the participant/non-participant difference is wide (for advocacy, $[-0.10, +0.11]$), so the data are

consistent with substantial gaps in either direction. The within-T3 exposure split—which compares groups inside the same arm—rejects equality on three of four outcomes (Table 3).

What the evidence supports is the weaker, and still consequential, statement: girls who never attended a session moved with the announcement, by amounts that survive composition trimming. Together with the dose gradient within T3, the pattern indicates a channel operating beyond the classroom, though it does not by itself locate that channel. We are explicit about what these comparisons identify. Non-participants live in the same villages, households, and gatekeeper networks that received the announcement, so the non-participant, ineligible-girl, and household-head contrasts establish community-wide exposure to the arm. They rule out a classroom-only account and a girl-specific reporting account; they do not identify transmission from participants to others, nor distinguish exposure through households from exposure through local leaders. Gender attitudes show no detectable announcement margin in either group, consistent with the insignificant T2–T3 difference on that index (Table S14), so the gender-attitude gains belong to the training tier rather than to the announcement.

An item-level decomposition shows what moves inside the counts (Online Appendix Table S18). The decline in perceived advantages is broad-based across all three coded items—*finding a husband* (−16.6 percentage points, the one item surviving a Holm correction across the ten items), *having children* (−16.3), and *being able to work* (−9.7)—and the rise in perceived disadvantages spreads across the welfare-coded items (loss of childhood, child health, mother’s health, domestic violence), with clean nulls on STDs and education. No single item drives the counts, so the aggregate shift is not an artifact of one salient answer; and because the instrument codes no directly economic advantage of early marriage (such as bride-price receipt), the item pattern cannot separate a change in norms from a change in the perceived economics of marriage—both readings survive it. Item-level movement is similar for participants and non-participants.

The strongest corroboration comes from a report the girls do not control: their household heads’ own early-marriage attitudes (Table 6). Two features of the head module matter before the estimates. First, its coverage: heads answer the household questionnaire even when the girl herself is not re-interviewed, so the head outcomes are observed for 86 percent of the baseline roster (6,642 of 7,719 girls; 5,875 distinct heads; 4,915 girls who responded at follow-up plus 1,727 who did not), against 64 percent for the girl-reported outcomes—the paper’s least attrition-exposed measurements. Second, availability does not differ by arm (joint $F = 0.92$, $p = 0.43$).

Under T3, heads are 22.5 percentage points more likely to see any disadvantage in a girl marrying before 18 (control mean 50 percent; randomization-inference $p = 0.002$ at 10,000 draws, Holm $p = 0.005$ across the five head outcomes). They are 16.8 points less likely to see any advantage ($p = 0.009$). They are also 23.7 points more likely to see a disadvantage in a birth before 18 ($p < 0.001$, Holm $p = 0.003$), and report an ideal marriage age for girls 0.49 years higher ($p = 0.001$).

The announcement specificity is formally tested in the heads and survives familywise correction. The Wald tests of equal T2 and T3 coefficients reject with randomization-inference $p < 0.001$ on both disadvantage outcomes (Holm $p = 0.0005$ and 0.004), sharper than the corresponding girl-level

contrasts, while the advantage and ideal-age contrasts are individually significant at 5–6 percent. Marginal Lee bounds on both the T3–control and T3–T2 head contrasts are one-sided for all four attitude and ideal-age outcomes.

Because heads—not girls—report these outcomes, a girl-level reporting response cannot produce them; and because marriage timing in these communities is decided largely by household heads, the head-level movement indicates that the announcement reached the decision-makers whose choices the girls’ attitudes are meant to predict. A head-level reporting response to the announced grant remains logically possible, but it would have to operate on a different respondent, in a different survey module, with the same arm pattern. The specificity is itself informative: the heads’ omnibus gender-norms index does not move in any arm (Table 2), so the head-level response is confined to the early-marriage domain the announcement made salient rather than reflecting a generalized shift in reported gender attitudes. We read the head results as corroborating a household- and community-level response to the announcement rather than internalization of curriculum content by participants.

Grant eligibility and dose. If the announcement moved attitudes through anticipated household resources, the response should be graded by whether the household plausibly stood to receive the grant. Eligibility for the entrepreneurship grant required age 14 by the program’s start, so T3 villages contain girls above and below the eligibility margin. We state the design’s power first: with 75 ineligible girls (aged 12 or younger at baseline) against 1,357 eligible T3 respondents, the design detects an eligibility contrast of roughly 0.20 on the binary outcomes at conventional power—about the size of the full advocacy effect—so the test is informative against the strong form of the anticipation reading, under which the shift concentrates in eligible households.

The gradient is flat: the T3 index effect is +0.63 SD for ineligible girls against +0.48 for eligible girls (difference -0.16 , $p = 0.35$), netting out the corresponding age gradient in the training-only arm leaves a triple difference of +0.03 ($p = 0.88$), and a wealth-dose gradient—the transfer is larger relative to poorer households’ resources—is right-signed but null (poorest-versus-richest tercile difference $p = 0.30$; triple difference $p = 0.88$; Online Appendix Table S19). Girls with no personal prospect of the grant shifted fully. The ineligible group is small (75 girls), and the 95 percent interval on the eligibility difference, roughly $[-0.50, +0.18]$ SD, admits a gradient as large as the training-only arm’s entire effect, so the test is informative against the strong form of the anticipation reading and suggestive against weaker forms. The pattern is consistent with community-wide exposure and weighs against a private income-expectation channel; it does not separate village-level normative change from village-level affirmation of an announced program—the residual named below.

Experimenter demand and social desirability. Because the normative outcomes are self-reports collected from girls who, in the announced-grant arm, were awaiting a real transfer, survey response bias deserves direct scrutiny: respondents who associate the interview with the program could report the attitudes they believe the program wants to hear (de Quidt et al., 2018; Bursztyn

et al., 2020), a concern documented for school-based attitude interventions in particular (Dhar et al., 2022).

Three features of the survey apparatus limit the channel from program to interviewer. Both the baseline and follow-up surveys were fielded by the Institut National de la Statistique et de la Démographie (INSD), the national statistical office, rather than by program implementers; the questionnaire did not mention or discuss the grant in either round; and the grant was communicated to T3 participants by program staff during the business-skills training, not through the survey. Survey responses played no role in program eligibility or in the grant’s allocation, which was tied to program participation.

The elicitation format also offers less purchase for acquiescence than agree–disagree scales. The attitude outcomes are a yes/no question on whether marriage before 18 has disadvantages (advantages), followed by counts of the specific disadvantages (advantages) the respondent names, seven and three coded items respectively. Inflating the count therefore requires naming concrete drawbacks rather than assenting to a statement.

The pattern of results is also difficult to reconcile with a reporting artifact. If girls in treated villages were tailoring self-reports to please, the response should appear wherever pleasing is cheap. Yet the self-reported outcomes on marriage, fertility, SRH knowledge and behavior, finance, menstrual hygiene, and decision-making power are all null in every arm, as is the household head’s omnibus gender-norms index (Table 2). And reported emotional violence, if anything, moves adversely in point terms under the training arms (it is not distinguishable from zero under the paper’s randomization-inference standard)—in either case not a favorable-presentation pattern.

One cheap-to-please self-report does move: gender attitudes improve under T2 and T3 (+0.32 and +0.43 SD). Its arm pattern, however, tracks the training tier rather than the announcement—the T2–T3 difference on that index is insignificant—so it does not fit an announcement-driven pleasing response. A training-content demand effect on that index remains possible, and we rest no claim on it.

The T2 arm, surveyed with the same instrument by the same enumerators after the same curriculum, shows smaller and individually insignificant normative shifts, roughly half the T3 point magnitudes on the advocacy counts. Contact with the program and its content therefore does not by itself generate the full reporting pattern. And within T3, the participation split above cuts against the most natural demand channel—direct program contact—since non-participants, who never sat in the training sessions where the grant was announced, shift almost as much as participants and, in point terms, more than T2 participants.

The household-head results above are the sharpest of these checks. The head-reported early-marriage attitudes move with the same arm pattern as the girls’, and no girl-level reporting response can generate a head-level report. The survey instrument carries no item measuring awareness of the grant, so a salience gradient across girls or villages cannot be tested; we note this as a design gap.

What survives all of these checks is a narrower residual. Girls and heads in T3 villages knew,

through community channels, that a real transfer was coming. A village-wide impulse to affirm the program’s message in that setting—now necessarily operating on two different respondent types in two different modules—cannot be fully separated from a genuine normative change by any comparison available in these data. We therefore read the normative results as arm-level effects of the announcement on stated attitudes, while noting that stated attitudes are themselves a policy-relevant margin in a setting where marriage decisions respond to community expectations (Bursztyn et al., 2020).

There are no detectable effects on Finance, SRH knowledge, SRH behavior, MHM, autonomous decision-making, household head opinion, and social skills under any arm by RI. Taken together, the intermediate outcome results paint a coherent picture. Program reach was substantial and robust across all arms. The specific training content reached girls in T2 and T3 at detectable but low rates. The early-marriage normative outcomes moved only in the full-package arm, whereas the broader gender-attitude index improved in both training arms; the announced cash transfer is thus the component associated with the shift in stated early-marriage preferences specifically, not with attitudinal change in general. On mental health, the full package produces no broad average deterioration; what the data support is a marginal decline concentrated in two self-worth items. The paragraphs that follow take up the mechanism, and Section 6.2 examines whether the normative gains and the self-worth costs are concentrated where social-norm conservatism is highest.

Attrition robustness. The normative attitude effects under T3 are robust to worst-case *differential* selection: the covariate-conditional bounds lie entirely on one side of zero for all four endpoints (e.g., *advocates early marriage* under T3 is bounded in $[-0.27, -0.06]$; Online Appendix Table S13). As for the main outcomes, these are trimming bounds for the surveyed subpopulation: they address selective response between arms (trimming 0.2–9.0 percent), not the 36 percent follow-up loss common to all arms (Section 6.1). A notable finding from the same bounds is that SRH knowledge improves robustly under T2 and T3 (bounds entirely above zero) despite null ITT estimates, implying that the zero ITT reflects take-up dilution rather than a genuine absence of impact.

Distinguishing T2 and T3. T2 and T3 received the same safe-space and training content, and only T3 carried the announced grant; T3 also realized higher take-up (33.3 against 24.3 percent) and higher training uptake (Table 2), so the contrast is between arms rather than between the promise and its absence. A natural concern is whether T2 and T3 estimates simply re-label the same human-capital channel. Table S14 reports Wald tests of $H_0 : \beta_{T2} = \beta_{T3}$ for every outcome. For all nine main outcomes equality cannot be rejected at the 5 percent level: on the marriage and fertility margins the data do not distinguish the two arm effects.

For intermediate outcomes, however, equality is rejected for take-up ($p = 0.032$), vocational training ($p = 0.091$), and—most importantly—both binary early-marriage normative outcomes (*disagrees with early marriage*: any $p = 0.011$; *advocates early marriage*: any $p = 0.042$). The corresponding count measures point the same way (disadvantage count $p = 0.032$; advocacy count

$p = 0.080$). Because the T2–T3 contrast is a family of tests, we also report familywise corrections (Online Appendix Table S15): within the seven-outcome family the contrast rests on (the four normative outcomes, gender attitudes, take-up, and vocational training), the smallest Holm-adjusted p is 0.077, and across all 26 Wald tests it is 0.29, so no individual pairwise difference is familywise-robust.

The T3-specificity of the normative shift therefore rests on the joint pattern rather than on any single difference test. The T3-versus-control advocacy effects survive the Holm correction at the five-percent level (familywise $p = 0.025$ and 0.029) while the T2-versus-control effects are null. The pairwise T2–T3 differences are same-signed on all four normative outcomes and individually significant on three (the advocacy count at $p = 0.080$). And the single index-level test of the announcement margin survives randomization inference on its own ($p = 0.029$; Table 5). We read this as consistent with the grant announcement operating above and beyond what training alone delivers, subject to the caveat noted earlier that the two arms may also have differed in delivery intensity and salience. The multi-country companion evaluation provides an external check on the training-only tier: in its Burkina Faso estimates—the same T1 villages and girls analyzed here, under its own index construction and covariate-selection procedure—the safe-spaces offer leaves the early-marriage attitude indicator unchanged (Bertelli et al., 2026). The attitude movement this paper documents therefore appears only in the arms that no other analysis of the program covers, and in the only arm carrying the announced grant.

Mental health. The average mental-health effect is statistically indistinguishable from zero: the eight-item GHQ index specified in the pre-analysis plan yields a T3 coefficient of -0.08 control SDs (SE 0.15; randomization-inference $p = 0.55$; Holm $p = 1.0$), the GHQ-12 and caseness scorings agree, and the marginal Lee bounds span zero ($[-0.28, 0.08]$; Table S9), so no deterioration is identified under worst-case differential selection either.

What the data support is narrower. A two-item self-worth subindex (*lost self-confidence, considered self worthless*) declines by 0.24 SD ($p = 0.066$; randomization-inference $p = 0.054$). Neither item survives a Holm correction across the eight items.¹⁷

The mechanism evidence points away from cognitive dissonance and toward a village-level channel. The decline is not steeper among participants (non-participants -0.33 SD, $p = 0.006$, versus participants -0.24 SD, $p = 0.095$, on the subindex; no participant deficit at all on the full index), and the within-person coupling between attitude change and mental health is absent or positive. The cost concentrates sharply in the most conservative social-norms tercile (-0.75 SD on self-worth, $p = 0.012$; -0.76 on the full index, $p = 0.043$; null elsewhere). Community reaction to the program’s normative push—or tension surrounding the announced grant—fits this pattern better than internalization through the girl’s own participation,¹⁸ though the design cannot separate these

¹⁷The module was introduced at follow-up, so no baseline index exists and a change-on-change specification is not feasible. Index regressions use the published PDS-LASSO specification; the exposure-split, coupling, and tercile regressions use strata and age controls with village-clustered standard errors. The items form a coherent scale ($\alpha = 0.81$); item-level effects, alternative scorings, and scale diagnostics are in Online Appendix S5.

¹⁸The companion multi-country evaluation reports, for the safe-spaces-only villages of this same trial under its

channels, and differential reporting cannot be excluded for a follow-up-only self-report. Online Appendix Tables S29 and S30 report the full set of mental-health analyses.

6.2 Heterogeneity analysis

Comparative Static 2 (social norms) is theoretically sign-ambiguous; Proposition 1 (SRH access) is signed. We examine each in turn before turning to the data-driven analysis.¹⁹ Because Proposition 1 delivers a pre-signed prediction, we treat the SRH-access heterogeneity tests as confirmatory and hold them to a familywise standard across the relevant outcome family (pregnancy and fertility outcomes).²⁰

Table S36 reports heterogeneity across social-norm tertiles, testing the sign of Comparative Static 2. No mental-health cell in this table is individually significant, and there is no significant average effect for a tertile decomposition to account for. The concentration of the self-worth decline in the most restrictive villages rests instead on the dedicated mental-health analysis (-0.76 SD on the full index, $p = 0.043$; -0.75 SD on the self-worth subindex, $p = 0.012$; Online Appendix Table S30).

The clearest gradients in Table S36 are on the marriage and pregnancy margins. Relative to the most restrictive tertile, safe spaces in the least restrictive villages are associated with a *lower* probability of remaining unmarried (-0.111 , $p < 0.10$) and of never having been pregnant (-0.327 , $p < 0.05$). A comparable pattern holds under the training arm (-0.185 , $p < 0.05$; -0.243 , $p < 0.10$). The same gradient is absent among girls unmarried at baseline ($+0.005$, insignificant), so the marriage-margin result is carried by the full sample rather than by the never-married subpopulation.

That the self-worth cost falls hardest in the most constrained communities—and on non-participants as much as participants—points to a village-level channel operating where community expectations are most conservative (Section 6.1.2), rather than to internalization of new norms

own specification (a GHQ-12 score with double-lasso covariate selection), a decline of 0.15 SD ($p = 0.10$) in Burkina Faso (Bertelli et al., 2026). A cost of similar sign in an arm with no announced grant weighs toward the program’s normative push, rather than grant-specific tension, as the source.

¹⁹In the tertile-interaction tables in this subsection and Online Appendix S7, outcomes are shown in the desirable direction (remaining unmarried, never pregnant, not currently pregnant), so a negative interaction indicates a smaller gain relative to the omitted first tertile.

²⁰Table S40 reports raw p -values only, and the tertile decompositions here should be read as raw and uncorrected. None of the SRH-access tertile interactions on the pregnancy and fertility outcomes reaches the 5 percent level even uncorrected, so none would survive a Holm step-down correction across that family at conventional levels. (The design-based randomization-inference analysis of Table C.2 implements the familywise correction for its declared focal family—the social-norms and economic-infrastructure moderators—which is post-hoc and treated as exploratory; see Section 6.3.) The tertile decomposition for social norms is treated as characterizing the sign of a comparative static (Comparative Static 2) rather than as a hypothesis test; raw p -values are reported because no signed prediction is being tested.

The exploratory heterogeneity analysis of Section 6.3 and the GML estimates of Appendix C are exploratory throughout (the only confirmatory test embedded in the CLAN framework is the sign-reversal prediction of Proposition 2, for which Holm-adjusted p -values are reported in the exogamy-falsification table of Appendix C).

This confirmatory/exploratory division tracks the study’s registration (RIDIE-STUDY-ID-5f8441974e6ed; Section 2.3): the four village-level heterogeneity dimensions were registered before follow-up data collection, whereas the GML implementation and the participation-barrier decomposition are post-registration methodological choices.

through the girl’s own participation. The net effect on marriage in tertile 1 is nonetheless ambiguous, consistent with Comparative Static 2. Conservatism raises both the marriage-market premium on skilled wives and the outside-option cost of remaining unmarried, leaving the sign undetermined without further data.

Table S40 reports heterogeneity across SRH access tertiles, testing Proposition 1. The proposition’s signed prediction is that treatment-induced delays in marriage and fertility are *larger* where SRH access is *greater*. On the pregnancy and fertility outcomes—the family to which the confirmatory standard applies—no tertile interaction reaches the 5 percent level even before correction; the one marginal cell (never pregnant, middle tertile, +0.228, $p < 0.10$) does not form a monotone access gradient. The confirmatory test therefore fails: the framework’s one pre-signed prediction is not supported on the family it was registered against. On the full-sample marriage indicator, likewise, neither the main effect (0.024) nor any tertile interaction is significant.

A separate, exploratory pattern appears on one marriage margin. Among girls unmarried at baseline, safe spaces raise the probability of remaining unmarried by 0.126 ($p < 0.10$) in the *least* accessible tertile, and that gain is smaller in the middle (-0.173 , $p < 0.10$) and most accessible (-0.130 , $p < 0.10$) tertiles. This gradient runs from access toward scarcity, opposite to the predicted direction, but it sits outside the confirmatory family and is significant only at the 10 percent level, so we record it as a lead rather than as a reversal of the proposition.

The knowledge margin points to what the program did instead. Table S41 shows that safe spaces raise SRH knowledge most in the lowest-access villages, where the information baseline is lowest (+0.417, $p < 0.10$), with a significantly smaller gain in the middle tertile (-0.635 , $p < 0.05$): the program’s information content substitutes for scarce local SRH services rather than amplifying existing access. That substitution channel operates through the program’s content and lies outside Proposition 1, which varies the outside-option term while holding the transmission of program content fixed; the failed test indicates that the outside-option mechanism the proposition isolates is not the margin along which SRH access operates in these data.

Heterogeneity across transportation access tertiles (Table S38) does not yield a coherent pattern, consistent with the theoretically ambiguous sign of Comparative Static 1. When both Π and Ω shift with L_v and the cross-tercile cells covary with social norms, the parametric tertile decomposition cannot identify the sign. We rely on the data-driven approach in the next section to isolate the dominant mechanisms jointly.

6.3 Exploratory heterogeneity: village structure and the participation pattern

We examined, as exploratory analysis, whether treatment responses vary with a fixed set of seven baseline village structural characteristics: social norms, SRH access, economic infrastructure, security, and three exogamy margins. The first approach uses design-based assignment-by-structure interactions with no machine-learning first stage. The second uses generic machine learning (BLP, GATES, CLAN) with a decomposition of village-level CATEs.

A participation-barrier pattern appeared: assignment raises take-up most in villages with per-

missive norms and stronger economic infrastructure, while reduced-form gains on pregnancy prevention and income generation are largest where the take-up response is weakest. The asymmetry has an institutional reading—girl-level take-up may gate the pregnancy and income channels more directly than marriage timing, which is largely decided by household heads.

CLAN connects these patterns to the framework by comparing baseline village characteristics across the highest- and lowest-ranked quintiles of each outcome’s treatment-effect proxy (Figure C.2). In every arm, the highest-ranked group for remaining never pregnant comes from more conservative villages, whereas under T3 the highest-ranked take-up group comes from more permissive villages. The marriage-timing contrasts under T3 are instead small and imprecise. A separate within-village analysis under T3 shows positive co-movement of take-up and marriage-timing proxies, consistent with—but not identifying—selection on gains. Exogamy provides a distinct connection to Proposition 2: Under T2, higher-ranked IGA effects are associated with lower female exogamy. The paired pattern is not consistently replicated across arms. Together, the results characterize outcome-specific heterogeneity without identifying the mechanisms behind the average (ITT) nulls.

The participation-barrier interpretation, however, does not survive the audit we imposed on it. No interaction survives familywise correction (smallest Holm-adjusted $p = 0.26$). A joint randomization test puts the probability that search alone produces a pattern this extreme at 0.103 within the T3 family and 0.260 when the search spans all three arms. The norms take-up gradient is absorbed by controls for documented delivery disruption, and the infrastructure-IGA gradient reverses in high-retention villages and rests on the one tercile where T3 assignment marginally raised follow-up response. Finally, on the income-generation channel, the barrier reading is observationally equivalent to the program substituting for absent local services, with opposite policy implications, so we draw no recommendation from it.

The full apparatus is in Appendix C: the interaction tables, randomization inference and multiplicity accounting, the attrition and implementation audit, the CATE decompositions, the exogamy falsification, and a mechanical scaling exercise. Further robustness is in Online Appendix S11. The framework’s predictions are summarized in Appendix A.

7 Conclusion

We study a government-implemented empowerment program for out-of-school girls in a bride-price setting, exploiting a design in which the same girl-targeted curriculum was delivered with and without an announced \$400 grant. The announcement roughly doubled the curriculum-only point effect on stated attitudes: endorsement of early marriage fell by roughly 20 percentage points relative to control in the announced-grant arm, and the index-level announcement margin—the announced-grant arm relative to the training-only arm—is +0.26 SD (randomization-inference $p = 0.03$; the smallest familywise-adjusted p across the index’s components is 0.08). The arms also differed in realized take-up, but that difference accounts for at most a tenth of the margin.

The response extends beyond participants. Girls who never attended a session—two-thirds of the arm—moved about 70 percent as much as attendees, and girls too young for grant eligibility moved as much as eligible girls. Household heads—whose reports no girl controls, and who are observed for 86 percent of the baseline roster—shift 17 to 24 percentage points on the same early-marriage items, an announcement-specific pattern that survives familywise correction at the head level. Head effects under the training-only arm are near null. Because no cash had been paid, the reach beyond participants cannot be an income spillover.

The results are difficult to attribute to survey response bias. Three features weigh against it: the independent survey apparatus, the null pattern across self-reports where pleasing answers would be cheap, and the replication of the arm pattern in a different respondent’s module. Whether they reflect normative change or village-wide affirmation of an announced program is the residual the design cannot resolve, and we say so.

Reported emotional violence rises in point terms under both training arms (roughly five percentage points on an 8.5 percent base, though not distinguishable from zero under randomization inference). We flag it as a potential adverse margin deserving monitoring rather than a finding. The advocacy declines survive familywise correction at the five-percent level, and all four normative endpoints remain one-sided under the Lee and Semenova differential-attrition bounds (bounds that address selective response between arms, for the surveyed subpopulation).

Because the community-sensitization arm reached every village in the sample, the experimental contrast identifies the *incremental* effect of the randomized girl-targeted components above this common baseline, not the gross effect of the program relative to zero exposure. Under that estimand, no treatment arm produces a significant effect on marriage or fertility that survives family-wise correction—nulls consistent with the offsetting supply- and demand-side responses a bride-price marriage market admits (Appendix A), and equally with effects of the common outreach baseline, incomplete delivery, or modest true effects below the design’s detectable range. Because follow-up loss is common to all arms and concentrated in the marriage domain, these behavioral nulls are the paper’s weakest evidence; the attitudinal results are its strongest.

Mental health shows no broad deterioration. What the data support is a marginal two-item self-worth decline that is concentrated in the most conservative villages, reaches non-participants as much as participants, and carries Lee bounds that span zero. That pattern fits a community-level reaction to the program’s normative push—or tension surrounding the announced grant—better than individual cognitive dissonance, though the design cannot separate these channels. (Online Appendix S6 collects the benefit, null, and cost sides into a consolidated welfare accounting.)

Beyond these average effects, the exploratory heterogeneity analysis leaves two leads for future designs rather than a finding. The audit comes first. No interaction survives familywise correction (smallest Holm-adjusted $p = 0.26$); a joint randomization test puts the probability that a pair this extreme emerges from search over the interaction family alone at roughly one in ten for the T3 family ($p = 0.103$) and roughly one in four when the search spans all three arms ($p = 0.260$; Section 6.3); the norms gradient in take-up is absorbed by controls for documented delivery disruption; and the

infrastructure gradient in IGA reverses sign in high-retention villages.

The key insights are as follows. On the pregnancy channel, conservative villages combine the largest predicted pregnancy returns with the smallest take-up response—the outside-option-cost reading of Comparative Static 2. On the income-generation channel, villages with weak economic infrastructure combine the largest predicted IGA gains with the smallest take-up response—the substitution reading of Comparative Static 1. Both appear in design-based assignment interactions with no machine-learning first stage, are corroborated by the GML decomposition, and are sign-stable in leave-one-village-out re-estimation. The pregnancy-channel gradient survives every check on its outcome leg, but a barrier reading needs both legs, and its take-up leg—the norms gradient in take-up—is the one the disruption controls absorb.

Because the moderators are not randomized and observed non-participation mixes individual choice with implementation disruption and selective follow-up—attrition is highest precisely in weak-infrastructure villages—the barrier interpretation remains a hypothesis rather than an established mechanism. The dispersion at stake is nonetheless large: across arms, the GATES decomposition predicts up to a 42 percentage-point reduction in the risk of ever being pregnant for the highest-effects group (the maximum, under T2; 28 points under T3). The marriage-delay channel exhibits no analogous pattern, and neither the emotional-violence effects nor the self-worth decline tracks the participation-barrier pattern, so the two patterns are channel-specific rather than a unitary asymmetry.

Two limitations should be noted. First, bride prices and direct measures of match quality are not observed, preventing formal tests of the demand-side channel. Second, follow-up coincided with the 2019–2021 escalation of insecurity in the study regions—distress marriages are a documented coping response to economic and security stress (Corno et al., 2020)—so the incremental null is estimated against a moving equilibrium, and external validity to a peacetime rollout should be assessed with that in mind.

The announcement margin also carries a cost interpretation, which we state carefully. At the time outcomes were measured no transfer had been paid, so the normative shifts and the roughly 9.5 percentage-point take-up gain of the announced-grant arm over the training-only arm were achieved at zero marginal transfer cost *at measurement*. The announcement was not free in any policy-relevant sense, however: the commitment was genuine and was honored after the survey, and a program repeating this design carries the \$400-per-beneficiary liability—plausibly, the announcement moved behavior precisely because it was credible. What the design shows is that the normative response precedes receipt, not that it can be obtained without payment.

Two hypotheses for future program design follow. First, complementary norm-change interventions may need to precede girl-targeted programs in conservative regions, whether through engagement with religious and traditional leaders or through household-head training. In those regions the take-up response is smallest precisely on the outcomes the program is predicted to improve most (Bicchieri and Mercier, 2014).

Second, the SRH-access results caution against treating girl-targeted programming and SRH

services as interchangeable: the model’s pre-signed prediction that treatment-induced delays would be larger where SRH access is greater (Proposition 1) fails in the data—the detectable marriage-margin and knowledge gains concentrate where access is scarcest—suggesting that the program’s information content substitutes for absent local services rather than amplifying existing ones. Under that substitution reading, girl-targeted content has the most room to matter in the least-served villages, but this is an interpretive step rather than an implication of the framework. Both are hypotheses, not estimated policy effects: baseline-context interactions identify heterogeneity across village types, not the effect of changing those conditions, and testing either complement requires cross-randomizing it against girl-targeted content.

A third, operational implication does not depend on mechanism: given the self-worth decline concentrated in the most conservative villages under the announced-grant arm, and its reach beyond program participants, empowerment programs that shift girls’ aspirations in conservative settings should incorporate mental-health monitoring and safeguarding for all girls in program communities from the outset, on precautionary grounds. More broadly, program evaluations in low-income settings should systematically characterize local social and economic conditions. The heterogeneity this study can detect is organized along village-level dimensions, and individual and household characteristics predict it no better.

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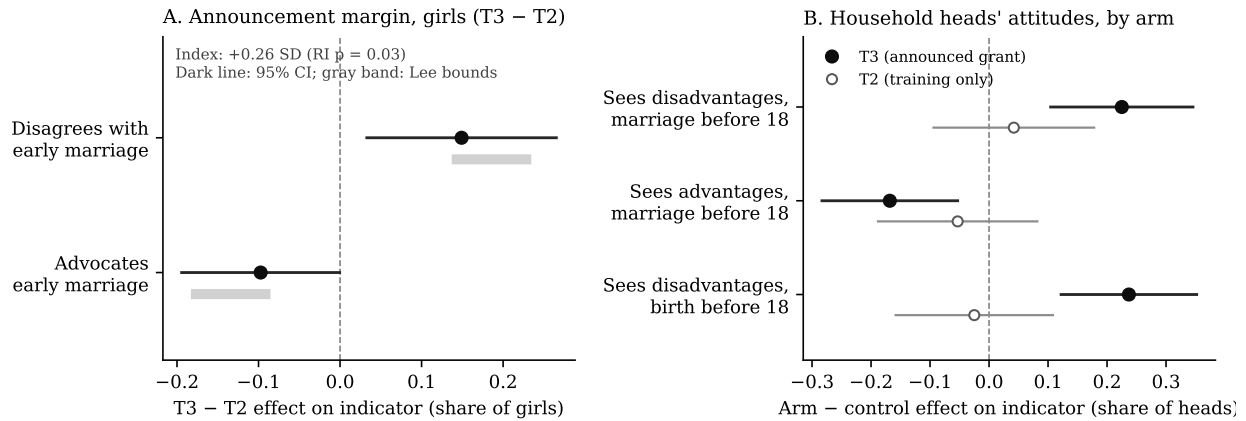
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Figures and Tables

Figure (1) The announcement margin: girls and household heads



Notes: Panel A: T3-T2 coefficients on the two binary normative outcomes from the ANCOVA specification of Table 5 (dark line: 95% confidence interval, village-clustered; gray band: T3-T2 Lee bounds). The annotated index is the single declared test of the announcement margin. Panel B: household-head-reported binary attitude items (Table 6); filled markers: T3 vs. control with 95% confidence intervals; open markers with light lines: T2 vs. control with 95% confidence intervals (all individually insignificant; exact inference in the table).

Table (1) Treatment Effects on Main Outcomes (ITT with RI)

Outcome	T1: Safe Spaces	T2: SS + Training	T3: SS+Tr.+ Ann. Grant	Control Mean	N
Ever Married	0.010 (0.017) [0.579] [1.000]	-0.006 (0.025) [0.834] [1.000]	0.001 (0.020) [0.950] [1.000]	0.799	4935
<i>RI p-val</i>					
<i>Holm p-val</i>					
Age at Mar. [†]	0.185 (0.107) [0.105] [0.948]	0.061 (0.103) [0.572] [1.000]	0.224 (0.110) [0.058] [0.520]	17.656	3686
<i>RI p-val</i>					
<i>Holm p-val</i>					
Mar. bf18	-0.004 (0.030) [0.912] [1.000]	0.016 (0.031) [0.628] [1.000]	-0.033 (0.028) [0.270] [1.000]	0.400	3597
<i>RI p-val</i>					
<i>Holm p-val</i>					
Ever Pregnant	-0.009 (0.042) [0.842] [1.000]	-0.056 (0.045) [0.241] [1.000]	-0.006 (0.039) [0.879] [1.000]	0.631	4936
<i>RI p-val</i>					
<i>Holm p-val</i>					
Curr. Pregnant	-0.018 (0.015) [0.251] [1.000]	-0.011 (0.016) [0.523] [1.000]	0.001 (0.017) [0.971] [1.000]	0.086	4601
<i>RI p-val</i>					
<i>Holm p-val</i>					
Preg. bf18	-0.009 (0.011) [0.463] [1.000]	-0.002 (0.013) [0.884] [1.000]	-0.012 (0.011) [0.297] [1.000]	0.214	4087
<i>RI p-val</i>					
<i>Holm p-val</i>					
Emo. Violence	0.032 (0.032) [0.355] [1.000]	0.049* (0.029) [0.124] [1.000]	0.051* (0.030) [0.104] [0.835]	0.085	3951
<i>RI p-val</i>					
<i>Holm p-val</i>					
IGA	0.206 (0.163) [0.241] [1.000]	-0.010 (0.138) [0.948] [1.000]	0.114 (0.148) [0.465] [1.000]	-0.007	4935
<i>RI p-val</i>					
<i>Holm p-val</i>					
Mental Health	0.073 (0.157) [0.664] [1.000]	-0.111 (0.168) [0.526] [1.000]	-0.090 (0.147) [0.560] [1.000]	0.003	4936
<i>RI p-val</i>					
<i>Holm p-val</i>					

Notes: Cluster-robust SEs in parentheses.

RI p-val: randomization inference p-value (10,000 within-strata permutations of the four-category village assignment, joint regression, PDS-selected controls held fixed).

Holm p-val: Holm (1979) step-down FWER-adjusted p-value applied to RI p-values; family = all 9 main outcomes per arm.

[†] Age at first marriage is observed only for girls married by follow-up, a post-treatment-selected subpopulation; the row is read descriptively and carries no significance stars.

Table (2) Treatment Effects on Intermediate Outcomes (ITT with RI)

Outcome	T1: Safe Spaces	T2: SS + Training	T3: SS+Tr.+ Ann. Grant	Control Mean	N
Prog. take up	0.253*** (0.040) [0.000]	0.198*** (0.036) [0.000]	0.293*** (0.040) [0.000]	0.039	4936
<i>RI p-val</i>	[0.002]	[0.003]	[0.002]		
<i>Holm p-val</i>	0.020	0.067***	0.114***	0.038	4936
Voc tr.	(0.021) [0.375]	(0.025) [0.013]	(0.028) [0.000]		
<i>RI p-val</i>	[1.000]	[0.192]	[0.006]		
<i>Holm p-val</i>	0.028	0.063***	0.075***	0.040	4908
Profes. tr.	(0.026) [0.323]	(0.021) [0.005]	(0.021) [0.001]		
<i>RI p-val</i>	[1.000]	[0.085]	[0.015]		
<i>Holm p-val</i>	0.037	0.049	0.119	0.003	4936
Finance	(0.110) [0.748]	(0.134) [0.738]	(0.103) [0.270]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	0.046	0.323***	0.428***	-0.005	4935
Gender Att.	(0.119) [0.714]	(0.121) [0.015]	(0.118) [0.001]		
<i>RI p-val</i>	[1.000]	[0.207]	[0.021]		
<i>Holm p-val</i>	-0.200*	0.109	0.134	0.000	4936
Aspirations	(0.117) [0.114]	(0.129) [0.431]	(0.125) [0.325]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	0.120	0.103	0.175	-0.000	4136
SRH Know.	(0.159) [0.483]	(0.149) [0.518]	(0.140) [0.239]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	0.068	0.175	0.157	-0.006	4136
SRH Behav.	(0.130) [0.641]	(0.108) [0.128]	(0.107) [0.173]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	0.033	-0.062	0.043	0.539	4181
MHM	(0.107) [0.765]	(0.102) [0.561]	(0.100) [0.678]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	0.062	0.032	-0.073	0.007	4936
Decision	(0.152) [0.700]	(0.138) [0.822]	(0.142) [0.623]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	0.163	0.181	0.007	0.002	2942
Decision(C)	(0.203) [0.462]	(0.143) [0.249]	(0.151) [0.970]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	-0.070	-0.102	-0.056	0.001	4917
HH Opinion	(0.153) [0.669]	(0.168) [0.568]	(0.165) [0.746]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	-0.066	0.034	-0.034	0.011	4936
Soc. Skills	(0.164) [0.707]	(0.176) [0.859]	(0.173) [0.848]		
<i>RI p-val</i>	[1.000]	[1.000]	[1.000]		
<i>Holm p-val</i>	0.032	0.055	0.204***	0.521	4935
Dis M. Early Any	(0.080) [0.716]	(0.077) [0.504]	(0.070) [0.010]		
<i>RI p-val</i>	[1.000]	[1.000]	[0.111]		
<i>Holm p-val</i>	0.053	0.112	0.436**	1.184	4935
Dis M. Early Nb	(0.212) [0.817]	(0.200) [0.592]	(0.184) [0.032]		
<i>RI p-val</i>	[1.000]	[1.000]	[0.325]		
<i>Holm p-val</i>	-0.057	-0.100	-0.206***	0.439	4935
Adv M. Early Any	(0.080) [0.502]	(0.067) [0.167]	(0.058) [0.002]		
<i>RI p-val</i>	[1.000]	[1.000]	[0.025]		
<i>Holm p-val</i>	-0.127	-0.237	-0.414***	0.807	4935
Adv M. Early Nb	(0.170) [0.489]	(0.146) [0.135]	(0.128) [0.002]		
<i>RI p-val</i>	[1.000]	[1.000]	[0.029]		
<i>Holm p-val</i>					

Notes: Cluster-robust SEs in parentheses; * p<0.10, ** p<0.05, *** p<0.01.
RI p-val: randomization-inference p-value (permutation of assignment, joint regression). *Holm p-val*: Holm (1979) step-down FWER adjustment applied to RI p-values; family = all 17 intermediate outcomes per arm.

Table (3) Normative attitude effects by program participation

	Non-participants	Participants	p (equal)	Control mean
<i>Panel A: T3 (Safe Spaces + Training + Announced Grant), N = 2,596</i>				
Dis M. Early Any	0.180** (0.071)	0.289*** (0.070)	0.026	0.521
Dis M. Early Nb	0.399** (0.189)	0.604*** (0.192)	0.105	1.184
Adv M. Early Any	-0.172*** (0.055)	-0.251*** (0.060)	0.029	0.439
Adv M. Early Nb	-0.362*** (0.114)	-0.507*** (0.132)	0.040	0.807
<i>Panel B: T2 (Safe Spaces + Training), N = 2,308</i>				
Dis M. Early Any	0.046 (0.077)	0.085 (0.076)	0.491	0.521
Dis M. Early Nb	0.103 (0.197)	0.181 (0.201)	0.562	1.184
Adv M. Early Any	-0.103 (0.064)	-0.134* (0.070)	0.413	0.439
Adv M. Early Nb	-0.246* (0.143)	-0.324** (0.154)	0.418	0.807

Notes: Each panel reports, for one treated arm, coefficients on indicators for non-participants and participants (relative to control girls) from one regression per outcome on the arm-and-control sample, with strata fixed effects and age dummies; village-clustered standard errors in parentheses. Participation is safe-space take-up, which is itself a choice, so the split is descriptive rather than experimentally identified. “Dis M. Early” denotes disagreement that early marriage is appropriate (Any = indicator; Nb = count of perceived disadvantages); “Adv M. Early” denotes advocacy of early marriage (Any = indicator; Nb = count of perceived advantages). The p (equal) column reports the p -value from the test that the participant and non-participant coefficients are equal. Raw take-up rates among take-up-observed respondents: 33.3 percent (T3), 24.3 percent (T2), 3.9 percent (control); panel N s can differ by one observation from the sum of arm counts because of item missingness on single outcomes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table (4) The announcement margin: T3–T2 gap by program participation

Outcome	Non-participants	Participants	p (equal)
Dis M. Early Any	0.131* (0.071)	0.181*** (0.061)	0.506
Dis M. Early Nb	0.278 (0.177)	0.430** (0.164)	0.406
Adv M. Early Any	-0.094* (0.052)	-0.100 (0.063)	0.921
Adv M. Early Nb	-0.172 (0.109)	-0.131 (0.122)	0.736
Gender attitudes	0.117 (0.140)	0.033 (0.136)	0.565

Notes: Coefficients on $T3 \times$ non-participant and $T3 \times$ participant indicators from one regression per outcome on the pooled T2-and-T3 sample ($N = 2,750$), with a participant main effect, strata fixed effects, and age dummies; village-clustered standard errors in parentheses. T2 and T3 delivered the same curriculum and differed only in the announced grant, so each column is the announcement margin within a participation status; participation is safe-space take-up, which is itself a choice, so the split is descriptive rather than experimentally identified. The p (equal) column tests equality of the two coefficients; the associated confidence intervals are wide, so equality is not established by non-rejection (see the text). N can differ by one observation from the sum of the arm counts because of item missingness. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table (5) The announcement margin: T3–T2 contrasts on the normative outcomes

	T3–T2	SE	p_{RI}	Holm	Lee bounds	IM 95% CI
Anti-early-marriage index	0.265**	(0.109)	0.029	—	[0.195, 0.482]	[−0.017, 0.737]
Dis M. Early Any	0.149**	(0.059)	0.021	0.083	[0.137, 0.235]	[0.024, 0.380]
Dis M. Early Nb	0.331**	(0.155)	0.049	0.148	[0.186, 0.528]	[−0.118, 0.896]
Adv M. Early Any	−0.097*	(0.049)	0.079	0.158	[−0.183, −0.085]	[−0.298, 0.001]
Adv M. Early Nb	−0.165	(0.097)	0.128	0.158	[−0.357, −0.138]	[−0.552, 0.025]

Notes: Coefficients on the T3 indicator from regressions on the pooled T2-and-T3 sample with the baseline outcome and its missingness indicator, strata fixed effects, and age dummies; village-clustered standard errors in parentheses. The index is an inverse-covariance-weighted (Anderson) aggregate of the four normative outcomes, each oriented so that higher values are anti-early-marriage, standardized by the control group; the four outcome rows decompose it. The index test was specified after the component arm-level results were known, so it is post-hoc; the Holm column corrects across the four components. p_{RI} : randomization inference from 10,000 within-strata permutations of the four-category village assignment ($|t|$ criterion, plus-one convention). Lee bounds: manual marginal trimming bounds for the T3–T2 contrast on unadjusted means—so they bracket a different estimand than the covariate-adjusted point estimates—trimming the higher-response arm (T3) by the response differential (8.9 percent of its respondents); they address only differential response between the two arms, not the common follow-up loss. IM 95% CI: Imbens–Manski confidence intervals for the partially identified contrast, with endpoint standard errors from a village-cluster bootstrap (200 replications, resampling within strata). Every identified set excludes zero; accounting for sampling uncertainty, the disagreement indicator remains strictly positive, the advocacy indicator is borderline, and the remaining intervals include zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (analytic).

Table (6) Household heads' early-marriage attitudes, by arm

Outcome (head-reported)	T1	T2	T3	p_{RI} (T3)	Holm (T3)	p_{RI} (T2=T3)	Holm	N
Head: any disadv. mar. bf18	0.061 (0.078)	0.042 (0.069)	0.225*** (0.062)	0.002	0.005	0.001	0.004	6,642
Head: any adv. mar. bf18	-0.051 (0.076)	-0.053 (0.069)	-0.168*** (0.059)	0.009	0.018	0.049	0.146	6,645
Head: any disadv. birth bf18	0.029 (0.078)	-0.025 (0.068)	0.237*** (0.059)	0.001	0.003	0.000	0.000	6,645
Head: ideal marriage age	0.192 (0.221)	0.161 (0.164)	0.494*** (0.145)	0.001	0.005	0.059	0.146	6,645
Head: late-marriage stigma	-0.343 (0.429)	-0.151 (0.356)	-0.192 (0.337)	0.598	0.598	0.916	0.916	3,788

Notes: Outcomes are reported by the girl's household head in the household module, reconstructed from the raw household files and merged on the household identifiers. The head module reaches households even when the girl herself is not re-interviewed: it covers 6,642–6,645 girls (5,875 distinct heads) of the 7,719-girl baseline roster (86 percent, versus 64 percent for the girl module), comprising 4,915 girls who responded and 1,727 who did not; arm counts for the first outcome are 1,465 (control), 1,527 (T1), 1,682 (T2), and 1,968 (T3). Head-module availability does not differ by arm (joint $F = 0.92$, $p = 0.43$). Coefficients from joint four-arm regressions at the girl level with strata fixed effects and age dummies; village-clustered standard errors in parentheses; stars on the T3 randomization-inference p -values (10,000 within-strata permutations of the four-category assignment). Holm columns correct across the five head outcomes. p_{RI} (T2=T3): randomization inference for the Wald test of equal T2 and T3 coefficients—the announcement-specificity test. Marginal Lee bounds for the T3–control and T3–T2 contrasts are one-sided for all four attitude and ideal-age outcomes (replication output). The stigma score has substantial item nonresponse ($N = 3,788$) and is the one null head outcome; we flag rather than interpret it. These outcomes are not reported by the girls, so girl-level reporting responses cannot generate them; a head-level reporting response remains logically possible. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix

A The Conceptual Framework: Model, Derivations, and Predictions

A.1 The Marriage-Market and Participation Model

We develop a simple model of adolescent empowerment in a bride-price marriage market. The model has two parts. The first is a marriage-market module: the girl-targeted components can raise the value of remaining unmarried, but they can also raise the perceived value of a trained bride. These two margins can offset, generating small average effects on marriage even when the program changes human capital and preferences. The second is a participation module: program gains are realized only when girls take up the girl-targeted components, and the villages where the gains are largest may be the villages where take-up is hardest. This participation barrier is the object examined, as an exploratory hypothesis, in Section 6.3. The objects Π_v and Ω_v are not separately identified in the data; they are conceptual objects used to organize signed heterogeneity predictions that are tested through reduced-form assignment-by-structure interactions and CATE decompositions.

Two features of the experimental setting guide the model. First, all villages received community sensitization, so the treatment coefficients identify the *incremental* effects of the randomized girl-targeted components relative to outreach-only exposure. Second, the cash grant in the full package had not been disbursed by follow-up. We therefore interpret the full-package arm as the effect of safe spaces and training combined with the incentive and credibility created by an anticipated transfer, not as the effect of cash receipt.

A.2 Marriage market

Consider a young woman in village v . Her human capital is h , and a denotes age at marriage. If she marries, marital surplus is

$$S_v(a, h) = U_v^m(a, h) + U_v^w(a, h) - V_v^w(a, h),$$

where U_v^m is the groom-side value of marriage, U_v^w is the woman's value inside marriage, and V_v^w is her outside option. In a bride-price setting, the transfer is negotiated between households and flows from the groom's household to the bride's household. We write the woman's participation constraint as a reduced-form representation of the household-mediated outside option: the transfer required to clear the match reflects the value of delaying marriage for the woman and her family. When this participation constraint binds, the bride price is

$$p_v(a, h) = V_v^w(a, h) - U_v^w(a, h).$$

The groom side chooses marriage timing taking this price schedule as given:

$$\max_a U_v^m(a, h) - p_v(a, h).$$

The first-order condition is

$$U_{v,a}^m(a, h) + U_{v,a}^w(a, h) - V_{v,a}^w(a, h) = 0, \tag{3}$$

or equivalently $S_{v,a}(a, h) = 0$. Let

$$\Delta_v \equiv S_{v,aa} < 0$$

be the curvature of surplus in age. The treatment affects marriage timing by changing human capital, so totally differentiating equation (3) gives

$$\frac{\partial a_v^*}{\partial T_z} = -\frac{S_{v,ah}}{S_{v,aa}} \frac{\partial h}{\partial T_z} = -\frac{\Pi_v - \Omega_v}{\Delta_v} \frac{\partial h}{\partial T_z}, \quad (4)$$

where

$$\Pi_v \equiv U_{v,ah}^m + U_{v,ah}^w, \quad \Omega_v \equiv V_{v,ah}^w.$$

The term Π_v is the marriage-market premium to skills: it captures how much human capital raises the marginal value of waiting inside marriage. The term Ω_v is the outside-option or bride-price channel: it captures how much human capital raises the marginal value of waiting outside marriage, and hence how much the groom side must compensate the bride's household for delaying marriage. Since $\Delta_v < 0$, the sign of the marriage response is governed by $\Pi_v - \Omega_v$: if $\Pi_v > \Omega_v$ the program delays marriage, and if $\Omega_v > \Pi_v$ it can accelerate marriage. The average ITT on marriage timing is $\mathbb{E}_v[p_{vz} q_z (\Pi_v - \Omega_v)]$, where p_{vz} is the village take-up rate and q_z the effective content intensity of arm z (Appendix A.6 derives this in closed form), so near-zero average effects require only that the *take-up-weighted* average of $\Pi_v - \Omega_v$ be close to zero—because the channels offset within villages, because opposite-signed villages balance, or because take-up is concentrated where $\Pi_v - \Omega_v$ is small—even when the program changes human capital or preferences. This ambiguity is specific to the bride-price regime: Proposition 3 in Appendix A.6 shows that under dowry Ω_v drops out of the timing condition, so the model predicts an unambiguous delay whenever $\Pi_v > 0$.

A.3 An announced transfer

The announced-grant arm calls for a special case in which a credible future transfer, not yet received, enters current stated attitudes and current marriage timing. Let the grant G be paid at a later date to a girl who has participated, whether or not she has married by then. Its value to the bride side depends on who controls it once she marries: let $\kappa(a) \in [0, 1]$ be the share of G that remains with the girl and her natal household if she marries at age a , with $\kappa'(a) \geq 0$, so that an older bride retains more control over resources she brings into marriage. The grant then raises the outside option by G and the woman's value inside a marriage contracted at age a by $(1 - \kappa(a))G$.

Two objects respond differently. The first is the bride side's reservation transfer at a given early age a_0 , $R_v \equiv V_v^w(a_0, h) - U_v^w(a_0, h)$, the natural model counterpart of a stated attitude toward marrying the girl early: the household objects to an early marriage when R_v is high, and its stated ideal marriage age is the age at which R_v falls below what the market pays. The announcement raises R_v by $\kappa(a_0)G > 0$ for any $\kappa(a_0) > 0$. The second object is realized timing, which follows the groom side's first-order condition $S_{v,a} = 0$. Because G enters surplus as a level, it moves timing only through the age-dependence of control: $\partial a_v^* / \partial G = \kappa'(a) / |\Delta_v|$, which is zero when control over the grant does not depend on the age at marriage and small when it depends weakly. (With the quadratic surplus of Appendix A.7 and $\kappa(a) = \kappa_0 + \kappa_1 a$, $a_v^* = \bar{a}_v + (\Pi_v - \Omega_v)h + \kappa_1 G$.)

The special case delivers three predictions. First, stated attitudes and stated ideal ages move with the announcement for eligible households, while marriage timing moves only if control over the grant rises with age at marriage—a pattern of attitude movement without timing movement that the model produces without offsetting. Second, households with no prospect of the grant—those of ineligible girls, and of girls who do not participate—respond only if the announcement is a public signal that updates their beliefs about the market value of an unmarried skilled girl, that is, about Ω_v . Such a signal raises R_v for every household in the village, matching the flat eligibility gradient of Section 6.1.2; but because it operates through Ω_v it does not delay marriage in this model and can accelerate it through the bride-price channel of equation (4). Third, the household head reports the bride side's reservation, so head-reported attitudes should move with the girls' and by the same mechanism, as they do. What the special case does not deliver is a signed prediction on realized

marriage timing, and it says nothing about affirmation of an announced program as an alternative reading of the same reports; that residual is empirical.

A.4 Participation and realized treatment effects

Let $z \in \{T1, T2, T3\}$ index the three randomized arms. The girl-targeted components raise human capital only if the girl participates:

$$h_{iv}(z) = h_{iv0} + D_{ivz}q_z, \quad 0 < q_{T1} < q_{T2} \leq q_{T3}, \quad (5)$$

where D_{ivz} is take-up and the ordering reflects the program design: T1 provides safe spaces and life skills, T2 adds vocational and business training, and T3 adds the full package with the anticipated grant.

A key implication is that participation can gate some outcomes more directly than others. Pregnancy-related behavior, reproductive-health knowledge, and income-generating activities are margins on which girls' own program exposure can plausibly change information, preferences, and actions. Marriage timing is more household-mediated: parents, household heads, and negotiating families play a central role, and bride-price transfers are bargained between households. Thus a girl-level participation barrier can be central for pregnancy and IGA without generating the same opposite-signed take-up/outcome pattern for marriage timing.

Participation is costly. Let X_v summarize village structure, including progressive norms or autonomy (N_v , the Unrestricted index), access to birth-control and reproductive-health services (R_v), economic infrastructure (K_v), security (G_v), and exogamy measures (σ_v^g, σ_v^b). A parsimonious latent-index rule is

$$D_{ivz} = \mathbb{1}\{\mu_z + \lambda_N N_v + \lambda_R R_v + \lambda_K K_v + \lambda_G G_v - \eta_i \geq 0\}, \quad (6)$$

where η_i is individual resistance to participation. The first-stage evidence implies $\lambda_N > 0$, $\lambda_R > 0$, and $\lambda_K > 0$: take-up is higher in less restrictive, better connected villages. Security is included as a structural village condition without a signed take-up loading; empirically it appears primarily as a returns shifter for IGA rather than as a take-up barrier.

For any outcome Y , write the assignment effect in village v as

$$m_Y^z(X_v) \equiv \mathbb{E}[Y(1) - Y(0) \mid X_v, Z = z] = p_z(X_v)\tau_Y^z(X_v), \quad (7)$$

where $p_z(X_v) = \Pr(D_{ivz} = 1 \mid X_v)$ is the take-up response and $\tau_Y^z(X_v)$ is the conditional return to participation among participants. Equation (7) is a *restricted* identity, not a generic decomposition: it imposes that assignment moves the outcome only through participation. We maintain that restriction for the human-capital channel outcomes (pregnancy, IGA, marriage timing) as a stated assumption, and deliberately do not apply it to mental health or emotional violence, for which the paper documents effects among nonparticipants in treated villages (Section 6.1.2); for those outcomes the assignment effect contains an additional exposure term for nonparticipants. The paper's own attitude results show that assignment reaches non-participants and household heads, so an exposure channel outside formal take-up exists for at least one outcome family. For the human-capital outcomes we assume that channel is second-order; the assumption is not tested, and the participation-barrier decomposition below should be read as conditional on it. Under the restriction, the ITT can be small either because $p_z(X_v)$ is low, because $\tau_Y^z(X_v)$ is low, or because positive and negative effects across contexts offset in the aggregate.

Participation-barrier condition. A village feature X_k is a participation barrier for outcome Y when it raises take-up but lowers realized assignment gains:

$$\frac{\partial p_z}{\partial X_k} > 0 \quad \text{and} \quad \frac{\partial m_Y^z}{\partial X_k} < 0. \quad (8)$$

The condition is defined for sign-normalized beneficial outcomes: Y is coded so that higher values are more desirable (the convention of Section 6.3; for the one non-inverted outcome, pregnancy before 18, the inequality on m_Y^z flips), and it is channel-specific—a village feature can be a barrier for one outcome and not for another. This is the empirical sign pattern in Section 6.3: favorable village conditions make participation easier, but the outcome gains are largest where conditions are unfavorable. Appendix A.6 derives the closed-form condition under which realized effects fall as conditions improve (conditional returns must decline proportionally faster than participation rises).

A.5 Predictions

Appendix A.6 develops the framework’s testable content in detail—two formal propositions whose signs follow from primitive assumptions, two comparative statics whose signs the data determine, and one definitional identity—together with the closed-form illustrations and the per-channel empirical mapping. Five implications organize the empirical analysis. First, average marriage and fertility ITTs can be near zero despite a strong first stage, through take-up dilution and the offsetting of Π_v and Ω_v (Tables 1 and 2); scaling ITTs by the first stage carries a complier interpretation only where the exclusion restriction is plausible, which excludes mental health and emotional violence. Second, the pregnancy channel has a social-norm participation barrier if conservative villages combine lower take-up with larger conditional pregnancy-prevention returns (Comparative Static 2). Third, the IGA channel has an infrastructure participation barrier if the marginal return to business training is largest where infrastructure is weak (Comparative Static 1). Fourth, marriage timing—partly chosen by parents and mediated by bride-price negotiation—need not display any opposite-signed take-up/outcome pattern even when participation matters for pregnancy or IGA. Fifth, exogamy generates the model’s most distinctive falsification test, a sign reversal across outcome families: male exogamy should strengthen marriage-delay effects (outside grooms cannot observe and price the girl’s program-induced outside-option gain) while female exogamy weakens IGA effects (girls anticipating out-marriage invest less locally) (Proposition 2). Mental health sits outside this mapping: when private preferences move faster than household and community norms, girls can experience short-run dissonance that need not operate through formal take-up, so we treat mental-health effects as an interpretation consistent with norm change in restrictive settings rather than as a test of the model.

The framework nests the standard monotone-delay logic of empowerment programs as a special case: with no outside-option channel ($\Omega_v = 0$), or under dowry (Proposition 3), any program that raises human capital unambiguously delays marriage, and under full take-up the participation-barrier condition cannot hold. Its distinctive content—sign-ambiguous marriage effects and opposite-signed reach-versus-gains patterns—therefore arises only from the combination of a bride-price regime and incomplete, structured participation. Throughout, the model is a mechanism-guided reduced-form theory of incidence, not a structural estimation exercise: bride prices and match quality are unobserved, so Π_v and Ω_v are disciplined indirectly through reduced-form sign patterns across outcomes and village structures, and the barrier mapping is channel-specific—social norms bind the pregnancy channel, economic infrastructure binds the IGA channel, and the marriage-delay channel shows no analogous barrier.

A.6 Theoretical Predictions

In this appendix, we develop the comparative statics of the framework in detail, distinguishing between two formal propositions whose signs follow from defensible primitive assumptions, two comparative statics whose signs are determined by the data, and one definitional identity that motivates the participation-barrier test of Section 6.3.

To start, recall that the treatment effect on equilibrium marriage age is

$$\frac{da}{dT} = -\frac{\Psi}{\Delta} \frac{\partial h}{\partial T}, \quad \Psi \equiv \Pi - \Omega, \quad \Delta \equiv U_{aa}^m + U_{aa}^w - V_{aa}^w < 0. \quad (9)$$

Since $\partial h/\partial T > 0$ by assumption and $\Delta < 0$ under standard concavity, the sign of the treatment effect is governed by the relative strength of the marriage-market channel Π and the outside-option channel Ω :

$$\text{sign}\left(\frac{da}{dT}\right) = \text{sign}(\Pi - \Omega).$$

A direct corollary is that when the two channels approximately offset ($\Pi \approx \Omega$), $da/dT \approx 0$ even though $\partial h/\partial T > 0$. The framework therefore admits null average treatment effects on marriage timing as one possibility consistent with successful human-capital transmission; the heterogeneity analysis is what distinguishes such offsetting from a null caused by treatment failure ($\partial h/\partial T = 0$).

The remainder of this appendix derives signed predictions from defensible assumptions on $\partial\Pi/\partial R_v$ (Proposition 1) and $\partial\Omega/\partial\sigma_v^g$ (Proposition 2), then states two comparative statics whose sign the theory leaves ambiguous and which must therefore be assessed empirically (Comparative Statics 1 and 2), and finally states the take-up identity that grounds the participation-barrier test (Definition 1).

Comparative Static 1 (Labor-market access shifts the difference $\Pi - \Omega$). *Let L_v denote village-level labor-market access, encompassing market connectivity, proximity to urban centers, and the density of income-generating opportunities. Better labor-market access can raise both channels: $\partial\Pi/\partial L_v \geq 0$ (skills are more productive inside marriage where local markets are accessible) and $\partial\Omega/\partial L_v \geq 0$ (skilled unmarried women can deploy human capital in the local labor market). The sign of the heterogeneous effect therefore reduces to the sign of the difference:*

$$\frac{\partial}{\partial L_v} \left(\frac{da}{dT} \right) = -\frac{1}{\Delta} \frac{\partial(\Pi - \Omega)}{\partial L_v} \frac{\partial h}{\partial T}.$$

Hence the comparative static is signed only conditional on which channel labor-market access shifts more, and the data identify the regime.

Interpretation. The proposition's sign turns on whether better labor markets strengthen the age-human-capital complementarity *inside* marriage more than they strengthen it *outside* marriage. If the marriage-market channel shifts up more (e.g., complementarities in joint household production with skilled spouses dominate), then $\partial(\Pi - \Omega)/\partial L_v > 0$ and treatment-induced delay is larger in better-connected villages. If the outside option shifts up more (e.g., women's labor-force opportunities expand more rapidly with L_v than within-marriage returns), then $\partial(\Pi - \Omega)/\partial L_v < 0$ and the delay is smaller. The CLAN analysis adjudicates.

Empirical test. The sign of heterogeneity with respect to the economic-infrastructure index in the CLAN analysis identifies which channel dominates.

Comparative Static 2 (Social-norm conservatism creates opposing pressures). *Let C_v denote a village-level index of social-norm conservatism, including the prevalence of early-marriage norms and barriers to female autonomy. Conservatism plausibly raises both channels:*

$$\frac{\partial\Pi}{\partial C_v} > 0 \quad \text{and} \quad \frac{\partial\Omega}{\partial C_v} > 0,$$

while Δ and $\partial h/\partial T$ remain unchanged. Then

$$\frac{\partial}{\partial C_v} \left(\frac{da}{dT} \right) = -\frac{1}{\Delta} \left(\frac{\partial \Pi}{\partial C_v} - \frac{\partial \Omega}{\partial C_v} \right) \frac{\partial h}{\partial T}.$$

Hence the sign of $\partial(da/dT)/\partial C_v$ is theoretically ambiguous and is identified by the data, not the model.

Interpretation. Conservative norms raise the marriage-market premium on skilled wives ($\partial \Pi/\partial C_v > 0$): in settings where marriage is a primary social institution, the age-human-capital complementarity inside marriage is stronger, pulling toward later marriage. At the same time, conservative norms strengthen the age-human-capital complementarity in the outside option ($\partial \Omega/\partial C_v > 0$): in restrictive environments, skilled older unmarried women face heightened social pressure, raising their outside option valuation at older ages and increasing the bride-price cost of waiting. If the marriage market premium effect dominates ($\partial \Pi/\partial C_v > \partial \Omega/\partial C_v$), conservative villages exhibit larger treatment-induced delays. If the bride-price effect dominates, conservative villages exhibit smaller delays or even acceleration.

Empirical test. The sign of heterogeneity with respect to the social-norm index in the CLAN analysis identifies which mechanism dominates empirically. Positive heterogeneity estimates favor the marriage-market premium interpretation; negative estimates favor the outside-option cost interpretation.

Empirical test. Under the model, the CLAN contrasts on the social-norms index provide evidence on which channel conservatism strengthens more. The empirical index increases with permissiveness, the reverse of the theoretical conservatism measure C_v . For outcomes coded so that higher values represent delayed marriage or pregnancy, a negative CLAN contrast indicates larger predicted gains in more conservative villages and favors the marriage-market premium interpretation. Conversely, a positive contrast indicates larger predicted gains in more permissive villages and favors the outside-option-cost interpretation.

Proposition 1 (Reproductive-health access decouples age from human capital in the outside option). *Let R_v denote village-level access to sexual and reproductive health (SRH) services. Suppose that better SRH access weakens the outside-option channel,*

$$\frac{\partial \Omega}{\partial R_v} < 0,$$

while Π , Δ , and $\partial h/\partial T$ remain unchanged. Then

$$\frac{\partial}{\partial R_v} \left(\frac{da}{dT} \right) > 0.$$

Thus, treatment-induced delays in marriage and fertility are larger where SRH services are more accessible.

Proof sketch. From (9), holding Π , Δ , and $\partial h/\partial T$ fixed, a decrease in Ω increases $\Psi = \Pi - \Omega$ and therefore increases da/dT because $\Delta < 0$. $\square$

Interpretation. When contraception and related services are available regardless of marital status, reproductive timing no longer penalizes older unmarried women. Human capital is then equally deployable across ages in the outside option (Ω falls), reducing the bride-price cost of skilled older women and allowing the marriage-market channel to dominate more easily. Conversely, where SRH access is limited, age and human capital remain strong complements in the outside option: skills

matter more the older a woman is, since the reproductive cost of remaining unmarried falls with age. This keeps Ω high, dampening the treatment-induced delay.

Empirical test. Treatment effects on pregnancy and fertility outcomes should be larger in villages with greater SRH access, especially under treatment arms that directly improve knowledge and agency.

Proposition 2 (Marriage regime moderates the demand-side channel through groom observability). *Measure the local marriage regime by two continuous variables constructed from village informants' recall of the 10 most recent unions: the male exogamy rate $\sigma_v^g \in [0, 1]$, the share of grooms who originate from outside the village, and the female exogamy rate $\sigma_v^b \in [0, 1]$, the share of brides who originate from outside the village. Overall exogamy is $\sigma_v = (\sigma_v^g + \sigma_v^b)/2$.*

Let $\iota_v = 1 - \sigma_v^g \in [0, 1]$ denote groom information about program participation. When σ_v^g is high, outside grooms do not observe the woman's program participation and therefore do not adjust the bride price upward to reflect her increased outside option valuation at older ages. They effectively underestimate V_{ah}^w , lowering the effective outside-option channel. Suppose that

$$\frac{\partial \Omega}{\partial \iota_v} > 0, \quad \text{equivalently} \quad \frac{\partial \Omega}{\partial \sigma_v^g} < 0,$$

while Π , Δ , and $\partial h/\partial T$ remain unchanged. Then

$$\frac{\partial}{\partial \sigma_v^g} \left(\frac{da}{dT} \right) > 0.$$

Treatment-induced marriage delay is therefore increasing in the male exogamy rate.

Proof sketch. From (9), $da/dT = -(\Pi - \Omega)/\Delta \cdot \partial h/\partial T$. Holding Π , Δ , and $\partial h/\partial T$ fixed, a rise in σ_v^g lowers ι_v , which lowers the effective Ω and therefore increases $\Psi = \Pi - \Omega$. Since $\Delta < 0$, $-1/\Delta > 0$, and a higher Ψ increases da/dT . $\square$

Interpretation. In endogamous villages, local grooms observe the program directly and can infer the woman's increased outside option value at older ages. They incorporate this into the bride price, which fully transmits Ω into the timing equilibrium and partially offsets the treatment-induced delay. In exogamous villages, outside grooms do not observe the program and therefore do not raise the bride price to reflect the woman's higher outside option at older ages. The effective Ω is muted, $\Pi - \Omega$ rises, and the treatment-induced delay is amplified.

The female exogamy rate σ_v^b generates a separate and opposing prediction for income-generating activities. When girls anticipate marrying outside the village, the local labor market is less relevant to their outside option, so $\partial \Omega^{\text{IGA}}/\partial \sigma_v^b < 0$. A rise in σ_v^b therefore reduces the IGA treatment effect while a rise in σ_v^g increases the marriage-delay effect. This sign reversal across outcome families (from the same village-level exogamy measures) is arguably a strong falsification test of the groom-observability mechanism since no village-level characteristic that shifts all outcomes uniformly can account for the pattern observed.

Empirical test. In the CLAN analysis, σ_v^g should load positively for marriage-delay and pregnancy outcomes, particularly under T2 and T3, the training-intensive arm that produces the human capital outside grooms cannot observe. The overall exogamy rate σ_v should load negatively for IGA outcomes.

Proposition 3 (The equilibrium ambiguity is specific to bride-price settings). *Consider the mirror-image dowry regime, in which the transfer flows from the bride's household to the groom's side and clears the groom side's participation constraint,*

$$d_v(a, h) = V_v^m(a) - U_v^m(a, h),$$

where $V_v^m(a)$ is the groom-side outside option, which does not vary with the bride's human capital. The bride's side chooses marriage timing taking the dowry schedule as given, $\max_a U_v^w(a, h) - d_v(a, h)$, so its objective is $\tilde{S}_v(a, h) = U_v^w(a, h) + U_v^m(a, h) - V_v^m(a)$, with curvature $\tilde{\Delta}_v \equiv \tilde{S}_{v,aa} < 0$. Then

$$\frac{\partial a_v^*}{\partial T_z} = -\frac{\Pi_v}{\tilde{\Delta}_v} \frac{\partial h}{\partial T_z} > 0 \quad \text{whenever } \Pi_v > 0 :$$

under dowry, a treatment that raises human capital unambiguously delays marriage, and the outside-option channel Ω_v does not enter the timing condition at all.

Proof sketch. Totally differentiating the timing first-order condition $\tilde{S}_{v,a} = 0$ gives $\partial a_v^* / \partial T_z = -\tilde{S}_{v,ah} / \tilde{S}_{v,aa} \cdot \partial h / \partial T_z$. Because V_v^m is invariant to the bride's h , the cross-partial is $\tilde{S}_{v,ah} = U_{v,ah}^w + U_{v,ah}^m = \Pi_v$, and $\tilde{\Delta}_v < 0$ delivers the sign. In the quadratic illustration, $\tilde{S}_v = -\frac{1}{2}(a - \bar{a}_v)^2 + \Pi_v h(a - \bar{a}_v) + \kappa_v h$ gives $a_v^* = \bar{a}_v + \Pi_v h$ and $\partial a_v^* / \partial T_z = D_{ivz} q_z \Pi_v$. The algebra is verified symbolically in the replication materials. $\square$

Interpretation. The asymmetry has a simple source: a market-clearing transfer transmits into the timing condition the outside option of the party it compensates. Under bride price, the groom side pays the bride's household for her forgone outside option, so V_v^w —and with it Ω_v , which rises with the girl's human capital—enters the equilibrium and can offset the premium channel Π_v . Under dowry, the bride's side compensates the groom's forgone outside option, which does not respond to her human capital, so only Π_v governs timing. Training also lowers the dowry level itself ($\partial d_v / \partial h = -U_{v,h}^m < 0$), relaxing the bride household's budget constraint without affecting the timing margin. The equilibrium ambiguity that rationalizes null average effects in our setting is therefore not a generic feature of marriage-market models: it is specific to settings where the transfer flows toward the bride's household. The framework accordingly assigns different diagnoses to null results across regimes: in a dowry setting, a null marriage effect of a human-capital intervention cannot reflect equilibrium offsetting and must instead indicate a weak first stage on the relevant human capital (a reading consistent with the null empowerment effects in Bangladesh reported by Buchmann et al., 2023), whereas in a bride-price setting a null is consistent with a strong first stage whose two demand- and supply-side channels cancel. Two caveats apply: the statement is partial-equilibrium (it holds the distribution of h , and hence V_v^m , fixed), and it maintains $\Pi_v > 0$.

Definition 1 (Take-up identity). *Extending the representative-pair model to a population of girls indexed by i , let*

$$\frac{\partial h_i}{\partial T} = D_i g_i,$$

where $D_i \in \{0, 1\}$ indicates program take-up and $g_i > 0$ is the conditional gain from participation. Then the individual-level treatment effect is

$$\frac{da_i}{dT} = -\frac{\Psi_i}{\Delta_i} D_i g_i,$$

and the village-level ITT averages this over D_i , so non-take-up dilutes the realized effect mechanically. Small village-level ITTs are therefore consistent with large conditional effects when take-up is selective in g_i .

Implication. Definition 1 is an algebraic identity, not a comparative static; it does not by itself predict the direction of selection. The substantive question is whether girls (or villages) with high g_i are systematically those with $D_i = 0$ — a participation-barrier asymmetry — or whether high- g_i girls self-select into participation. The participation-barrier test of Section 6.3 adjudicates this directly by regressing the village-level take-up CATE on each outcome's CATE.

Taken together, the framework yields four kinds of empirical content. First, the channel-decomposition identity $\text{sign}(da/dT) = \text{sign}(\Pi - \Omega)$ admits null average treatment effects when the two channels

approximately offset, distinguishing offsetting from treatment failure as alternative interpretations of small ITTs. Second, Proposition 1 (SRH access) is signed under the defensible assumption that contraceptive access weakens the outside-option channel by decoupling reproductive risk from age, leaving the marriage-market channel approximately unchanged (empirically the prediction is not supported: the tertile analysis of Section 6.2 finds the detectable gains concentrated where SRH access is scarcest, the reverse of the predicted gradient). Third, Proposition 2 (exogamy with sign reversal) is signed under the groom-observability assumption: higher male exogamy mutes the effective Ω because outside grooms cannot price the woman’s program-induced outside-option gain, while higher female exogamy weakens incentives to invest in the local labor market. The same village-level exogamy structure therefore predicts a positive gradient on marriage-delay outcomes and a negative gradient on IGA outcomes — a sign reversal across outcome families that no uniform village-level confound can replicate, providing a demanding falsification test of the framework (the discriminating power of the test also requires the two exogamy margins to covary positively across villages, which they do, $\hat{\rho} = 0.39$; empirically the reversal is supported at conventional significance for the T2 arm only; see Section 6.3). This finding also inverts the endogamy result of Hotte and Lambert (2023): whereas they show that ethnic endogamy reduces the cadeau (marriage gift) because grooms need not signal to a familiar bride, we show that endogamy strengthens the Ω channel by allowing local grooms to observe and fully price the woman’s outside-option gain.

Fourth, Comparative Statics 1 (labor-market access) and 2 (social-norm conservatism) are theoretically sign-ambiguous: each village characteristic shifts both channels, and the sign of the comparative static reduces to the sign of $\partial(\Pi - \Omega)/\partial L_v$ or $\partial(\Pi - \Omega)/\partial C_v$ respectively. The data select the regime empirically, and the CLAN analysis in Section 6.3 reports which channel appears to dominate in each case rather than confirming a pre-signed prediction.

Finally, Definition 1 (take-up identity) frames the participation-barrier test of Section 6.3: small village-level ITTs are mechanically consistent with large conditional treatment effects when take-up is selective in g_i , but the substantive question is whether the selection is toward the participation barrier (high- g_i girls do not take up) or against it (high- g_i girls self-select into participation). That is an empirical question, not a theoretical one.

Overall, this exercise helps clarify that child marriage reduction programs (targeting out-of-school girls) that work through human capital accumulation in bride-price settings can generate mixed average effects in general. The distribution of local labor-market and social conditions determines whether supply or demand shifts dominate. Proposition 3 makes the distinction from dowry settings formal: when the transfer flows from the bride’s household to the groom’s side, the woman’s outside option does not enter the timing condition, so a human-capital intervention delays marriage unambiguously whenever the marriage-market premium is positive. The equilibrium ambiguity—and the null average effects it rationalizes—is therefore specific to bride-price settings, which helps explain why the same type of intervention can produce starkly different results across contexts.

A.7 Closed-form illustrations and empirical mapping

This subsection collects the closed-form versions of the two modules of Appendix A and the per-channel empirical mapping.

Closed-form marriage-timing illustration. Let marital surplus be quadratic:

$$S_v(a, h) = -\frac{1}{2}(a - \bar{a}_v)^2 + (\Pi_v - \Omega_v)h(a - \bar{a}_v) + \kappa_v h. \quad (10)$$

Then $S_{v,a} = -(a - \bar{a}_v) + (\Pi_v - \Omega_v)h$, so

$$a_v^*(h) = \bar{a}_v + (\Pi_v - \Omega_v)h.$$

If assignment to arm z changes girl i 's human capital by $\partial h_{iv}/\partial T_z = D_{ivz}q_z$, where D_{ivz} is take-up and q_z is the effective content intensity of the arm, then her marriage-timing response is

$$\frac{\partial a_{iv}^*}{\partial T_z} = D_{ivz}q_z(\Pi_v - \Omega_v), \quad (11)$$

and averaging over girls within the village, the village-level assignment effect is $p_{vz}q_z(\Pi_v - \Omega_v)$, where $p_{vz} = \Pr(D_{ivz} = 1 \mid v)$ is the village take-up rate. This expression is the closed-form version of equation (4): marriage effects are diluted by incomplete take-up and can be small when the marriage-market premium and outside-option channel offset. The expression also clarifies why the model does not require identical effects across arms: safe spaces, training, and the announced grant can have different effective intensities q_z .

Is the offsetting region a knife-edge? A natural objection is that rationalizing a null marriage effect by $\Pi_v \approx \Omega_v$ requires a coincidence. A simple simulation of equation (11) shows that it does not, but also shows why the argument has little bite at the study's take-up. Draw Π_v and Ω_v independently and uniformly on $[0, 1.5]$ years of desired-age shift per standard deviation of h , draw village take-up uniformly on the observed arm range $[0.20, 0.29]$, and set the participant gain q_z to one standard deviation. The share of draws in which the village-level marriage effect is smaller than 0.23 years, twice the standard error of the T3 age-at-marriage estimate, is 0.86; with $q_z = 0.5$ it is 1.00, and allowing Π_v and Ω_v to be positively dependent raises it further. The offsetting region is therefore not thin. The same exercise at full take-up gives shares of 0.28 and 0.52, which is the honest reading: at this study's participation rates an undetectable marriage effect is what almost any (Π_v, Ω_v) pair delivers, so the null does not discriminate between offsetting, low effective dosage, and small true effects.

Closed-form participation-barrier illustration. Consider one village index X , oriented so higher values are more favorable. Let

$$p(X) = p_0 + p_1X, \quad p_1 > 0,$$

and let the conditional return be higher in less favorable villages:

$$\tau_Y(X) = \tau_0 - \tau_1X, \quad \tau_1 > 0.$$

Then the assignment CATE is

$$m_Y(X) = p(X)\tau_Y(X) = (p_0 + p_1X)(\tau_0 - \tau_1X).$$

At $X = 0$,

$$\left. \frac{\partial m_Y}{\partial X} \right|_{X=0} = p_1\tau_0 - p_0\tau_1. \quad (12)$$

Thus realized treatment effects fall as conditions improve when

$$\frac{\tau_1}{\tau_0} > \frac{p_1}{p_0}. \quad (13)$$

The condition has a direct interpretation: easier contexts raise participation, but if conditional returns decline faster than participation rises, the largest assignment gains are found precisely where take-up is lowest.

Average ITTs. Equation (7) maps to the ANCOVA specification in equation (1). The coefficients β_d are incremental ITTs relative to outreach-only villages. The model predicts that average marriage and fertility effects can be close to zero even when the program has a first stage, because (i) take-up is incomplete and (ii) the marriage-market premium Π_v and outside-option channel Ω_v can offset. This is one interpretation of the broadly null marriage and fertility ITTs, alongside strong take-up and attitude effects, in Tables 1 and 2; low effective dosage, compression by the common outreach baseline, and true effects below the detectable range are observationally equivalent in these data, and the framework does not adjudicate among them.

Compliance-adjusted effects. Since $m_Y^z(X_v) = p_z(X_v)\tau_Y^z(X_v)$, scaling ITTs by the take-up first stage recovers a complier effect only for outcomes whose exclusion restriction is plausible. This is most credible for the human-capital channel outcomes: marriage, fertility, and IGA. It is not credible for mental health or emotional violence, where assignment can change exposure and dissonance even without formal participation. The T3 LATE on age at marriage is therefore naturally larger than the ITT, consistent with take-up dilution.

Pregnancy barrier: norms. Let Y be a desirable pregnancy outcome, such as Never Pregnant, and let N_v be the Unrestricted index, where higher values mean less restrictive norms. The first stage implies

$$\frac{\partial p_z}{\partial N_v} > 0.$$

The pregnancy channel has a social-norm participation barrier if conservative villages have larger conditional pregnancy-prevention returns:

$$\frac{\partial \tau_{\text{NeverPreg}}^z}{\partial N_v} < 0 \quad \text{and} \quad \left| \frac{\partial \tau_{\text{NeverPreg}}^z}{\partial N_v} \right| \text{ is large enough to satisfy (13).}$$

The predicted sign pattern is therefore

$$\frac{\partial p_z}{\partial N_v} > 0, \quad \frac{\partial m_{\text{NeverPreg}}^z}{\partial N_v} < 0. \tag{14}$$

This is exactly the headline barrier result: Unrestricted loads positively on take-up and negatively on the Never Pregnant CATE, and the same sign reversal appears in the design-based assignment-by-Unrestricted interaction.

IGA barrier: economic infrastructure. Let K_v be economic infrastructure, oriented so higher values mean better market access. The first stage implies

$$\frac{\partial p_z}{\partial K_v} > 0.$$

The IGA channel has an infrastructure participation barrier if the marginal return to business training is largest where infrastructure is weak:

$$\frac{\partial \tau_{\text{IGA}}^z}{\partial K_v} < 0 \quad \text{and} \quad \left| \frac{\partial \tau_{\text{IGA}}^z}{\partial K_v} \right| \text{ is large enough to satisfy (13).}$$

The predicted sign pattern is

$$\frac{\partial p_z}{\partial K_v} > 0, \quad \frac{\partial m_{\text{IGA}}^z}{\partial K_v} < 0. \quad (15)$$

This maps to the strongest empirical result in the paper: economic infrastructure raises take-up but lowers the IGA CATE, and the pattern is replicated by design-based interactions with no machine-learning first stage. In the language of the model, business training substitutes for weak local economic infrastructure more than it complements strong infrastructure.

Marriage timing: no necessary participation barrier. Marriage timing is partly chosen by parents and households and is mediated by bride-price negotiation. A girl-level participation barrier need not appear even if participation matters for pregnancy or IGA. In the model, marriage timing is governed by $D_{ivz}q_z(\Pi_v - \Omega_v)$, but D_{ivz} is only one margin: the sign and magnitude also depend on whether local marriage markets price the girl’s new human capital and outside option. The model therefore does not predict an opposite-signed take-up/outcome pattern for marriage delay. The observed pattern—positive or ambiguous CATE gradients on Age at Marriage and Married after 18 in more favorable villages—is consistent with self-selection of high-return girls into participation rather than a participation barrier.

Exogamy sign reversal. If many grooms come from outside the village, they may not observe local program participation and may underprice the girl’s program-induced outside-option gain. This lowers the effective Ω_v and can strengthen marriage-delay effects:

$$\frac{\partial(\Pi_v - \Omega_v)}{\partial \sigma_v^g} > 0.$$

By contrast, if girls expect to marry out, investments tied to the local labor market have lower returns, so the IGA effect can fall with female exogamy:

$$\frac{\partial m_{\text{IGA}}^z}{\partial \sigma_v^b} < 0.$$

This sign reversal across outcome families is a demanding falsification test because a generic village-level omitted variable would tend to move outcomes in the same direction—provided the two exogamy margins covary positively across villages, which they do ($\hat{\rho} = 0.39$; Section 6.3). We treat it as a secondary test rather than a headline result: the sign reversal is strongest for T2 and directionally consistent elsewhere, but the full pattern is not uniformly statistically significant.

Mental health and normative dissonance. The model above is designed for human-capital channel outcomes. Mental health and emotional violence require a different interpretation. The full package changes attitudes toward early marriage, and the announced grant may make the outside option feel credible. When private preferences move faster than household and community norms, girls can experience short-run dissonance or conflict. This exposure mechanism need not operate through formal take-up alone and should not be given a clean LATE interpretation. The framework therefore treats mental-health deterioration as an interpretation consistent with norm change in restrictive settings, not as a directly identified mechanism and not as a test of the participation-barrier model.

Nesting. The framework nests the standard monotone-delay model of empowerment programs as a special case. When the outside-option channel is absent ($\Omega_v = 0$), equation (4) reduces to $\partial a^*/\partial T_z = -(\Pi_v/\Delta_v)\partial h/\partial T_z > 0$ whenever $\Pi_v > 0$: any program that raises human capital unambiguously delays marriage, which is the implicit prediction of most program logic models.

The same collapse occurs under dowry, where the market-clearing transfer compensates the groom side's outside option and Ω_v drops out of the timing condition (Proposition 3). On the participation side, under full take-up ($p_z(X_v) = 1$) the assignment effect in equation (7) equals the conditional return $\tau_Y^z(X_v)$ and the participation-barrier condition (8) cannot hold, since $\partial p_z / \partial X_k = 0$.

B Additional Balance and Attrition Analysis

Table (B.1) Balance Table: Non-Attrited Sample

Variable	(1) Full Sample	(2) Control	(3) T1: Safe Space	(4) T2: SS + Training	(5) T3: Full Package	(6) p-value (F-test)
<i>Panel A1: Main Outcomes</i>						
Ever Married	0.658 [0.474]	0.681 [0.466]	0.684 [0.465]	0.556 [0.497]	0.706 [0.456]	0.105
Age at Mar.	17.301 [1.969]	17.303 [1.848]	17.071 [1.988]	17.158 [1.982]	17.549 [2.000]	0.108
Mar. bf18	0.449 [0.497]	0.450 [0.498]	0.514 [0.500]	0.437 [0.496]	0.411 [0.492]	0.291
Ever Pregnant	0.586 [0.493]	0.585 [0.493]	0.612 [0.488]	0.542 [0.498]	0.602 [0.490]	0.257
Currently Pregnant	0.081 [0.273]	0.083 [0.275]	0.074 [0.262]	0.073 [0.261]	0.091 [0.287]	0.707
Preg. bf18	0.104 [0.306]	0.123 [0.328]	0.107 [0.310]	0.095 [0.293]	0.097 [0.296]	0.528
Emo. Violence	0.122 [0.328]	0.141 [0.349]	0.091 [0.288]	0.144 [0.351]	0.114 [0.318]	0.536
Inc. Gen. Act.	-0.061 [0.892]	0.045 [1.010]	-0.178 [0.724]	-0.058 [0.903]	-0.053 [0.896]	0.289
<i>Panel A2: Intermediate Outcomes</i>						
Finance	-0.035 [0.998]	0.050 [1.020]	-0.117 [0.972]	-0.063 [1.010]	-0.011 [0.987]	0.359
Gender Att.	0.057 [0.920]	0.041 [0.971]	0.109 [0.915]	-0.083 [0.955]	0.143 [0.840]	0.263
Aspirations	-0.105 [0.966]	-0.066 [0.945]	-0.124 [0.949]	-0.154 [1.015]	-0.079 [0.950]	0.742
SRH Know.	0.069 [1.001]	0.019 [1.007]	0.166 [0.901]	-0.091 [1.038]	0.165 [1.016]	0.174
SRH Behav.	-0.032 [0.996]	-0.050 [1.040]	0.114 [0.794]	-0.103 [1.080]	-0.067 [1.015]	0.077
MHM	0.094 [1.131]	0.014 [1.022]	0.033 [1.049]	0.183 [1.238]	0.122 [1.167]	0.900
Decision	9.267 [4.413]	9.688 [4.232]	8.602 [4.443]	9.084 [4.176]	9.602 [4.635]	0.373
Decision (Couple)	0.003 [1.004]	-0.040 [1.063]	-0.246 [0.953]	0.234 [0.837]	0.024 [1.052]	0.046
HH Opinion	0.197 [1.083]	-0.035 [0.989]	0.086 [0.975]	0.480 [1.176]	0.212 [1.093]	0.124
Any Disadv. Early Mar.	0.393 [0.489]	0.369 [0.483]	0.404 [0.491]	0.334 [0.472]	0.450 [0.498]	0.273
Disv M. Early Nb	0.382 [0.486]	0.357 [0.479]	0.401 [0.490]	0.319 [0.466]	0.437 [0.496]	0.246
Any Adv. M. Early	0.391 [0.488]	0.446 [0.497]	0.401 [0.490]	0.321 [0.467]	0.402 [0.490]	0.399
Adv M. Early Nb	0.190 [0.393]	0.232 [0.422]	0.245 [0.431]	0.121 [0.326]	0.176 [0.381]	0.062
<i>Panel B: Adolescent Covariates</i>						
Age	19.074 [3.652]	19.216 [3.775]	18.954 [3.585]	18.912 [3.582]	19.190 [3.663]	0.351
Ever School	0.164 [0.370]	0.172 [0.377]	0.170 [0.376]	0.147 [0.354]	0.167 [0.373]	0.728
Wish Mar. Age	19.954 [2.817]	19.969 [3.094]	19.556 [2.418]	19.927 [2.807]	20.259 [2.849]	0.543
Ideal Mar. Age	18.559 [1.916]	18.743 [2.108]	18.356 [1.801]	18.488 [1.994]	18.636 [1.770]	0.574
Gender Equal.	2.387 [1.902]	2.465 [1.818]	2.621 [2.102]	2.034 [1.910]	2.446 [1.758]	0.375

Continued on next page

Table B.1 – continued from previous page

Variable	(1) Full Sample	(2) Control	(3) T1: Safe Space	(4) T2: SS + Training	(5) T3: Full Package	(6) p-value (F-test)
Ideal Children	14.090 [7.075]	14.413 [7.087]	13.718 [6.790]	14.067 [7.422]	14.152 [6.980]	0.521
Ideal Girl Edu	4.456 [0.903]	4.408 [0.910]	4.559 [0.772]	4.430 [0.962]	4.433 [0.933]	0.402
FP Knowledge	0.757 [0.429]	0.734 [0.442]	0.810 [0.392]	0.722 [0.448]	0.762 [0.426]	0.209
Saves Money	0.191 [0.393]	0.250 [0.433]	0.116 [0.320]	0.176 [0.381]	0.215 [0.411]	0.006
Hours Worked	0.750 [2.615]	0.938 [2.857]	0.728 [2.412]	0.741 [2.607]	0.639 [2.579]	0.661
Days Worked	1.646 [5.836]	1.960 [6.329]	2.005 [6.540]	1.331 [4.849]	1.416 [5.637]	0.745
<i>Panel C: Household Covariates</i>						
HH Size	7.337 [5.896]	7.091 [5.056]	7.300 [5.993]	7.633 [6.289]	7.299 [6.044]	0.844
Female Head	0.019 [0.137]	0.018 [0.132]	0.014 [0.119]	0.029 [0.168]	0.016 [0.125]	0.210
Polygamous	0.298 [0.458]	0.255 [0.436]	0.334 [0.472]	0.336 [0.473]	0.273 [0.445]	0.109
Asset Index	0.208 [0.120]	0.201 [0.112]	0.208 [0.127]	0.215 [0.122]	0.206 [0.117]	0.920
Food Insecure	0.117 [0.321]	0.108 [0.310]	0.136 [0.343]	0.085 [0.279]	0.134 [0.341]	0.648
Food Insec. Index	0.117 [0.321]	0.108 [0.310]	0.136 [0.343]	0.085 [0.279]	0.134 [0.341]	0.648
<i>Panel D: Village Covariates</i>						
Prohib Eth In	0.469 [0.501]	0.500 [0.507]	0.421 [0.500]	0.525 [0.506]	0.435 [0.501]	0.768
Prohib Eth Out	0.321 [0.468]	0.342 [0.481]	0.316 [0.471]	0.300 [0.464]	0.326 [0.474]	0.972
Prohib Rel In	0.049 [0.217]	0.079 [0.273]	0.053 [0.226]	0.050 [0.221]	0.022 [0.147]	0.704
Prohib Rel Out	0.074 [0.263]	0.105 [0.311]	0.079 [0.273]	0.050 [0.221]	0.065 [0.250]	0.856
Ideal Age Eth	17.130 [1.291]	17.105 [1.203]	16.842 [1.603]	17.125 [1.202]	17.391 [1.125]	0.289
Ideal Age Vill	16.414 [1.151]	16.417 [1.168]	16.180 [1.227]	16.384 [1.102]	16.640 [1.106]	0.306
Eligible Pop	77.142 [40.213]	81.026 [51.103]	82.184 [45.184]	70.650 [31.178]	75.413 [32.287]	0.510
Eth Frac	0.228 [0.209]	0.236 [0.217]	0.217 [0.209]	0.216 [0.214]	0.242 [0.204]	0.845
Rel Frac	0.367 [0.060]	0.372 [0.059]	0.364 [0.056]	0.368 [0.068]	0.363 [0.059]	0.916
Bride Promises	0.331 [0.381]	0.295 [0.379]	0.242 [0.297]	0.406 [0.404]	0.376 [0.421]	0.327
Promise Broken	0.363 [0.482]	0.378 [0.492]	0.316 [0.471]	0.368 [0.489]	0.386 [0.493]	0.911
Norms Index	-0.034 [0.841]	0.023 [0.996]	0.015 [0.938]	-0.051 [0.704]	-0.108 [0.741]	0.795
Birth-control Access	0.083 [1.064]	-0.029 [1.003]	-0.018 [1.157]	0.050 [1.094]	0.287 [1.011]	0.493
Econ. Infra. Index	0.133 [0.759]	0.127 [0.766]	0.106 [0.849]	0.159 [0.777]	0.139 [0.679]	0.987
Transport Index	0.054 [0.857]	0.107 [0.720]	-0.098 [1.296]	0.136 [0.574]	0.063 [0.711]	0.619
N (Adolescents)	4936	1077	1108	1232	1519	

Clusters	162	38	38	40	46
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Notes: Standard deviations in brackets. Panel D computed at village level.
p-values from F-test of joint equality with strata FE and cluster-robust SEs.

Table (B.2) Balance Table: Full Baseline Sample

Variable	(1) Full Sample	(2) Control	(3) T1: Safe Space	(4) T2: SS + Training	(5) T3: Full Package	(6) p-value (F-test)
<i>Panel A1: Main Outcomes</i>						
Ever Married	0.603 [0.489]	0.620 [0.485]	0.641 [0.480]	0.508 [0.500]	0.642 [0.479]	0.055
Age at Mar.	17.267 [1.953]	17.321 [1.844]	17.080 [1.905]	17.010 [2.018]	17.563 [1.982]	0.037
Mar. bf18	0.453 [0.498]	0.443 [0.497]	0.531 [0.499]	0.439 [0.496]	0.409 [0.492]	0.061
Ever Pregnant	0.524 [0.499]	0.513 [0.500]	0.547 [0.498]	0.482 [0.500]	0.550 [0.498]	0.138
Currently Pregnant	0.085 [0.278]	0.091 [0.288]	0.083 [0.275]	0.077 [0.266]	0.087 [0.282]	0.908
Preg. bf18	0.067 [0.250]	0.077 [0.267]	0.063 [0.244]	0.060 [0.238]	0.067 [0.250]	0.755
Emo. Violence	0.134 [0.340]	0.157 [0.364]	0.107 [0.310]	0.147 [0.354]	0.127 [0.333]	0.538
Inc. Gen. Act.	-0.071 [0.890]	0.005 [1.008]	-0.180 [0.722]	-0.097 [0.850]	-0.012 [0.944]	0.260
<i>Panel A2: Intermediate Outcomes</i>						
Finance	-0.060 [0.978]	0.007 [1.005]	-0.136 [0.921]	-0.090 [0.978]	-0.022 [1.000]	0.364
Gender Att.	0.032 [0.928]	-0.006 [0.995]	0.041 [0.899]	-0.055 [0.950]	0.133 [0.867]	0.363
Aspirations	-0.034 [1.000]	-0.011 [0.988]	-0.008 [0.978]	-0.086 [1.054]	-0.029 [0.978]	0.804
SRH Know.	0.043 [1.003]	0.002 [0.997]	0.117 [0.910]	-0.081 [1.032]	0.122 [1.044]	0.254
SRH Behav.	0.009 [0.960]	0.005 [0.996]	0.160 [0.719]	-0.057 [1.052]	-0.058 [1.013]	0.010
MHM	0.051 [1.074]	-0.001 [0.998]	-0.031 [0.950]	0.152 [1.203]	0.073 [1.104]	0.830
Decision	9.244 [4.499]	9.729 [4.191]	8.692 [4.584]	8.952 [4.248]	9.600 [4.795]	0.338
Decision (Couple)	-0.001 [0.996]	-0.003 [1.002]	-0.230 [0.955]	0.157 [0.888]	0.049 [1.054]	0.100
HH Opinion	0.199 [1.102]	0.006 [0.994]	0.077 [1.055]	0.459 [1.180]	0.222 [1.104]	0.111
Any Disadv. Early Mar.	0.385 [0.487]	0.364 [0.481]	0.390 [0.488]	0.347 [0.476]	0.428 [0.495]	0.556
Disv M. Early Nb	0.373 [0.484]	0.354 [0.478]	0.383 [0.486]	0.332 [0.471]	0.415 [0.493]	0.546
Any Adv. M. Early	0.386 [0.487]	0.422 [0.494]	0.412 [0.492]	0.337 [0.473]	0.380 [0.486]	0.581
Adv M. Early Nb	0.181 [0.385]	0.212 [0.409]	0.233 [0.423]	0.126 [0.332]	0.161 [0.368]	0.104
<i>Panel B: Adolescent Covariates</i>						
Age	18.529 [3.769]	18.576 [3.882]	18.470 [3.685]	18.379 [3.731]	18.677 [3.783]	0.472
Ever School	0.178 [0.382]	0.180 [0.384]	0.184 [0.388]	0.157 [0.364]	0.188 [0.391]	0.616
Wish Mar. Age	19.912 [2.752]	19.811 [2.929]	19.504 [2.363]	19.935 [2.700]	20.319 [2.907]	0.361
Ideal Mar. Age	18.566 [1.953]	18.671 [2.070]	18.342 [1.835]	18.557 [2.046]	18.684 [1.857]	0.521
Gender Equal.	2.320 [1.933]	2.461 [1.836]	2.295 [2.096]	2.023 [1.927]	2.495 [1.832]	0.331
Ideal Children	13.257 [7.194]	13.351 [7.224]	13.119 [6.846]	13.046 [7.485]	13.485 [7.200]	0.788
Ideal Girl Edu	4.409 [0.920]	4.351 [0.970]	4.467 [0.799]	4.401 [0.963]	4.411 [0.936]	0.689

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Table B.2 – continued from previous page

Variable	(1) Full Sample	(2) Control	(3) T1: Safe Space	(4) T2: SS + Training	(5) T3: Full Package	(6) p-value (F-test)
FP Knowledge	0.738 [0.440]	0.729 [0.444]	0.789 [0.408]	0.707 [0.455]	0.727 [0.446]	0.187
Saves Money	0.178 [0.382]	0.229 [0.420]	0.123 [0.329]	0.157 [0.364]	0.202 [0.402]	0.039
Hours Worked	0.760 [2.651]	0.862 [2.703]	0.675 [2.307]	0.706 [2.468]	0.801 [3.017]	0.792
Days Worked	1.606 [5.693]	1.833 [6.060]	1.825 [6.208]	1.212 [4.520]	1.593 [5.855]	0.656
<i>Panel C: Household Covariates</i>						
HH Size	7.732 [6.040]	7.473 [5.266]	7.744 [6.193]	7.899 [6.324]	7.775 [6.209]	0.881
Female Head	0.023 [0.149]	0.024 [0.153]	0.015 [0.121]	0.031 [0.173]	0.021 [0.145]	0.068
Polygamous	0.321 [0.467]	0.276 [0.447]	0.352 [0.478]	0.348 [0.477]	0.303 [0.460]	0.110
Asset Index	0.206 [0.119]	0.201 [0.118]	0.212 [0.121]	0.209 [0.123]	0.202 [0.115]	0.430
Food Insecure	0.119 [0.324]	0.104 [0.306]	0.148 [0.356]	0.083 [0.276]	0.138 [0.345]	0.329
Food Insec. Index	0.119 [0.324]	0.104 [0.306]	0.148 [0.356]	0.083 [0.276]	0.138 [0.345]	0.329
<i>Panel D: Village Covariates</i>						
Prohib Eth In	0.439 [0.498]	0.475 [0.506]	0.452 [0.504]	0.442 [0.502]	0.396 [0.494]	0.868
Prohib Eth Out	0.318 [0.467]	0.350 [0.483]	0.381 [0.492]	0.256 [0.441]	0.292 [0.459]	0.529
Prohib Rel In	0.040 [0.198]	0.050 [0.221]	0.071 [0.261]	0.023 [0.152]	0.021 [0.144]	0.568
Prohib Rel Out	0.081 [0.274]	0.100 [0.304]	0.119 [0.328]	0.047 [0.213]	0.062 [0.245]	0.570
Ideal Age Eth	17.121 [1.452]	17.250 [0.981]	16.690 [1.645]	16.930 [1.737]	17.562 [1.201]	0.034
Ideal Age Vill	16.411 [1.147]	16.370 [1.161]	16.127 [1.214]	16.441 [1.083]	16.678 [1.101]	0.174
Eligible Pop	77.087 [40.253]	82.925 [52.453]	82.333 [43.483]	69.116 [30.760]	74.771 [32.162]	0.272
Eth Frac	0.233 [0.213]	0.239 [0.220]	0.208 [0.203]	0.228 [0.223]	0.254 [0.210]	0.748
Rel Frac	0.367 [0.060]	0.371 [0.059]	0.361 [0.056]	0.369 [0.067]	0.365 [0.059]	0.870
Bride Promises	0.330 [0.382]	0.305 [0.389]	0.270 [0.316]	0.387 [0.398]	0.358 [0.419]	0.635
Promise Broken	0.345 [0.477]	0.359 [0.486]	0.286 [0.457]	0.366 [0.488]	0.370 [0.488]	0.853
Norms Index	-0.046 [0.841]	-0.000 [1.000]	-0.047 [0.950]	-0.056 [0.685]	-0.075 [0.742]	0.955
Birth-control Access	0.088 [1.040]	-0.000 [1.000]	-0.011 [1.114]	0.041 [1.063]	0.291 [0.989]	0.490
Econ. Infra. Index	0.069 [0.856]	-0.000 [1.000]	0.028 [0.991]	0.113 [0.769]	0.122 [0.677]	0.876
Transport Index	0.015 [0.925]	0.000 [1.000]	-0.173 [1.297]	0.131 [0.564]	0.087 [0.705]	0.439
N (Adolescents)	7719	1704	1881	1946	2188	
Clusters	173	40	42	43	48	

Notes: Standard deviations in brackets. Panel D computed at village level.
p-values from F-test of joint equality with strata FE and cluster-robust SEs.

Table (B.3) Omnibus Orthogonality Test

Comparison	(1) F-statistic	(2) p(obs)	(3) p(RI)	(4) N
<i>Panel A: Full Baseline Sample</i>				
T1: Safe Space vs. Control	4.419	0.0000	1.0000	3585
T2: SS + Training vs. Control	3.638	0.0000	0.5860	3650
T3: Full Package vs. Control	1.553	0.0486	0.7480	3892
Any Treatment vs. Control	1.055	0.3955	0.1200	7719
<i>Panel B: Non-Attrited Sample</i>				
T1: Safe Space vs. Control	5.359	0.0000	0.6040	2185
T2: SS + Training vs. Control	3.294	0.0000	0.6440	2309
T3: Full Package vs. Control	1.449	0.0832	0.1600	2596
Any Treatment vs. Control	1.020	0.4476	0.5480	4936
Notes: F-statistic from joint test of 38 baseline covariates.				
p(obs) = sampling-based p-value; p(RI) = randomization inference (500 permutations).				
Missing covariate values imputed with sample means.				

Table (B.4) Differential Attrition by Treatment Arm

	(1)	(2)	(3)
	Adolescent	Household	Village
T1: Safe Space	0.0400 (0.0539)	0.0155 (0.0108)	0.0222 (0.0520)
T2: SS + Training	0.0134 (0.0468)	0.0082 (0.0085)	0.0221 (0.0488)
T3: Full Package	-0.0546 (0.0398)	0.0208 (0.0120)	0.0126 (0.0428)
F-stat (T1=T2=T3)	3.382	0.998	0.034
p-value	0.0362	0.3708	0.9662
N	7719	4293	173
Notes: Cluster-robust SEs in parentheses. Strata FE included.			
Dependent variable: attrition indicator (1 = attrited).			

C GML Heterogeneity Estimates

This section reports the generic machine learning (GML) heterogeneity estimates summarized in Section 6.3 of the paper: the BLP, GATES, and CLAN estimates, the decomposition of village-level CATEs onto the seven baseline structural predictors, the exogamy falsification test, and a mechanical scaling exercise. The estimation framework is described in the paper’s empirical-strategy section and in Online Appendix S8; the following paragraphs restate the implementation for completeness.

For each arm, we randomly split the data into training and test samples and train a machine learner on the training data to form a proxy s_i for the true CATE $s_0(Z) = \mathbb{E}[Y|T = 1, Z] - \mathbb{E}[Y|T = 0, Z]$; cross-fitting makes the subsequent inference orthogonal to first-stage overfitting. Three features of the CATE are then targeted in the test sample through propensity-weighted regressions. The *best linear predictor* (BLP) regresses the outcome on demeaned assignment and its interaction with the centered proxy score: the interaction coefficient β_2 is the best linear predictor of treatment-effect heterogeneity, so $\beta_2 \neq 0$ indicates that heterogeneity is present even when the average effect β_1 is small or zero. The *sorted group average treatment effects* (GATES) are arm-specific treatment effects for quintiles of the proxy score. The sorting is on the *proxy*, not on the true effect: when $\beta_2 \neq 0$ the true group effects covary positively with the proxy ranking, so a broadly increasing profile from the least-affected group (G1) to the most-affected group (G5) is expected, but monotonicity of the estimated GATES does not follow from $\beta_2 \neq 0$, and local departures from monotonicity are compatible with a noisy proxy.²¹ *Classification analysis* (CLAN) is descriptive: for any baseline variable X_k , $\delta(X_k)$ is the difference in its mean between the most- and least-affected quintiles, $\mathbb{E}[X_k | G_5] - \mathbb{E}[X_k | G_1]$. We focus the CLAN analysis on indexes of local conditions (social-norm indices, SRH access, transportation access), so $\delta(X_k)$ characterizes the role of local conditions in the pattern of treatment effects across the score distribution.

To rule out “lucky” splits, we repeat the process above for 200 random splits of the data and then median-aggregate point estimates, confidence intervals, and p-values over the splits.²² For the BLP estimates, p-values are multiplied by two and can thus be interpreted at the 5% level; for GATES and CLAN, we median-aggregate the 95% confidence bounds without further adjustment, leading to a $1 - 2\alpha = 90\%$ coverage region (assuming a dominant learner prevails). Four machine learners are used – random forest, gradient boosting, elastic net, and partial least squares, implemented via the `caret` package in R (Kuhn and Max, 2008) – and the best learner is selected agnostically on the median-aggregated goodness-of-fit metrics of Online Appendix S8, as suggested by Chernozhukov et al. (2025). All results are reproducible with the scripts in the replication package.²³

C.1 Design-based interactions and the participation pattern

The parametric analysis above offered partial evidence on Comparative Static 2 and evidence against Proposition 1, but could not resolve cases where multiple mechanisms operate simultaneously. This subsection reports the design-based evidence on the participation pattern that organizes the paper’s exploratory heterogeneity analysis: individual-level regressions of outcomes on assignment interacted with a fixed set of baseline village structural predictors, with no machine-learning first stage. The generic machine learning (GML) estimates of Chernozhukov et al. (2025) that motivated and corroborate this pattern—BLP, GATES, and CLAN, the decomposition of village-level CATEs, and the exogamy falsification of Proposition 2—are reported in Appendix C and summarized at the end of this subsection. All results here are exploratory.

A direct decomposition of the participation barrier. A descriptive CLAN contrast (Appendix C, Figure C.2) first suggested the pattern: the girls predicted to benefit most are not those most likely to participate. But that contrast does not isolate which structural village features predict the gap between predicted return and predicted take-up (a village-level analogue of testing for selection on gains, Ito et al., 2023), and any test built on estimated CATEs inherits the first-stage estimation error of the ML proxies. Our primary evidence therefore uses no

²¹Groups are sorted by the proxy score S , so monotonicity holds by construction in S itself; it need not hold in the estimated group-specific effects $\hat{\gamma}_k$, which average the true CATE over proxy-defined groups and carry estimation error.

²²All GML-derived estimates in this paper—the BLP, GATES, and CLAN estimates and figures, the village-level CATE decompositions of Section 6.3, and the split-uncertainty tables of Online Appendix S11—come from a single estimation run of 200 splits per arm, with all four learners fitted on every split and the best learner selected per outcome-arm on the goodness-of-fit criteria of Online Appendix S8.

²³See the replication package for more details.

machine-learning first stage at all. Table C.1 reports individual-level regressions of each outcome on assignment, a standardized baseline village predictor, and their interaction ($y_i = \alpha + \beta T3 + \gamma (T3 \times X_v) + \delta X_v + \text{strata FE} + \text{village-clustered standard errors}$), for a fixed set of seven baseline village-level structural predictors: an index of women’s autonomy (Unrestricted), an index of birth-control infrastructure (Birthcontrols), an index of economic infrastructure (Econ. Infra.), a security index, and three exogamy indices (male, female, overall). The barrier-consistent pattern appears as opposite signs of γ on take-up and on the outcome: assignment raises participation most where it raises the outcome least. Because these regressions use only assignment, baseline predictors, and observed outcomes, their coefficients are design-based with respect to assignment; the interactions highlighted here are nonetheless drawn from a larger family of moderators, outcomes, and arms, a selection and multiplicity caveat that applies to all interaction results in this section. Strata fixed effects enter every design-based interaction model reported in this section and in Online Appendix S11.9, with the same specification across arms, moderators, and outcomes; no other covariates are selected or added cell by cell. Two further limits—that the moderators are not randomized and that observed non-participation mixes individual choice with implementation disruption and access—are taken up in the closing discussion of this section.

Randomization inference and multiplicity for the interaction family. All interaction results in this section are exploratory: the interactions were not registered, and the family highlighted here was assembled after the exploratory analyses that suggested the pattern. We discipline inference over them in three steps, and the first carries the most weight. First, we test the headline pattern against specification search directly: under re-randomization, the probability that *some* (take-up, outcome) pair within the T3 family of seven moderators and eight outcomes exhibits an opposite-signed pattern at least as extreme as the observed best pair—measured by $\min(|t_{\text{take-up}}|, |t_{\text{outcome}}|)$ —is 0.103 (0.260 when the search space spans all three arms). Because this test conditions on the full searched family rather than on any post-hoc selection, it is the paper’s primary statement about how seriously to take the pattern. Second, we declare a small focal family—the T3 interactions of the two moderators named by the barrier hypothesis (Unrestricted, Econ. Infra.) with take-up and the two named outcomes (Never Pregnant, IGA)—and hold it to a familywise standard. The declaration postdates the exploratory analyses, so the correction disciplines multiplicity within the family without conferring pre-specified status. For each of the six cells, Table C.2 reports the analytic cluster p -value alongside a randomization-inference p -value (10,000 within-strata permutations of the four-category village assignment, holding arm counts fixed)²⁴ and Holm familywise-corrected RI p -values. Randomization inference leaves the infrastructure take-up cell individually significant at the five-percent level (Econ. Infra. $\times$ Take-up: analytic $p = 0.016$, RI $p = 0.043$) and the infrastructure IGA cell at the ten-percent level (Econ. Infra. $\times$ IGA: 0.028/0.086), while the norms cells are individually insignificant under RI (Unrestricted $\times$ Take-up: 0.122/0.162; Unrestricted $\times$ Never Pregnant: 0.065/0.110); no cell survives the Holm correction across the family (the smallest familywise-adjusted p is 0.26). Third, the point pattern is stable to single-village influence: in leave-one-village-out (88 re-estimations per cell) and leave-one-stratum-out sweeps, all four headline coefficients retain their sign in every re-estimation—a check that should be read beside the high-retention reversal reported below, which removes 41 percent of villages at once and does flip the infrastructure-IGA cell.²⁵ The conclusion we draw is deliberately restrained but two-sided: the joint randomization test puts the probability that search over the T3 interaction family alone produces an opposite-signed pattern this extreme at roughly one in ten, yet no individual cell survives familywise correction, and the norms cells are fragile on their own. We therefore present the participation barrier as a hypothesis the data make attractive—with the search-artifact explanation unlikely though not rejected at conventional levels—rather than a finding the experiment establishes.

Attrition and implementation sensitivity. Two selection channels could generate the interaction pattern mechanically: differential follow-up response and differential program delivery. We audit both (Online Appendix S11.10). On attrition: because a linear assignment-by-moderator interaction on follow-up response (Online

²⁴The permutation count follows the randomization-inference benchmark of Young (2019), who uses 10,000 draws, and the guidance of Athey and Imbens (2017) on computing Fisher-exact p -values by simulation. At 10,000 draws the Monte Carlo standard error of a reported p -value near 0.10 is roughly 0.003, so the reported digits are not simulation noise.

²⁵Village counts differ across analyses because each reports the villages meeting its own data requirements: 173 villages enter the baseline roster and 162 have at least one follow-up respondent. The T3-and-control pair comprises 88 baseline-roster villages—hence 88 leave-one-village-out re-estimations—of which 84 have follow-up respondents, the count in the village-level CATE analyses; tables that impose further estimation requirements (subsample outcomes, non-missing moderators) can report marginally fewer clusters, such as the 83 in the tertile ITT tables.

Appendix Table S174) can miss gaps concentrated in one tercile, we also parameterize the response margin by moderator tercile, with randomization inference (Online Appendix Table S175). On the norms dimension the T3 response margin is clean: the strata-adjusted T3–control response gap in the most conservative tercile is +0.031 (RI $p = 0.74$), and equality across terciles is not rejected ($p_{\text{RI}} = 0.63$)—the raw +13.5 percentage-point tercile-1 gap visible in Online Appendix Table S176 is absorbed by strata. On the infrastructure dimension a gap survives adjustment: T3 raises follow-up response by 20.8 percentage points in the least-connected tercile (analytic $p = 0.042$, RI $p = 0.086$; equality across terciles $p_{\text{RI}} = 0.17$)—precisely the cell that identifies the Econ. Infra. $\times$ IGA gradient, a selection caution that compounds the high-retention reversal reported below. T1 exhibits significant differential response across norms terciles (equality $p_{\text{RI}} = 0.018$), a caution that applies to heterogeneity claims in that arm. Reweighting respondents by an inverse-probability response model leaves the focal cells essentially unchanged (norms $\times$ Never Pregnant -0.097 ; Econ. Infra. $\times$ IGA -0.223 ; Online Appendix Table S177), and adversarial imputation of attritors’ outcomes overturns the two key cells only at shifts of 0.4 within-cell standard deviations. Restricting to the 102 of 173 villages with follow-up response of at least 65 percent, however, *reverses* the Econ. Infra. $\times$ IGA interaction (-0.242 to $+0.093$) while leaving the norms $\times$ Never Pregnant cell stable (-0.082): the IGA gradient is concentrated in low-retention villages, and we cannot distinguish selective observation from the loss of identifying variation the restriction imposes. On implementation: program records name four suspended safe-space villages and three provinces with documented delivery disruption (Gnagna, Tapoa, Sourou—68 villages, 34 in the T3 estimation sample), and disruption-province villages are strongly selected on the norms moderator ($p < 0.001$) and more weakly on economic infrastructure ($p = 0.09$; untabulated selection regressions, replication package). Dropping the four suspended villages changes nothing, but adding a T3 $\times$ disruption-province control produces the audit’s sharpest result (Online Appendix Table S178): the norms $\times$ take-up gradient collapses entirely ($+0.069$ to -0.019) while the outcome-side cells are unchanged (norms $\times$ Never Pregnant -0.077 ; Econ. Infra. $\times$ IGA -0.259). The take-up leg of the norms pattern is therefore not separable from documented supply-side disruption, while the outcome-side gradients are robust to it. The thirteen Est-region spaces displaced within zone are not individually identified in program records, so province-level controls are the feasible bound on their influence—a residual limitation. In sum, each barrier pair fails a different check: the norms pair keeps its outcome leg but loses its participation leg to supply-side disruption, and the infrastructure pair keeps its participation leg while its IGA leg reverses in high-retention villages and rests on the one tercile where T3 assignment marginally raised follow-up response. The demand-side “participation barrier” reading is accordingly one of several the data admit.

The IGA gradient in policy units. The IGA index aggregates four components (Anderson weighting): an indicator for any paid activity in the past 30 days (entry), IHS-transformed cash profits (income), the number of own-initiative activities (agency), and the number of activities over the past six months (diversification). Re-estimating the T3 $\times$ Econ. Infra. interaction component by component (Online Appendix Table S32), the gradient is negative on all four margins and is led by entry (-0.24 control SDs per SD of infrastructure, $p = 0.060$) and own-initiative activity (-0.24 , $p = 0.056$), with weaker gradients on diversification (-0.16 , $p = 0.16$) and realized profits (-0.12 , $p = 0.35$). The infrastructure gradient is therefore about whether girls start and initiate income-generating activity, not about differential earnings conditional on activity. It also does not reflect different starting levels: baseline activity is uniformly low (6–8 percent report any paid activity) and statistically indistinguishable across infrastructure terciles (Online Appendix Table S33). A substitution reading is consistent with this pattern: where market-provided opportunities and connections are scarce, the training’s content—basic business skills and the push to initiate an activity—has no local substitute, so its incremental effect on entry is larger, whereas better-connected villages offer those inputs through existing channels. The selection alternative documented above still applies with equal force: the gradient thins in high-retention villages, so part of it may reflect who is observed rather than latent returns. In levels, no component shows a significant average incremental effect in any arm, matching the null index ITTs.

Who decides matters. The channel asymmetry has a natural institutional reading. In the study communities, marriage timing is decided largely by parents and household heads, with bride-price transfers negotiated between the two families, whereas pregnancy-related knowledge and contraceptive use operate through the girl’s own information and behavior, and income generation through her access to local markets. Girl-level take-up can therefore gate the pregnancy and IGA channels—the program changes what the girl knows and does—but is less likely to gate directly a marriage decision made by her parents. A barrier defined over the girl’s participation is accordingly expected on the pregnancy and IGA channels and not on the marriage-delay channel, which is what the decomposition finds: on the marriage margin the program’s reach runs through slower-moving household and community channels, consistent with the normative shifts documented in Section 6.1.2. This institutional reading is interpretive rather than deductive: participation could still influence marriage timing through bargaining, information, networks, or

household expectations.

Implication. Taken at face value, the evidence points to two distinct constraints—*social norms* on the pregnancy channel and *economic infrastructure* on the income-generation channel—each appearing in the design-based assignment interactions of Table C.1, each aligning with one of the framework’s two unresolved-sign comparative statics, and each corroborated by the CATE decomposition. Three limits on this reading are central: the moderators are not randomized, so the interactions document heterogeneity in reach and in reduced-form assignment effects rather than identified returns to participation; observed non-participation mixes individual choice with implementation disruption and access, and the sensitivity audit bears this out concretely (the norms $\times$ take-up gradient is absorbed by disruption controls, and the Econ. Infra. $\times$ IGA gradient is concentrated in low-retention villages); and the highlighted contrasts are drawn from a larger searched family and do not survive familywise correction (Table C.2). The marriage-delay channel exhibits no analogous pattern; its heterogeneity reflects individual-level selection of high-return girls into participation. A final limit is observational equivalence on the income channel: the participation-barrier reading (weak infrastructure gates engagement with a program whose returns are high there) and the substitution reading that Comparative Static 1 assigns to the same pattern (the program’s content stands in for what strong infrastructure would otherwise provide) generate identical opposite-signed interactions, and nothing in these data separates them. The two readings carry opposite policy implications—a binding barrier implies that pairing the program with infrastructure investment would unlock high returns, whereas substitution implies those returns would shrink as infrastructure improves—so we draw no infrastructure recommendation from this pattern. If the barrier reading is correct on the pregnancy channel, the corresponding lever is a norm-change complement; testing either channel’s lever requires randomizing the complement itself.

GML corroboration and the exogamy falsification. Four summary facts from the machine-learning estimates (Appendix C; methodology in Online Appendix S8) matter for the argument here. First, the decomposition of village-level CATEs onto the same seven structural predictors corroborates—but does not independently establish—the two design-based patterns: take-up is structurally predictable from village conditions ($R^2 = 0.55$), and the norms and infrastructure reversals appear with the same signs, with sign-agreement rates near one across the 200 GML replications. Second, the pattern is moderator-specific rather than general: Security acts as a returns-shifter, not a barrier—secure villages support larger IGA gains (+0.484*** in the design-based interactions) without differentially gating participation—and Econ. Infra. enters Never Pregnant *positively* ($\gamma = +0.105$, $p = 0.005$), an anti-barrier alignment, so each channel’s opposite-signed pattern is specific to one moderator. Third, the dispersion at stake is large: the GATES decomposition predicts up to a 42 percentage-point reduction in the risk of ever being pregnant for the most-affected quintile (Online Appendix Table S180). Fourth, the exogamy sign-reversal falsification is one-legged outside T2: the income-generation leg is robust in every arm, the marriage-delay leg is detectable only in T2, and as a joint intersection–union prediction—requiring both legs to reject individually—the reversal is established in T2 alone; the weaker cross-family contrast test also rejects for T1 ($p_H = 0.015$) but T1’s delay leg is individually null (Online Appendix Table S179).

C.2 GML estimates

BLP: average and heterogeneous effects. BLP estimates are broadly consistent with Table 1, with some precision gains from variance reduction (Online Appendix Tables S42 – S45 contain all BLP results). Emotional violence rises under all arms, and mental health shows no significant average movement in any arm ($\hat{\beta}_1$ between -0.11 and $+0.06$, all $p > 0.16$), consistent with the null index ITT (Section 6.1.2). IGA increases under T1 ($\hat{\beta}_1 = 0.229$, $p = 0.004$ in BLP, compared to $p = 0.218$ on the unadjusted ITT), a discrepancy worth flagging.²⁶ The emotional-violence increase appears under all arms. The T2/T3 specificity rules out reporting effects generated by safe-space participation alone, which is common to all arms; it cannot, however, separate a true increase in violence induced

²⁶The BLP regresses the residualized outcome on the residualized treatment $\times$ ML proxy, so the precision of $\hat{\beta}_1$ reflects the variance of the proxy as well as the treatment indicator. When the proxy explains a non-trivial share of outcome variance, the residualized regression has lower error variance than the unadjusted ITT and the BLP p -value can be smaller. The IGA T1 finding should therefore be flagged as suggestive of an effect that the unadjusted ITT was underpowered to detect, rather than as evidence of an effect the ITT missed in the substantive sense. An additional reason for caution is that T1 is the one arm with attrition above control (+4.0 percentage points), so it is the arm for which the attrition-bound analysis is least reassuring (see the monotonicity discussion in Section 6.1). The strong heterogeneity loading for this cell ($\hat{\beta}_2 = 0.982$, $p < 0.001$) is consistent with this reading: the BLP-vs-ITT gap reflects heterogeneous effects whose average is borderline rather than a contradiction between the two estimators.

Table (C.1) Assignment $\times$ village-structure interactions (T3): design-based specification

Village predictor X_v	Coefficient on $T3 \times X_v$		
	Take-up	Never Pregnant	IGA
Unrestricted	0.069 (0.044)	-0.081* (0.043)	0.127 (0.129)
Birthcontrols	0.106*** (0.035)	0.007 (0.048)	0.252* (0.144)
Econ. Infra.	0.103** (0.042)	0.105*** (0.037)	-0.242*** (0.108)
Security	-0.031 (0.042)	0.067 (0.054)	0.484*** (0.101)
Male exog.	0.070* (0.038)	-0.043 (0.037)	-0.271* (0.151)
Female exog.	0.021 (0.041)	-0.031 (0.041)	-0.112 (0.170)
Overall exog.	0.055 (0.036)	-0.009 (0.045)	-0.241 (0.165)
N (individuals)	2596	2596	2595

Notes: Each cell is the coefficient γ on $T3 \times X_v$ from a separate individual-level regression $y_i = \alpha + \beta T3 + \gamma (T3 \times X_v) + \delta X_v + \text{strata FE}$ on the T3-union-control (non-attrited) sample, with no ML first stage. X_v is standardized to mean 0, SD 1 across the villages in the estimation sample, so γ is per village-SD. Standard errors clustered at the village level in parentheses. Outcomes are coded so that a positive coefficient is the favourable direction (Never Pregnant = 1 – Ever Pregnant). The barrier-consistent pattern is opposite signs of γ on take-up and on the outcome. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (analytic cluster p -values; randomization-inference and familywise-corrected p -values for this family are reported in Table C.2).

Table (C.2) Focal interaction family: assignment $\times$ village structure (T3)

Predictor X_v	Outcome	γ	SE	p (cluster)	p (RI)	p (Holm-RI)
Unrestricted	Take-up	0.069	(0.044)	0.122	0.162	0.346
Unrestricted	Never Pregnant	-0.081	(0.043)	0.065	0.110	0.346
Unrestricted	IGA	0.127	(0.129)	0.326	0.375	0.375
Econ. Infra.	Take-up	0.103	(0.042)	0.016	0.043	0.256
Econ. Infra.	Never Pregnant	0.105	(0.037)	0.005	0.045	0.256
Econ. Infra.	IGA	-0.242	(0.108)	0.028	0.086	0.346

Notes: The focal family was declared before this randomization-inference analysis but after the exploratory heterogeneity analyses and after registration; it is therefore post-hoc, and these six cells—like all interaction results in the paper—are exploratory. The family comprises $T3 \times \{\text{Take-up, Never Pregnant, IGA}\} \times \{\text{Unrestricted, Econ. Infra.}\}$, the two moderators and three outcomes named by the participation-barrier hypothesis; the Holm column disciplines multiplicity within this declared family but cannot confer pre-specified status. The family excludes Security $\times$ IGA (+0.484, RI $p = 0.001$), the strongest interaction in the wider table, because the barrier hypothesis names Security as a returns-shifter rather than a participation barrier; under a Holm correction across all 63 estimated interaction cells it is the only cell surviving at the ten-percent level. Each row is the coefficient γ on $T3 \times X_v$ from the design-based specification of Table C.1. p (cluster) is the analytic village-clustered p -value; p (RI) is the randomization-inference p -value from 10000 permutations of the four-category assignment across villages within randomization strata (arm counts fixed), comparing $|t|$; p (Holm-RI) applies the Holm step-down correction to the six RI p -values. No PDS-LASSO controls enter these regressions, so no selection step is held fixed under permutation.

by training content from improved recognition and reporting of violence induced by that same content, since both produce the identical arm pattern. We therefore treat improved recognition and reporting as a live alternative reading of the emotional-violence results throughout. The heterogeneity coefficient $\hat{\beta}_2$ is statistically significant for take-up, never-pregnant, emotional violence, IGA, mental health, and all early-marriage perception outcomes across all three arms, indicating the presence of heterogeneous treatment effects, consistent with Definition 1. For the marriage-timing outcomes (never married, age at marriage, marriage before 18), currently-pregnant, and pregnancy before 18, we cannot reject $\hat{\beta}_2 = 0$ in nearly every arm–outcome cell (the exception is never-married under T2), suggesting either treatment homogeneity or a noisy proxy along these margins.

GATES: the dispersion of treatment effects. GATES results (Figure C.1 for the pregnancy outcomes; Figure S2 for the economic and social outcomes; the extreme-quintile point estimates and coverage regions quoted below are tabulated in Online Appendix Table S180; the full set of GATES results are gathered in Tables S46 – S49) reveal stark dispersion across the proxy-score quintiles. Take-up contrasts are especially pronounced under T1 and T3: the bottom-quintile effect is indistinguishable from zero while the top-quintile effect reaches 53 and 68 percentage points, respectively. For never-pregnant, girls in the highest-effects group experience up to a 41 percentage-point reduction in the risk of ever being pregnant. A similar gradient appears for IGA. Negative effects for the lowest-effects quintile on pregnancy and emotional violence indicate that some girls are on average harmed under assignment, reinforcing that the near-zero ITT is a diluted average of large positive and small negative effects across subgroups—the pattern Definition 1 formalizes.

CLAN: the participation barrier. Figure C.2 reveals the paper’s central heterogeneity pattern: the girls predicted to benefit most from the program are *not* those most likely to participate. For take-up (T3), the highest-effect group comes from villages with more progressive social norms, better SRH access, and stronger economic infrastructure—favorable conditions for program engagement. In contrast, for childbearing the highest-effect group, those for whom the program is predicted to deliver the largest pregnancy reduction, comes from villages with the *most* conservative social norms for all three arms. The villages where predicted returns are largest are thus also those where the participation response is smallest. Read through the framework, this pattern points to $\partial(\Pi - \Omega)/\partial C_v < 0$ —the outside-option-cost channel dominating in conservative villages—though as an exploratory contrast it selects, rather than confirms, this sign of Comparative Static 2. All CLAN results are in Online Appendix Tables S50 – S119.

A corroborating decomposition of the village-level CATEs. The GML output permits a sharper but noisier version of the same test: regressing each village-level CATE (for take-up and for each main outcome) on the same seven standardized predictors, in the spirit of best-linear-projection summaries of machine-learning heterogeneity (Semenova and Chernozhukov, 2021; Athey et al., 2019). The CATEs are the GML S -statistic of Chernozhukov et al. (2025) averaged across girls within each cluster (84 villages in the T3 sample); the predictors are standardized to mean zero and unit standard deviation, so slopes are per-standard-deviation. Strata fixed effects and heteroskedasticity-robust standard errors are used throughout. For each dependent CATE, columns 1–7 of Online Appendix Tables S122, S123, and S124 report the seven univariate regressions; column 8 reports the joint specification with all seven predictors and an F -test of joint significance.²⁷ Because the dependent variables are themselves estimates, the analytic standard errors in these tables ignore first-stage estimation error; for that reason the CATE-regression tables carry no significance stars, we avoid significance language for individual slopes, and the across-split sign-agreement rate is our stability metric. We characterize their stability by re-estimating every regression once per GML replication and reporting the across-split distribution of slopes—median, empirical 90% interval, and the share of replications in which the sign agrees (Online Appendix S11). These are descriptive stability diagnostics, not tests: the replications share data and are not independent, so agreement rates carry no formal error-rate guarantee. We use them only to flag slopes whose sign is sensitive to the sample split—we quote the agreement rate alongside each headline estimate below and draw no conclusion from cells where it is low. No conclusion in this section rests on a CATE regression alone: every claim we carry forward is anchored in the design-based interactions of Table C.1 and corroborated—not established—by the decomposition.

²⁷Tables for the remaining six T3 dependents (Age at Mar., Mar. af18, Curr. Not Preg., Preg. bf18, No Emo. Viol., Mental Health), the T1 and T2 arms, the five family-of-outcomes indices (Online Appendix Table S158), and specifications without strata fixed effects are reported in Online Appendix S11. Online Appendix S11 also reports the split-uncertainty tables, the full design-based interaction tables for all arms and outcomes, and the original CATE-on-CATE reduced-form regression (Online Appendix Tables S159 and S160) for comparison.

In both designs the barrier-consistent pattern is *opposite signs* on take-up and on the outcome—for γ in the assignment interactions and for the slopes in the CATE decomposition: a predictor that raises participation should depress the outcome benefit, and conversely. For the four infrastructure-type predictors (Unrestricted, Birthcontrols, Econ. Infra., Security) we expect a positive slope on take-up and a negative slope on each desirable-positive outcome CATE (with the sign flipped on Preg. bf18, the only non-inverted outcome). For the three exogamy indices we expect positive slopes on marriage and fertility outcomes and a negative slope on the IGA outcome, reflecting the sign-reversal of Proposition 2.

Take-up. The design-based interactions document the gradient: assignment raises take-up by more in better-connected villages ($\gamma = +0.103$ per SD of Econ. Infra., $p = 0.016$; Table C.1), while the norms gradient is positive but individually insignificant ($\gamma = +0.069$, $p = 0.122$) and does not survive controls for documented implementation disruption (it falls to -0.019 ; see *Attrition and implementation sensitivity* below), so differential reach by norms cannot be attributed to demand-side constraints. The CATE decomposition corroborates: take-up is structurally predictable from village structure ($R^2 = 0.55$; joint $F(7, 68) = 3.78$, $p = 0.002$; Online Appendix Table S122), with the same infrastructure predictors standing out: Econ. Infra. (+0.065), Birthcontrols (+0.059), and Unrestricted (+0.033 univariately, +0.040 in the joint specification), each with a sign-agreement rate of at least 0.99 across the GML replications (Online Appendix S11). Security is null, and the exogamy indices are small univariately (overall exogamy +0.032) and jointly null. One caveat disciplines the headline number: under one-sided noncompliance the take-up CATE tracks the take-up level itself, so the decomposition establishes that *participation* is structurally predictable—the premise the barrier test requires—rather than uncovering treatment-effect heterogeneity per se. The infrastructure side of the framework therefore predicts the cross-village distribution of program engagement; the marriage-market side does not.

Pregnancy. On the autonomy dimension the reversal appears directly in the assignment interactions, though with modest individual significance: per SD of Unrestricted, T3 raises take-up by +0.069 ($p = 0.122$) while its effect on remaining never-pregnant declines by 0.081 ($p = 0.065$; Table C.1). The CATE decomposition shows the same reversal: Unrestricted is positive on the take-up CATE (+0.040 in the joint specification) and -0.061 on the Never Pregnant CATE (-0.064 jointly; Online Appendix Table S123), each slope with sign agreement of at least 0.99. Conservative villages thus combine the lowest take-up response with the largest pregnancy benefit, exactly the pattern Comparative Static 2 predicts. In the CATE decomposition, Econ. Infra. is positive on this outcome (+0.037 univariately, +0.040 jointly), Security is positive but small (+0.014), and the Birthcontrols slope is split-sensitive on this outcome (sign agreement 0.63 in the joint specification), so we draw no conclusion from it. In the design-based interactions, Econ. Infra. enters Never Pregnant *positively* ($\gamma = +0.105$, $p = 0.005$; Table C.1)—the same sign as its take-up interaction, an anti-barrier alignment. The opposite-signed barrier pattern on the pregnancy channel is therefore specific to the norms index, not to infrastructure more broadly.

Income-generating activity. The design-based interactions again provide the primary evidence: per SD of Econ. Infra., $\gamma = +0.103$ on take-up ($p = 0.016$) versus $\gamma = -0.242$ ($p = 0.028$) on IGA (Table C.1). The CATE decomposition shows the same reversal on the infrastructure dimension (Online Appendix Table S124): Econ. Infra. is +0.065 on the take-up CATE but -0.101 on the IGA CATE (-0.125 in the joint specification), with sign agreement 1.00: villages with weak economic infrastructure have low predicted take-up and large predicted IGA gains, and conversely. The slope is negative in every one of the 200 GML replications (Online Appendix S11), so we treat the sign as unambiguous. The joint specification reaches $R^2 = 0.56$ ($F(7, 68) = 7.56$, $p < 0.001$). Read through the framework, this is the cross-village pattern Comparative Static 1 assigns to substitution: in income generation the program substitutes for—rather than complements—market connectivity. An important competing explanation specific to this outcome is selection and delivery: weak-infrastructure villages experienced the greatest implementation disruption and the highest attrition (Section 6.1), so part of the gradient may reflect who is observed and what was delivered rather than latent returns. The sensitivity audit below quantifies this: the gradient survives disruption controls (-0.259) but reverses to +0.093 in villages with high follow-up retention. Security is null on take-up but strongly positive on the IGA gradient in both designs (+0.346 in the joint decomposition; +0.484*** in the interactions): a returns-shifter rather than a barrier, since secure villages support larger income-generation gains without differentially gating participation. The exogamy indices carry the predicted negative-on-IGA sign univariately (male exogamy -0.142 , overall exogamy -0.146); in the joint specification the male-exogamy slope flips sign while overall and female exogamy stay negative; the Unrestricted slope on this outcome is split-sensitive (sign agreement 0.84–0.89), and we draw no conclusion from it.

Other outcomes. Age at Marriage and Mar. af18 exhibit no barrier pattern on the infrastructure dimension: their strongest predictor is Security (+0.114 univariate and +0.161 joint on Age at Marriage, +0.020 univariate on

Mar. af18; joint $F = 7.39$ and $F = 14.12$), Unrestricted enters Age at Marriage *positively* (+0.077)—the same direction as its take-up slope, not the reversal—and the exogamy indices are negative on Age at Marriage univariately (−0.091 to −0.111). Selection of high-return girls into participation on the marriage-delay channel appears instead at the individual level: within villages, the take-up CATE rises with the Age-at-Marriage CATE (+0.037; Online Appendix Table S160). Emotional-violence returns load on Econ. Infra. in the anti-barrier direction (+0.029 univariately), with the seven predictors weakly related jointly ($F = 1.99$). Mental-health returns are not predictable from village structure ($F = 0.67$; no predictor stands out), so the mental-health cost is not concentrated where take-up is structurally lowest. Currently-not-pregnant is likewise unpredictable ($F = 0.70$). The early-pregnancy outcome Preg. bf18 is jointly predictable ($F = 3.28$), with Security (−0.020) and male exogamy (+0.015) the strongest slopes and Unrestricted smaller (−0.012): more secure villages show *larger* predicted reductions in early pregnancy—an anti-barrier alignment on this outcome rather than the flipped-sign barrier pattern the framework assigns to this non-inverted outcome.

Village versus individual heterogeneity. Apportioning treatment-effect heterogeneity between village and individual dimensions from the GML output requires care. The natural decomposition—the between-village share of variance in the individual-level proxy scores—is largely circular here: the proxies are functions of the learners’ covariate set, in which village-level variables feature prominently, so a high between-village share mostly reflects the information the learners were given rather than an established dominance of village over individual conditions, and split-level estimation error smoothed by median aggregation need not preserve the within-/between-village split. With that caveat stated, one feature of the decomposition is not circular: the learners observe the same covariate set for every outcome, so the *cross-outcome contrast* in where they place predicted heterogeneity is informative about the outcomes rather than the features. That contrast is sharp. The between-village share (ANOVA intraclass correlation across the 84 T3 villages) is 0.78–0.94 for the take-up, income-generation, normative-attitude, and mental-health scores, but falls to 0.37–0.39 for the marriage-timing outcomes (age at marriage, never married, pregnancy before 18)—the model allocates heterogeneity to the individual level precisely on the channel where the design-based evidence indicates self-selection of high-return girls rather than village structure. The allocation is also the within-country analog of the companion evaluation’s finding that country indicators dominate individual characteristics in predicting treatment-effect variation (Bertelli et al., 2026), with the difference that the community layer is measured here rather than absorbed in an indicator. Individual and household characteristics do not predict the estimated scores better than village structure: regressions of the individual CATEs on 19 individual and household baseline characteristics (plus age and strata dummies) achieve R^2 comparable to what the seven village-level structural predictors achieve on their own—with the circularity caveat applying symmetrically. The village-level framing of the participation-barrier analysis therefore rests on the design-based assignment interactions of Table C.1, which use no machine-learning first stage, not on a variance decomposition of estimated scores.

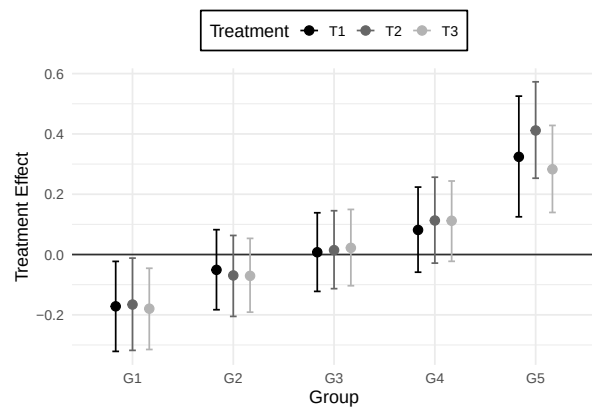
Exogamy falsification test (Proposition 2). Proposition 2 delivers the framework’s most distinctive testable prediction: the same village-level exogamy structure should produce a positive gradient for marriage-delay outcomes (under male exogamy σ_v^g , since outside grooms cannot observe or price the woman’s program-induced outside-option gain) and a negative gradient for IGA outcomes (under female exogamy σ_v^b , since girls anticipating out-marriage invest less in the local labor market). Because the prediction is a sign reversal across outcome families generated by the same village-level moderator, no uniform village-level confound can replicate it. One condition disciplines this argument: the two exogamy margins must covary positively across villages, since if $\text{Corr}(\sigma_v^g, \sigma_v^b)$ were negative, a confound loading positively on one margin would load negatively on the other and could generate the opposite-signed gradients mechanically. The condition holds in the data: the two margins are positively correlated ($\hat{\rho} = 0.39$ across the 162 study villages, $p < 0.001$; between 0.33 and 0.41, all $p < 0.01$, within each arm-specific estimation sample). A moderate positive correlation protects the test against confounds that load *uniformly* on both margins; it does not exclude a confound that loads differentially on the two margins, which could still generate opposite-signed gradients, so the insulation the sign-reversal argument provides is partial rather than complete. Online Appendix Table S179 reports the CLAN δ estimates and 90% confidence intervals for σ_v^g separately on the marriage-delay outcome family (Never Married, Age at Mar., Mar. af18) and for σ_v^b on the income-generation family (IGA). Holm-adjusted p -values are reported within each panel. The final row reports the cross-family test of $H_0 : \delta_{\sigma_v^g}^{\text{delay}} = -\delta_{\sigma_v^b}^{\text{IGA}}$ (i.e., the two coefficients have signs that are statistically distinguishable across families). The IGA gradient on σ_v^b is negative in every arm and survives the Holm correction in all three (T1: $\delta = -0.156$, $p_H = 0.015$; T2: -0.187 , $p_H = 0.015$; T3: -0.110 , $p_H = 0.020$), consistent with the prediction that girls anticipating out-marriage invest less in the local labor market. The marriage-delay gradient on σ_v^g is null in eight of the nine

outcome-by-arm cells under Holm correction; only $T2 \times \text{Never Married}$ is positive ($\delta = +0.309$, $p_H = 0.045$). The cross-family contrast supports the sign reversal at conventional significance for T1 and T2 ($\bar{\delta}^{\text{delay}} - \delta^{\text{IGA}} = +0.202$ and $+0.312$, both $p_H = 0.015$) and marginally for T3 ($+0.081$, $p_H = 0.100$). Because the prediction is a conjunction—it requires *both* a positive delay gradient and a negative IGA gradient—the appropriate joint test is intersection–union: the reversal is established only where both legs reject, which holds in T2 alone (Never Married $p_H = 0.045$ and IGA $p_H = 0.015$; the intersection–union p is the larger of the two, 0.045) and fails in T1 and T3, where only the IGA leg rejects. We therefore read the evidence as one-legged outside T2: the income-generation leg of the prediction is robust in every arm, while the marriage-delay leg is statistically detectable only in the safe-spaces-plus-training arm, so the sign-reversal pattern rests more on the income-generation side than on the marriage side.

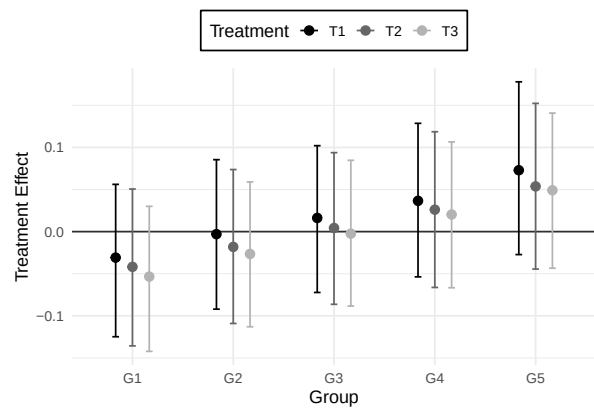
Taken together, BLP, GATES, and CLAN do four things. They (a) leave Proposition 1 unsupported: the signed SRH-access prediction fails in the parametric tertile analysis (Section 6.2), and nothing in the data-driven analysis reverses that conclusion, so the framework’s signed content rests on the exogamy falsification alone; (b) provide partial support for Proposition 2’s sign-reversal falsification, which is established as a joint (intersection–union) prediction in T2 only, with the income-generation leg robust in every arm and the cross-family contrast significant for T1 and T2 and marginal for T3 (Online Appendix Table S179); (c) point, when read through the framework, toward the outside-option-cost channel for Comparative Static 2 ($\partial(\Pi - \Omega)/\partial C_v < 0$) and toward substitution for Comparative Static 1 ($\partial(\Pi - \Omega)/\partial L_v < 0$ for IGA), with the design-based interactions and the structural-predictor decomposition (Table C.1 and Online Appendix Tables S122–S124) jointly singling out *Unrestricted* (norms) as the operative dimension for pregnancy and *Econ. Infra.* for IGA; and (d) document that participation is structurally predictable from village conditions ($R^2 = 0.55$, $F(7, 68) = 3.78$)—the premise the barrier test of Definition 1 requires, though under one-sided noncompliance this decomposition tracks take-up levels rather than treatment-effect heterogeneity per se. The marriage-delay channel shows no analogous pattern and instead self-selects high-return girls into participation.

A mechanical scaling exercise. A back-of-envelope exercise conveys the magnitudes at issue under strong, explicitly counterfactual assumptions (the exercise is illustrative only). Using tercile-specific ITTs estimated directly from assignment (untabulated; replication package), the T3 take-up ITT is 10.2 percentage points in the most conservative norms tercile against 38.7 in the middle tercile, and the implied per-participant ever-pregnant effect in the conservative tercile is -1.33 —a ratio outside the unit interval, itself a warning that the quotient divides a small, imprecisely estimated take-up ITT into a larger pregnancy ITT and should be read as an artifact of that division. Taken at face value, raising the conservative tercile’s take-up response to the middle tercile’s level with conditional returns held fixed would improve the aggregate ever-pregnant ITT by roughly 12.9 percentage points—moving an essentially null aggregate effect (-0.006 , Table 1) to about -0.14 , beyond the ex-post MDE for this outcome (0.109). The analogous calculation on the infrastructure margin (take-up ITTs of 20.1 versus 34.6 percentage points, implied per-participant IGA effect of 0.65 SD) raises the aggregate IGA ITT by about 0.03 SD, roughly a quarter of its current point estimate. These figures are mechanical, not estimates of the effect of raising participation: they assume assignment affects outcomes only through measured take-up, that conditional returns stay fixed as marginal participants enter, and that no equilibrium adjustment occurs—precisely the assumptions this design cannot test. We report the exercise only to convey what would be at stake if a participation constraint were binding and returns portable.

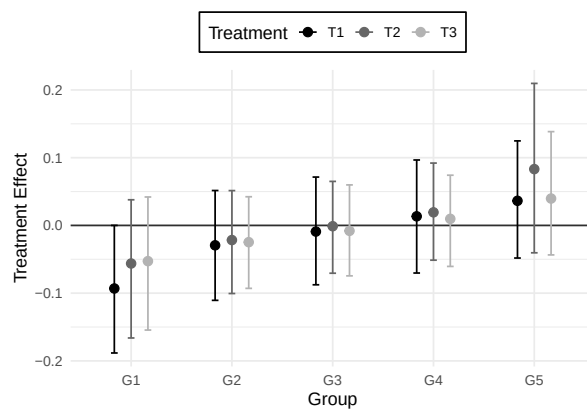
Figure (C.1) GATES – Pregnancy Outcomes



(a) Never Pregnant

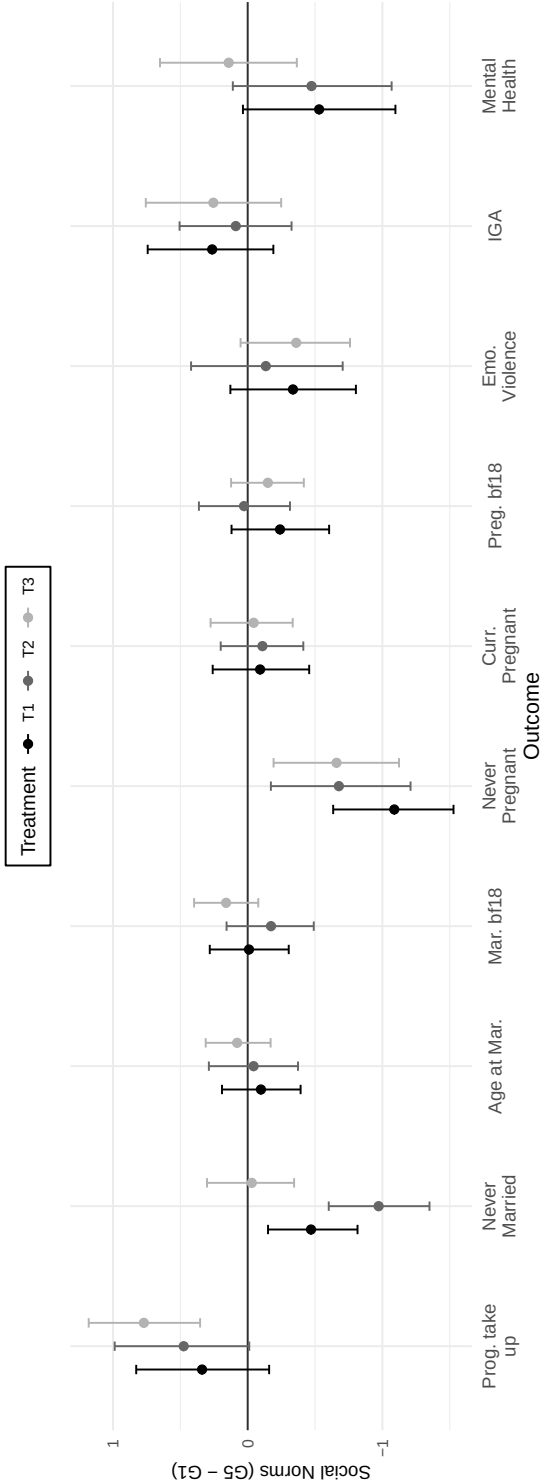


(b) Currently Pregnant

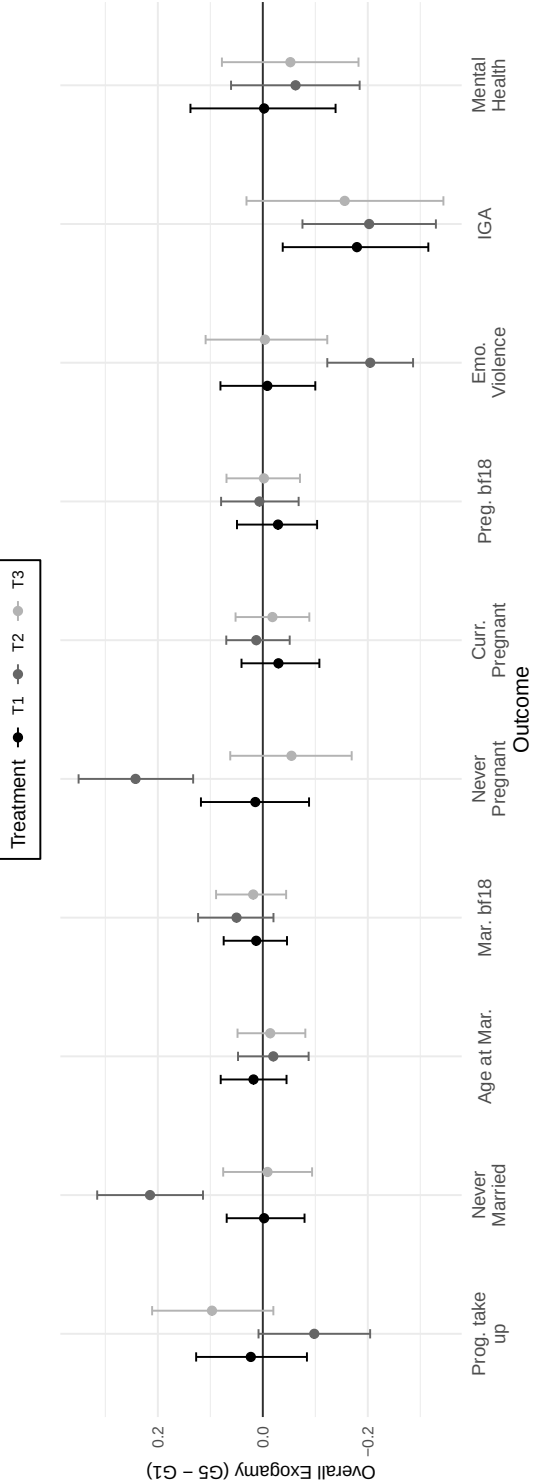


(c) Pregnancy before Age 18

Figure (C.2) Village-level CLAN Results for Main Outcomes

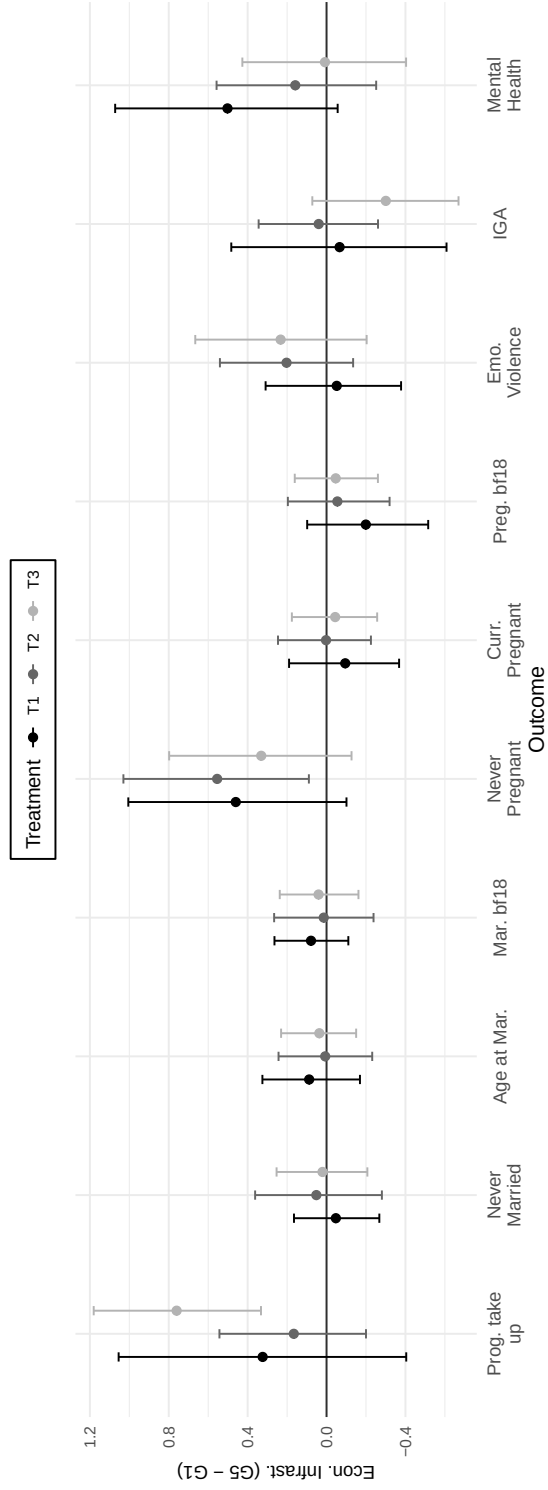


(a) Social norms index

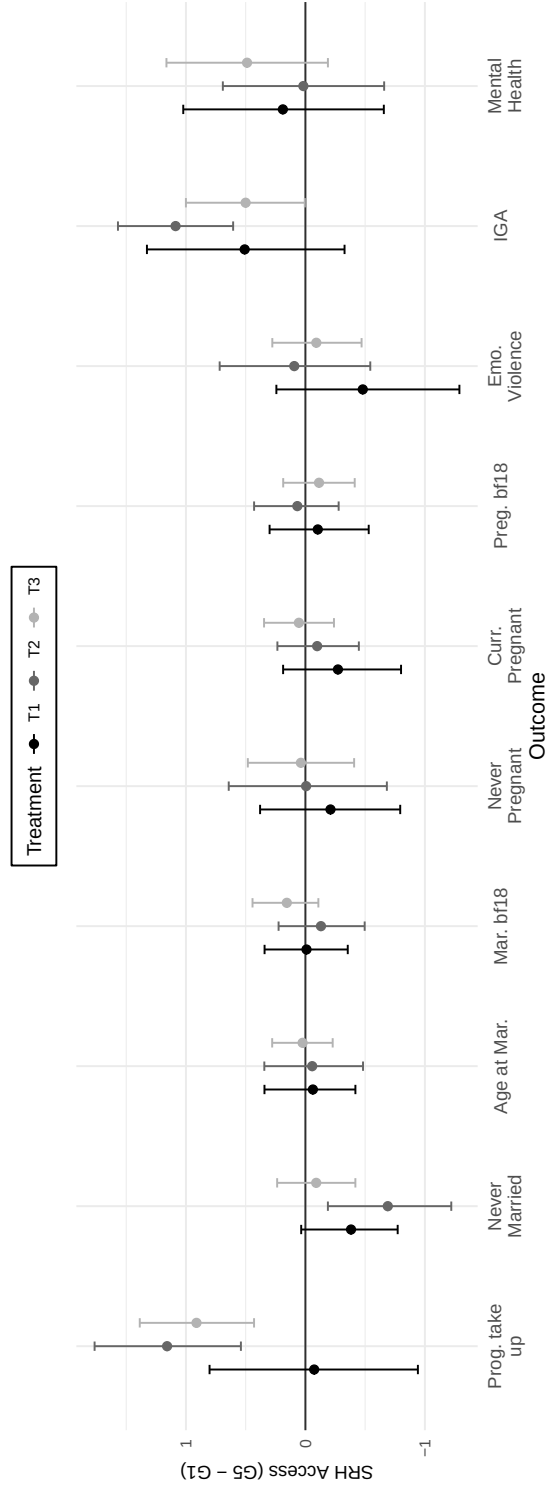


(b) Exogamy index

Figure (C.2) Village-level CLAN Results for Main Outcomes (continued)



(c) Economic infrastructure index



(d) Access to birth control index

This appendix collects the study timeline, the randomization design, and the map of the study villages. The participant-flow display is in Online Appendix S12.

Figure (D.1) Study Timeline

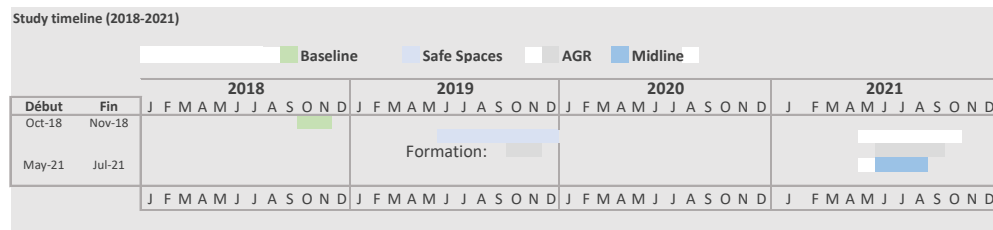


Figure (D.2) Randomization Design

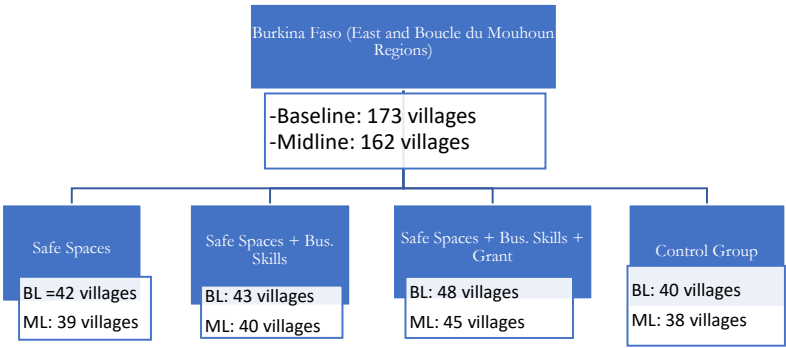
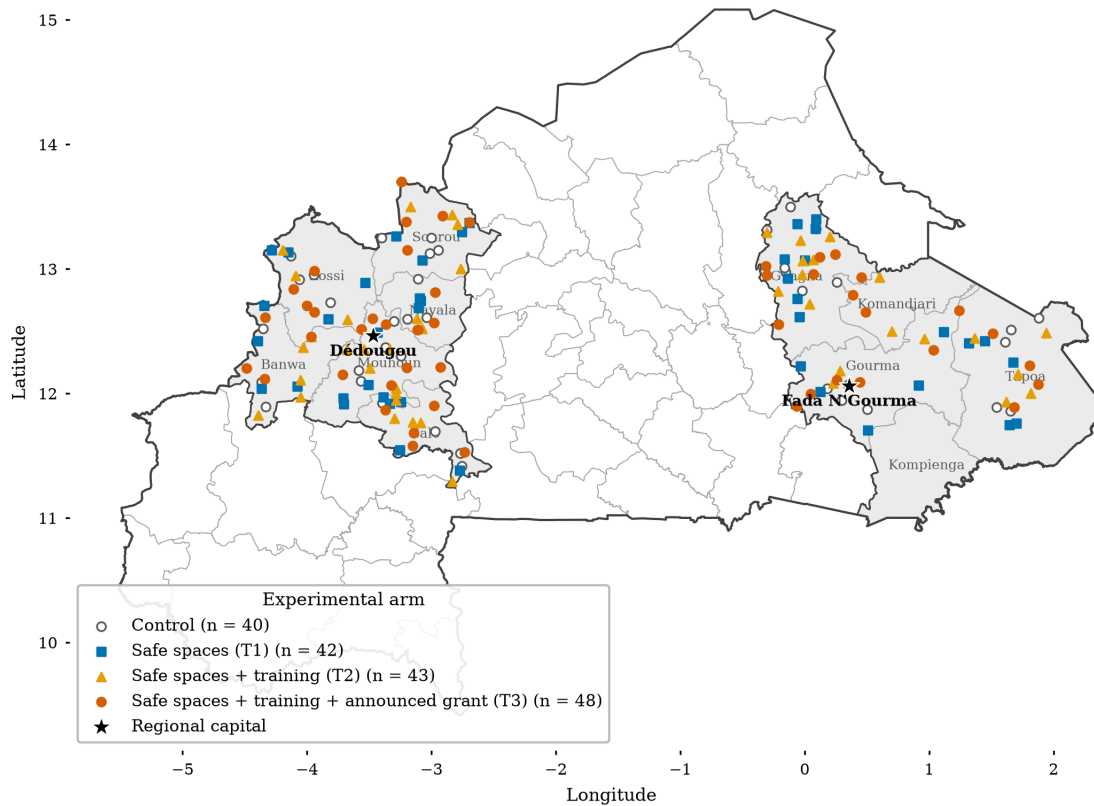


Figure (D.3) Map of the Study Villages



Notes: Study villages by experimental arm in the Boucle du Mouhoun and Est regions of Burkina Faso. Stars mark the regional capitals, Dédougou and Fada N'Gourma, where the public randomization lotteries were held (Section 2.3).

E Post-Mortem Power Analysis

Security-related attrition reduced the follow-up sample from 7,719 girls in 173 villages to 4,936 girls in 162 villages. This appendix summarizes an ex-post precision counterfactual that asks how precisely each treatment-arm coefficient of Tables 1 and 2 would have been estimated had the full baseline sample been retained. Point estimates are held at their realized values, so the exercise isolates precision; whether attrition moved the estimands themselves is the province of the bounds discussed in Section 6.1.

The follow-up loss was concentrated within villages: 36 percent of girls were lost, but 94 percent of the village clusters survived. With standard errors clustered at the village level and intra-cluster correlations of 0.04–0.54 across outcomes, additional respondents in already-observed villages contribute far less precision than additional villages would. Holding each outcome’s standard deviation and intra-cluster correlation at their realized values and rescaling the cluster-robust standard errors by the design-variance ratio, the counterfactual-to-realized standard-error ratio lies between 0.89 and 0.96 for all 26 outcomes: full-sample retention would have tightened the standard errors by only 4–11 percent. A companion adjustment that re-measures each outcome’s dispersion and clustering on the full baseline sample rather than the retained girls’ baseline records leaves this conclusion unchanged.

No coefficient changes inferential status under the counterfactual. The two borderline cells sharpen slightly—age at marriage under T3 moves from an implied p of 0.054 to 0.029, emotional violence under T3 from 0.09 to 0.07—while the marriage and fertility nulls remain firmly null, with counterfactual confidence intervals nearly as wide as the realized ones. The average nulls of Table 1 are therefore not an artifact of attrition-induced power loss. Online Appendix S4 presents the methods, the full coefficient-by-coefficient results (Online Appendix Tables S23, S24, and S25), the design-based minimum detectable effects under both designs (Online Appendix Table S26), and the underlying design parameters (Online Appendix Table S27).